HCIA-AI V4.0 Certification Practice Exam • Chapter 1

Module 1: AI Overview (100 Questions & Answer Key)

Curated by: Umed Jamal Nouri • v1.1 (Nov 2026) • github.com/UmedJaMala

Question 1

TRUE OR FALSE • 16 Points

Speech processing is an AI discipline that only handles optical text character recognition, and cannot process acoustic voice signals or speech synthesis.

A. True
B. False

Question 2

TRUE OR FALSE • 16 Points

Huawei's all-scenario AI solutions can be deployed in all kinds of environments, including public cloud, private clouds, various forms of edge computing, industrial IoT devices, and consumer devices.

A. True
B. False

Question 3

TRUE OR FALSE • 16 Points

Artificial Intelligence (AI) is a subset of Machine Learning, and Machine Learning is a subset of Deep Learning.

A. True
B. False

Question 4

TRUE OR FALSE • 16 Points

Foundation models require developers to train a completely new model from scratch with random initial weights for every distinct downstream task, completely prohibiting transfer learning.

A. True
B. False

Question 5

TRUE OR FALSE • 16 Points

An AI smart speaker relies only on computer vision algorithms and does not use speech processing or natural language understanding technologies.

A. True
B. False

Page Answers: **Q1**: False • **Q2**: True • **Q3**: False • **Q4**: False • **Q5**: False

Question 6

TRUE OR FALSE • 16 Points

Connectionism believes that intelligence originates from symbolic operations and knowledge representation systems rather than neural network connections.

A. True
B. False

Question 7

TRUE OR FALSE • 16 Points

Behaviorism asserts that intelligence is formed through continuous interaction, perception, and action with the surrounding environment.

A. True
B. False

Question 8

TRUE OR FALSE • 16 Points

The Turing Test was proposed by Alan Turing in 1950 to provide an operational definition of whether a machine can exhibit human-equivalent intelligence.

A. True
B. False

Question 9

TRUE OR FALSE • 16 Points

The term 'Artificial Intelligence' was officially coined at the Dartmouth Conference held in the summer of 1956.

A. True
B. False

Question 10

TRUE OR FALSE • 16 Points

Artificial General Intelligence (AGI) refers to narrow AI systems designed only to solve a single dedicated task, such as playing tic-tac-toe.

A. True
B. False

Page Answers: **Q6**: False • **Q7**: True • **Q8**: True • **Q9**: True • **Q10**: False

Question 11

TRUE OR FALSE • 16 Points

In Natural Language Processing, Part-of-Speech (POS) tagging and Named Entity Recognition (NER) are typical sequence labeling tasks.

A. True
B. False

Question 12

TRUE OR FALSE • 16 Points

Knowledge graphs represent structured real-world knowledge primarily in the form of entity-relation-entity triplets (head entity, relation, tail entity).

A. True
B. False

Question 13

TRUE OR FALSE • 16 Points

Semantic segmentation classifies each pixel in an image into a category label, while distinguishing different individual instances of the exact same category.

A. True
B. False

Question 14

TRUE OR FALSE • 16 Points

Optical Character Recognition (OCR) converts textual imagery captured by cameras or scanners into editable, machine-encoded text.

A. True
B. False

Question 15

TRUE OR FALSE • 16 Points

The 'Emergence' phenomenon in large foundation models refers to complex capabilities (such as multi-step arithmetic or analogical reasoning) that appear spontaneously once the model parameter scale and data size surpass certain critical thresholds.

A. True
B. False

Page Answers: **Q11**: True • **Q12**: True • **Q13**: False • **Q14**: True • **Q15**: True

Question 16

TRUE OR FALSE • 16 Points

DeepSeek models adopt Multi-Head Latent Attention (MLA) architecture primarily to significantly reduce KV cache memory consumption during inference while retaining performance.

A. True
B. False

Question 17

TRUE OR FALSE • 16 Points

DeepSeek-R1 relies purely on supervised instruction fine-tuning without using reinforcement learning to achieve its complex mathematical and logical reasoning capabilities.

A. True
B. False

Question 18

TRUE OR FALSE • 16 Points

In the Mixture of Experts (MoE) architecture used by advanced foundation models like DeepSeek-V3, all expert sub-networks are activated at the same time for every single token processed.

A. True
B. False

Question 19

TRUE OR FALSE • 16 Points

Huawei's Pangu-Weather model uses 3D neural networks for medium-range global weather forecasting, achieving higher prediction speeds than conventional numerical weather prediction (NWP) systems.

A. True
B. False

Question 20

TRUE OR FALSE • 16 Points

AI4Science refers to applying artificial intelligence techniques and large foundation models to fundamental scientific research fields such as drug discovery, materials science, and protein structure prediction.

A. True
B. False

Page Answers: **Q16**: True • **Q17**: False • **Q18**: False • **Q19**: True • **Q20**: True

Question 21

TRUE OR FALSE • 16 Points

Watermarking generated content and establishing cryptographic provenance are technical countermeasures to address the spread of unauthorized deepfakes and AI-generated disinformation.

A. True
B. False

Question 22

TRUE OR FALSE • 16 Points

Embodied AI focuses only on software chatbots living only in cloud servers with no connection to physical entities or robotic actuators.

A. True
B. False

Question 23

TRUE OR FALSE • 16 Points

Data security and privacy protection laws (such as GDPR) require that personal data used in AI training must comply with principles of lawfulness, user consent, and data minimization.

A. True
B. False

Question 24

TRUE OR FALSE • 16 Points

Chain of Thought (CoT) prompting strictly prohibits language models from generating intermediate reasoning steps and only allows direct one-word outputs.

A. True
B. False

Question 25

TRUE OR FALSE • 16 Points

Model explainability (or interpretability) is of negligible importance when deploying AI in high-stakes domains like clinical medical diagnosis, autonomous driving, and judicial sentencing.

A. True
B. False

Page Answers: **Q21**: True • **Q22**: False • **Q23**: True • **Q24**: False • **Q25**: False

Question 26

SINGLE-ANSWER QUESTION • 16 Points

Which of the following correct hierarchical relationships describes Artificial Intelligence, Machine Learning, and Deep Learning?

- **A.** Deep Learning (subset of) AI (subset of) Machine Learning
- **B.** Machine Learning (subset of) Deep Learning (subset of) AI
- **C.** Deep Learning (subset of) Machine Learning (subset of) AI
- **D.** AI (subset of) Machine Learning (subset of) Deep Learning

Question 27

SINGLE-ANSWER QUESTION • 16 Points

A manufacturer wants to produce virtual voice assistants for hospitals to handle patient inquiries and book appointments. Which of the following technical domains is primarily involved?

- **A.** Autonomous driving and SLAM navigation
- **B.** Computer vision and image reconstruction
- **C.** Natural language processing and speech recognition
- **D.** High-frequency RF antenna impedance tuning

Question 28

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT a typical use case for Natural Language Processing (NLP)?

- **A.** Text summarization of long legal documents
- **B.** Machine translation between English and French
- **C.** Sentiment analysis of product reviews
- **D.** Automatic image super-resolution upscaling

Question 29

SINGLE-ANSWER QUESTION • 16 Points

Which of the following schools of artificial intelligence relies on simulated biological neural networks and connection weights to realize cognitive capabilities?

- **A.** Logicism
- **B.** Connectionism
- **C.** Behaviorism
- **D.** Symbolism

Question 30

SINGLE-ANSWER QUESTION • 16 Points

Which of the following AI schools is rooted in cybernetics and evolutionary computing, arguing that intelligence develops through continuous interaction and feedback with the environment?

- **A.** Symbolism
- **B.** Structuralism
- **C.** Behaviorism
- **D.** Connectionism

Page Answers: **Q26**: C • **Q27**: C • **Q28**: D • **Q29**: B • **Q30**: C

Question 31

SINGLE-ANSWER QUESTION • 16 Points

What are the three fundamental cornerstones (essential elements) driving the modern development of Artificial Intelligence?

- **A.** Python, PyTorch, and MindSpore
- **B.** Deep Learning, Reinforcement Learning, and Supervised Learning
- **C.** Data, Algorithms, and Computing Power
- **D.** Sensors, Internet, and Cloud Storage

Question 32

SINGLE-ANSWER QUESTION • 16 Points

Which Computer Vision task outputs both the bounding box coordinates and the semantic category label for each detected item in an image?

- **A.** Image Classification
- **B.** Object Detection
- **C.** Style Transfer
- **D.** Image Denoising

Question 33

SINGLE-ANSWER QUESTION • 16 Points

In which historical event was the term 'Artificial Intelligence' first formally proposed?

- **A.** Dartmouth Summer Research Project on Artificial Intelligence
- **B.** The Turing Test Paper Publication
- **C.** The Victory of Deep Blue over Garry Kasparov
- **D.** The Invention of the Perceptron by Frank Rosenblatt

Question 34

SINGLE-ANSWER QUESTION • 16 Points

Which level of artificial intelligence possesses general cognitive abilities identical to or exceeding those of human beings across all intellectual domains?

- **A.** Artificial General Intelligence
- **B.** Expert Systems
- **C.** Artificial Narrow Intelligence
- **D.** Reactive Machines

Question 35

SINGLE-ANSWER QUESTION • 16 Points

What is the primary architectural innovation introduced by DeepSeek to compress KV Cache storage requirements during large model inference?

- **A.** Grouped-Query Attention
- **B.** Multi-Head Latent Attention
- **C.** Multi-Head Attention
- **D.** Multi-Query Attention

Page Answers: **Q31**: C • **Q32**: B • **Q33**: A • **Q34**: A • **Q35**: B

Question 36

SINGLE-ANSWER QUESTION • 16 Points

DeepSeek-V3 uses an MoE architecture. What does 'MoE' stand for in modern foundation models?

- **A.** Matrix of Evaluation
- **B.** Multi-objective Estimation
- **C.** Model of Extraction
- **D.** Mixture of Experts

Question 37

SINGLE-ANSWER QUESTION • 16 Points

Which of the following NLP tasks classifies an entire sequence of words (e.g., an email) into predefined categories like 'Spam' or 'Ham'?

- **A.** Sequence Labeling
- **B.** Named Entity Recognition
- **C.** Text Classification
- **D.** Machine Translation

Question 38

SINGLE-ANSWER QUESTION • 16 Points

A Knowledge Graph stores relationships between entities using triplets. Which of the following is the standard triplet format?

- **A.** (Head Entity, Relation, Tail Entity)
- **B.** (Primary Class, Feature Vector, Weight)
- **C.** (Input Tensor, Activation, Output Bias)
- **D.** (Source Node, Distance Metric, Target)

Question 39

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes the difference between Instance Segmentation and Semantic Segmentation?

- **A.** Semantic segmentation labels each pixel by category, whereas instance segmentation additionally separates each individual object instance
- **B.** Semantic segmentation isolates each individual instance separately, whereas instance segmentation aggregates all pixels into one object
- **C.** Semantic segmentation is restricted to binary foreground masks, whereas instance segmentation performs holistic scene categorization
- **D.** Semantic segmentation requires bounding box annotations, whereas instance segmentation relies only on image-level classification tags

Question 40

SINGLE-ANSWER QUESTION • 16 Points

Which of the following technologies enables an autonomous vehicle or surveillance camera to maintain continuous identity tracking of a pedestrian across sequential video frames?

- **A.** Single Image Bounding Box Detection
- **B.** Multiple Object Tracking Framework
- **C.** Static Semantic Foreground Segmentation
- **D.** Global Visual Scene Categorization

Page Answers: **Q36**: D • **Q37**: C • **Q38**: A • **Q39**: A • **Q40**: B

Question 41

SINGLE-ANSWER QUESTION • 16 Points

In the Scaling Law of large foundation models, what primarily determines the predictable improvement in model cross-entropy loss?

- **A.** Optimizer momentum decay factor, dropout rate, and weight initialization seed
- **B.** Activation function non-linearity degree and batch normalization epsilon
- **C.** Gradient clipping threshold, learning rate warmup steps, and L2 penalty
- **D.** Number of model parameters, training dataset size, and total compute budget

Question 42

SINGLE-ANSWER QUESTION • 16 Points

Which prompting technique instructs a large language model to articulate intermediate reasoning steps to dramatically improve deductive problem-solving?

- **A.** Data Normalization and Z-Score Scaling
- **B.** Zero-Shot Chain of Thought
- **C.** Parameter-Efficient Low-Rank Adaptation
- **D.** Stochastic Gradient Descent with Momentum

Question 43

SINGLE-ANSWER QUESTION • 16 Points

Huawei's AI-driven biodiversity monitoring initiative in Chile helped protect which endangered native species using acoustic pattern detection?

- **A.** African Elephant
- **B.** Giant Panda
- **C.** Polar Bear
- **D.** Darwin's fox

Question 44

SINGLE-ANSWER QUESTION • 16 Points

Which Huawei AI large model demonstrated remarkable precision in medium-range global meteorological forecasting, surpassing conventional Numerical Weather Prediction (NWP)?

- **A.** Pangu-Mine
- **B.** Pangu-Drug
- **C.** Pangu-Weather
- **D.** Pangu-Finance

Question 45

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is considered a primary ethical challenge associated with generative AI and synthetic media?

- **A.** Transient electromagnetic interference in optical transceivers
- **B.** Exponential expansion of physical datacenter floor space limits
- **C.** Temporary increases in round-trip fiber network routing latency
- **D.** Spread of fake videos, impersonation, and fraud

Page Answers: **Q41**: D • **Q42**: B • **Q43**: D • **Q44**: C • **Q45**: D

Question 46

SINGLE-ANSWER QUESTION • 16 Points

Which of the following AI technologies combines perception, physical reasoning, and real-time robotic motor control to act in physical environments?

- **A.** Embodied AI
- **B.** Expert systems
- **C.** Linear regression
- **D.** K-means clustering

Question 47

SINGLE-ANSWER QUESTION • 16 Points

What is the primary purpose of Automatic Speech Recognition (ASR) in an intelligent speech processing system?

- **A.** Compressing raw audio files without transcription
- **B.** Converting written text into synthesized audible human voice
- **C.** Converting spoken human acoustic audio signals into readable text
- **D.** Translating text directly between two different languages

Question 48

SINGLE-ANSWER QUESTION • 16 Points

What is the primary function of Text-to-Speech (TTS) technology?

- **A.** Translating spoken audio from one dialect to another
- **B.** Filtering out background noise from recorded audio
- **C.** Extracting keywords from audio transcripts
- **D.** Synthesizing spoken audio output from written text input

Question 49

SINGLE-ANSWER QUESTION • 16 Points

Which of the following AI pioneers proposed the famous concept of a 'Universal Computing Machine' and formulated the imitation game test for machine intelligence?

- **A.** Claude Shannon
- **B.** Alan Turing
- **C.** Norbert Wiener
- **D.** John von Neumann

Question 50

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT an AI for Science (AI4Science) application domain?

- **A.** Predicting 3D protein folding structures
- **B.** Developing fixed rule-based automation
- **C.** Discovering novel battery crystal materials
- **D.** Simulating high-energy molecular dynamics

Page Answers: **Q46**: A • **Q47**: C • **Q48**: D • **Q49**: B • **Q50**: B

Question 51

SINGLE-ANSWER QUESTION • 16 Points

What does a 'Token' typically represent in the context of large language model processing?

- **A.** A complete raw document file stored in the training dataset
- **B.** The basic atomic semantic text unit parsed by a model
- **C.** A fixed-length sentence paragraph separated by newline characters
- **D.** A trainable weight matrix inside the feed-forward network layer

Question 52

SINGLE-ANSWER QUESTION • 16 Points

What major open-source contribution did DeepSeek provide to the global AI development community?

- **A.** Mandating proprietary commercial licenses on all transformer attention weights
- **B.** Banning the deployment of mixture-of-experts architectures across private clouds
- **C.** Releasing open model weights and architectural reports achieving frontier performance
- **D.** Requiring static compilation of all dynamic computation graphs prior to execution

Question 53

SINGLE-ANSWER QUESTION • 16 Points

Which Computer Vision generative architecture uses a forward noise-adding process followed by a learned reverse denoising process to create high-fidelity images?

- **A.** Diffusion Models
- **B.** K-Nearest Neighbors
- **C.** Support Vector Regression
- **D.** Linear Discriminant Analysis

Question 54

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is an example of Narrow AI (Weak AI)?

- **A.** AlphaGo playing Go at superhuman levels
- **B.** A general robot capable of inventing scientific theories and raising human children seamlessly
- **C.** An omniscient superintelligence governing human civilization
- **D.** A conscious machine possessing autonomous subjective emotion and free will

Question 55

SINGLE-ANSWER QUESTION • 16 Points

Which phase is commonly known as an 'AI Winter' in the history of artificial intelligence development?

- **A.** Memory bus interface parity check error interruptions
- **B.** Operating system kernel freezes caused by unhandled memory leaks during neural training runs
- **C.** Periods when rule-based expert systems completely solved general human intelligence
- **D.** Prolonged periods when funding, commercial interest, and academic research declined sharply

Page Answers: **Q51**: B • **Q52**: C • **Q53**: A • **Q54**: A • **Q55**: D

Question 56

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes the 'Hallucination' problem in Large Language Models?

- **A.** The model generates plausible-sounding but factually false or unverifiable statements confidently
- **B.** The model encounters catastrophic memory overflow during tokenization of long-context windows
- **C.** The model produces fixed identical output embeddings regardless of the temperature parameter
- **D.** The model vocabulary lookup table fails to tokenize out-of-vocabulary mathematical operator symbols

Question 57

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of 'Computing Power' (e.g., AI accelerators like Huawei Ascend NPUs) in modern AI?

- **A.** Generating high-precision ground truth labels automatically without programmatic human review
- **B.** Providing massive parallel matrix calculation throughput required to train and infer deep models
- **C.** Executing optical surface contour scanning for geometric solid modeling
- **D.** Formulating core loss function equations mathematically without requiring human intervention

Question 58

SINGLE-ANSWER QUESTION • 16 Points

In Natural Language Processing, which task aims to identify boundaries of real-world entities (such as person names, organization names, and geographic locations)?

- **A.** Named Entity Recognition
- **B.** Part-of-Speech Tagging
- **C.** Word Stemming
- **D.** Machine Translation

Question 59

SINGLE-ANSWER QUESTION • 16 Points

What does the Dartmouth Workshop in 1956 signify in computer science history?

- **A.** The creation of the Linux operating system kernel
- **B.** The formulation of the World Wide Web by Tim Berners-Lee
- **C.** The invention of the world's first electronic computer
- **D.** The birth of Artificial Intelligence as a distinct academic discipline

Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding reasoning models like DeepSeek-R1 are TRUE? (Multiple Choice)

- **A.** They employ large-scale reinforcement learning to develop verifiable reasoning capabilities
- **B.** They autonomously generate and verify intermediate step-by-step chain-of-thought deductions
- **C.** They require supervised fine-tuning on millions of human-annotated reasoning trajectories
- **D.** They rely on static non-contextual n-gram frequency lookup tables without self-attention

Page Answers: **Q56**: A • **Q57**: B • **Q58**: A • **Q59**: D • **Q60**: AB

Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized AI ethical concerns widely debated internationally?

- **A.** Algorithmic bias and discriminatory outcomes in automated scoring systems
- **B.** Low-frequency audible vibration produced by server power transformer coils
- **C.** Spread of fake videos, synthetic impersonation, and automated fraud
- **D.** Data privacy violations from unauthorized scraping and mass biometric collection

Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Weak AI (ANI) and Strong AI (AGI) are correct?

- **A.** Strong AI has already been completely realized and deployed across enterprise cloud platforms
- **B.** Weak AI is designed and optimized to excel at specific, predefined narrow cognitive tasks
- **C.** All currently deployed commercial AI applications belong to the category of Weak AI
- **D.** Weak AI algorithms cannot be executed on modern multi-core GPU computing clusters

Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

In enterprise intelligence, which of the following represent typical deployment scenarios of Artificial Intelligence?

- **A.** Unmanned haulage trucks and visual conveyor belt inspection in Intelligent Mining
- **B.** Automated optical inspection for defect detection in Smart Manufacturing
- **C.** Replacing relational database transaction logs with unindexed raw text files
- **D.** Dynamic grid power dispatching and equipment fault prediction in Intelligent Energy

Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

In the AI industry ecosystem, which of the following components constitute the foundational Infrastructure and Framework layers?

- **A.** Specialized AI processors, NPUs, GPUs, and high-performance server clusters
- **B.** End-user domain applications, smart city solutions, and industry SaaS workflows
- **C.** Deep learning frameworks, operator acceleration libraries, and compiler toolchains
- **D.** Business process outsourcing, customer relationship management, and ERP portals

Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following generative neural network architectures and models are widely applied for synthetic data and media generation?

- **A.** Diffusion Models
- **B.** Generative Adversarial Networks
- **C.** Variational Autoencoders
- **D.** Autoregressive Transformer Models

Page Answers: **Q61**: ACD • **Q62**: BC • **Q63**: ABD • **Q64**: AC • **Q65**: ABCD

Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following AI schools of thought are primarily grounded in bionics and environmental interaction respectively?

- **A.** Symbolism
- **B.** Connectionism
- **C.** Behaviorism
- **D.** Functionalism

Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are major subfields and application domains of Artificial Intelligence?

- **A.** Speech Processing
- **B.** Knowledge Graph
- **C.** Natural Language Processing
- **D.** Computer Vision

Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are foundational Computer Vision (CV) visual perception tasks?

- **A.** Semantic and Instance Segmentation
- **B.** 2D and 3D Object Detection
- **C.** Syntactic Dependency Tree Parsing and Part-of-Speech Tagging
- **D.** Image Classification and Optical Character Recognition

Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

Natural Language Processing (NLP) is primarily divided into which core functional branches?

- **A.** Natural Language Understanding
- **B.** Speech Synthesis
- **C.** Natural Language Generation
- **D.** Image Classification

Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are core characteristics and advantages of large foundation models compared to traditional small models?

- **A.** Exact identical output reproduction across arbitrary random temperature values
- **B.** Emergent abilities at large parameter and data scales
- **C.** Stronger in-context learning and zero-shot reasoning capabilities
- **D.** Strong generalization across diverse downstream tasks through pre-training

Page Answers: **Q66**: BC • **Q67**: ABCD • **Q68**: ABD • **Q69**: AC • **Q70**: BCD

Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are core architectural innovations introduced in DeepSeek-V3 to optimize inference KV cache and sparse computation?

- **A.** Multi-Head Latent Attention
- **B.** Dense Convolutional Residual Stacking
- **C.** DeepSeekMoE architecture with shared experts
- **D.** Static Global Average Pooling

Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following factors contributed to the rapid rise and global popularity of DeepSeek?

- **A.** Open availability of model weights and full technical transparency
- **B.** Outstanding performance competitive with top proprietary frontier models
- **C.** Mandatory subscription fees exceeding $10,000 per user
- **D.** Greatly reduced training and inference costs

Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are landmark achievements of AI for Science (AI4Science) in biology and meteorology respectively?

- **A.** AlphaFold for 3D protein structure prediction
- **B.** Word2Vec for static lexical word embedding generation
- **C.** Pangu-Weather for global numerical weather forecasting
- **D.** AlexNet for 2D natural image category classification

Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are ethical, social, and legal issues raised by the rapid growth of AI?

- **A.** Privacy leakage and unauthorized profiling of individuals
- **B.** Algorithmic bias resulting from unrepresentative training data
- **C.** Potential disruption of traditional employment and labor markets
- **D.** Copyright infringement and intellectual property ownership of generated works

Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

In a Knowledge Graph, a basic semantic assertion is stored as a 'triplet'. Which types of primary components constitute the nodes and edges of this triplet?

- **A.** Entities
- **B.** Biases
- **C.** Relations
- **D.** Gradients

Page Answers: **Q71**: AC • **Q72**: ABD • **Q73**: AC • **Q74**: ABCD • **Q75**: AC

Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are the two main tasks of speech processing?

- **A.** Image Segmentation
- **B.** Speech Recognition
- **C.** Object Detection
- **D.** Speech Synthesis

Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are characteristics of Artificial Narrow Intelligence (Weak AI)?

- **A.** Lack of genuine self-awareness and conscious intentionality
- **B.** Highly effective at solving specific, well-defined problems
- **C.** Universal human-like general cognitive adaptability
- **D.** Inability to autonomously transfer learned skills to unrelated cognitive domains

Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

What types of data are typically used in the pre-training of multimodal foundation models?

- **A.** Text datasets from books, encyclopedias, and web documents
- **B.** Video clips and temporal action sequences
- **C.** High-resolution image collections with text descriptions
- **D.** Audio recordings and speech transcripts

Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following measures can reduce security and privacy risks when training and serving AI models?

- **A.** Data anonymization and de-identification
- **B.** Storing unencrypted raw biometric identifiers in public training repositories
- **C.** Differential privacy mechanisms during gradient updates
- **D.** Robust access control and model output content filtering

Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are standard implementations of Chain of Thought (CoT) prompting in Large Language Models?

- **A.** Zero-Shot-CoT
- **B.** Weight Pruning
- **C.** Few-Shot-CoT
- **D.** Model Quantization

Page Answers: **Q76**: BD • **Q77**: ABD • **Q78**: ABCD • **Q79**: ACD • **Q80**: AC

Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following hardware and software layers comprise Huawei's full-stack AI solution?

- **A.** Ascend AI processor hardware series
- **B.** Compute Architecture for Neural Networks chip operator library
- **C.** ModelArts one-stop AI development platform
- **D.** MindSpore AI framework

Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are examples of Generative AI applications in real life?

- **A.** Static relational database B-tree index rebuilding
- **B.** Text-to-Image creation of artworks and product designs
- **C.** AI-assisted code generation and completion tools
- **D.** Automated essay drafting and conversational assistants

Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are the primary computing chip architectures powering modern high-performance AI model training and inference?

- **A.** Legacy Mechanical Relays
- **B.** Graphics Processing Units
- **C.** Magnetic Tape Storage Controllers
- **D.** Neural Processing Units

Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are main features of the Behaviorism school of AI?

- **A.** Inspired by biological reflexes and adaptive behaviors
- **B.** Emphasis on perception-action loops with the environment
- **C.** Exclusive reliance on First-Order Predicate Logic theorem proving
- **D.** Foundation in reinforcement learning and evolutionary algorithms

Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are key milestones in the historical development of Artificial Intelligence?

- **A.** 2016: AlphaGo defeats world champion Lee Sedol
- **B.** 1950: Alan Turing proposes the Turing Test
- **C.** 1956: The Dartmouth Workshop coins the term AI
- **D.** 1997: IBM Deep Blue defeats world chess champion Garry Kasparov

Page Answers: **Q81**: ABCD • **Q82**: BCD • **Q83**: BD • **Q84**: ABD • **Q85**: ABCD

Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are major advantages of Mixture of Experts (MoE) models?

A. Specialized sub-networks focusing on distinct knowledge domains
B. Massive model parameter capacity without a proportional increase in FLOPs per token
C. Eliminating the need for training data completely
D. Lower inference computation latency compared to dense models of equivalent total parameter size

Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized challenges or technical limitations of current Large Language Models?

A. Hallucination
B. Inability to execute basic string concatenation
C. Total absence of zero-shot in-context adaptation
D. Massive computational costs during pre-training and scaling

Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Embodied AI are correct?

A. It requires close integration of perception, spatial reasoning, planning, and motor dynamics
B. Embodied AI agents perceive the physical world via multimodal sensors and act via physical bodies or manipulators
C. It is fundamental to the development of next-generation humanoid robotics
D. It operates strictly on text strings without any spatial or physical interaction

Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following methods can be used to improve the security and fairness of AI systems?

A. Testing model safety and output quality before deployment
B. Deploying medical models without testing their accuracy
C. Making models explainable and understanding their decisions
D. Protecting data privacy and checking datasets for bias

Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are downstream tasks enabled by pre-trained foundation models?

A. Multilingual document translation and summarization
B. Creative writing and programmatic code completion
C. Open-domain question answering and dialogue
D. Automated legal contract analysis

Page Answers: **Q86**: ABD • **Q87**: AD • **Q88**: ABC • **Q89**: ACD • **Q90**: ABCD

Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized as the core driving elements (cornerstones) of modern Artificial Intelligence?

- **A.** High-quality Big Data
- **B.** Massive Computing Power
- **C.** Advanced Machine Learning and Deep Learning Algorithms
- **D.** First-Order Predicate Logic Theorem Proving

Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following historical milestones are landmark events in the development of Artificial Intelligence?

- **A.** The 1956 Dartmouth Summer Research Project on Artificial Intelligence
- **B.** The 1997 Deep Blue victory over world chess champion Garry Kasparov
- **C.** The 2016 AlphaGo victory in the ancient board game of Go
- **D.** The 1943 McCulloch-Pitts formal mathematical artificial neuron model

Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

Artificial intelligence that is designed to perform a single specific dedicated task (such as facial recognition or playing chess) is known as ____ AI.

Answer: _____

Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

In natural language processing, NER stands for Named Entity ____.

Answer: _____

Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

Knowledge graphs represent structured relational knowledge primarily using ____, consisting of a head entity, a relation, and a tail entity.

Answer: _____

Page Answers: **Q91**: ABC • **Q92**: ABCD • **Q93**: Narrow (or Weak) • **Q94**: Recognition • **Q95**: triplets (or triples)

Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

The computer vision technology used to recognize and digitize printed or handwritten characters from images into machine-readable text is called ____.

Answer: _____

Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

In modern large language models, CoT stands for Chain of ____.

Answer: _____

Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

DeepSeek models use MLA, which stands for Multi-Head ____ Attention, to optimize KV cache memory during inference.

Answer: _____

Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

In MoE architecture, MoE stands for Mixture of ____.

Answer: _____

Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's large meteorological AI model designed for high-accuracy global weather forecasting is called Pangu-____.

Answer: _____

Page Answers: **Q96**: OCR (or Optical Character Recognition) • **Q97**: Thought • **Q98**: Latent • **Q99**: Experts • **Q100**: Weather

HCIA-AI V4.0 Certification Practice Exam • Chapter 2

Module 2: Machine Learning Overview (100 Questions & Answer Key)

Curated by: Umed Jamal Nouri • v1.1 (Nov 2026) • github.com/UmedJaMala

Question 1

TRUE OR FALSE • 16 Points

L1 regularization is commonly referred to as weight decay in machine learning optimization.

A. True
B. False

Question 2

TRUE OR FALSE • 16 Points

In a random search for hyperparameters, each setting item is sampled from possible parameter values independently based on a specified probability distribution.

A. True
B. False

Question 3

TRUE OR FALSE • 16 Points

Assuming a dataset contains the areas and historical sale prices of 21,613 houses, predicting the price of an unseen house based on its area is a typical regression task.

A. True
B. False

Question 4

TRUE OR FALSE • 16 Points

In machine learning regression tasks, Mean Squared Error (MSE) is one of the most commonly used loss functions.

A. True
B. False

Question 5

TRUE OR FALSE • 16 Points

Gradient descent is the only optimization method available for training machine learning algorithms.

A. True
B. False

Page Answers: **Q1**: False • **Q2**: True • **Q3**: True • **Q4**: True • **Q5**: False

Question 6

TRUE OR FALSE • 16 Points

If the hyperparameter search space is large and high-dimensional, random search is usually more computationally efficient than exhaustive grid search.

A. True
B. False

Question 7

TRUE OR FALSE • 16 Points

In supervised learning, the training dataset must contain only unlabelled inputs without any ground-truth target labels.

A. True
B. False

Question 8

TRUE OR FALSE • 16 Points

K-means clustering requires pre-labeled training samples to establish class boundaries.

A. True
B. False

Question 9

TRUE OR FALSE • 16 Points

High variance in a machine learning model typically indicates that the model is suffering from underfitting on the training dataset.

A. True
B. False

Question 10

TRUE OR FALSE • 16 Points

L1 regularization tends to produce sparse weight vectors by driving uninformative feature weights strictly to zero, effectively performing feature selection.

A. True
B. False

Page Answers: **Q6**: True • **Q7**: False • **Q8**: False • **Q9**: False • **Q10**: True

Question 11

TRUE OR FALSE • 16 Points

The k-Nearest Neighbors (KNN) algorithm is an instance-based lazy learning method that does not construct an explicit internal model during training.

A. True
B. False

Question 12

TRUE OR FALSE • 16 Points

A smaller value of k in KNN makes the decision boundary smoother and reduces sensitivity to noisy outlier points.

A. True
B. False

Question 13

TRUE OR FALSE • 16 Points

Naive Bayes assumes that all feature attributes are conditionally independent of each other given the class label.

A. True
B. False

Question 14

TRUE OR FALSE • 16 Points

Support Vector Machines (SVM) can solve linearly non-separable problems in original feature spaces by using kernel functions to map data into higher-dimensional spaces.

A. True
B. False

Question 15

TRUE OR FALSE • 16 Points

In Decision Trees, the ID3 algorithm uses the Gini Impurity criterion to select the optimal splitting attribute.

A. True
B. False

Page Answers: **Q11**: True • **Q12**: False • **Q13**: True • **Q14**: True • **Q15**: False

Question 16

TRUE OR FALSE • 16 Points

Random Forest builds multiple independent decision trees using bootstrap data samples and random feature subsets to reduce prediction variance.

A. True
B. False

Question 17

TRUE OR FALSE • 16 Points

In Boosting algorithms (such as AdaBoost and GBDT), base learners are trained in parallel completely independently of one another.

A. True
B. False

Question 18

TRUE OR FALSE • 16 Points

Standardization (Z-score normalization) scales feature values strictly between 0 and 1, forcing all values to be non-negative.

A. True
B. False

Question 19

TRUE OR FALSE • 16 Points

In Reinforcement Learning, the learning agent improves its policy by receiving immediate scalar reward signals from the environment based on its chosen actions.

A. True
B. False

Question 20

TRUE OR FALSE • 16 Points

One-Hot Encoding is primarily used to convert continuous floating-point target variables into discrete categories for regression tasks.

A. True
B. False

Page Answers: **Q16**: True • **Q17**: False • **Q18**: False • **Q19**: True • **Q20**: False

Question 21

TRUE OR FALSE • 16 Points

The Area Under the ROC Curve (AUC) is equal to 1.0 for a perfect binary classification model.

A. True
B. False

Question 22

TRUE OR FALSE • 16 Points

Stochastic Gradient Descent (SGD) updates model parameters using the entire training dataset at the same time in each single iteration step.

A. True
B. False

Question 23

TRUE OR FALSE • 16 Points

In cross-validation, the test dataset is merged into the training set to prevent the model from underfitting.

A. True
B. False

Question 24

TRUE OR FALSE • 16 Points

Linear regression models are naturally immune to multicollinearity among input features.

A. True
B. False

Question 25

TRUE OR FALSE • 16 Points

In Logistic Regression, the Sigmoid function maps any real-valued number into a probability range between 0 and 1.

A. True
B. False

Page Answers: **Q21**: True • **Q22**: False • **Q23**: False • **Q24**: False • **Q25**: True

Question 26

SINGLE-ANSWER QUESTION • 16 Points

Without considering any regularization terms, the support vectors of a Support Vector Machine (SVM) are composed of which data points?

- **A.** The points closest to the separating hyperplane
- **B.** Points of a certain arbitrary class
- **C.** The points farthest from the separating hyperplane
- **D.** Points on the separating hyperplane

Question 27

SINGLE-ANSWER QUESTION • 16 Points

A computer uses labeled images to learn and distinguish between cats and dogs. What type of machine learning does this scenario represent?

- **A.** Self-supervised clustering
- **B.** Supervised learning
- **C.** Reinforcement learning
- **D.** Unsupervised learning

Question 28

SINGLE-ANSWER QUESTION • 16 Points

Which of the following statements is TRUE regarding classification and regression in supervised learning?

- **A.** Classification predicts continuous numerical values, whereas regression predicts discrete categories
- **B.** Both classification and regression predict discrete categorical labels
- **C.** Both classification and regression predict continuous numerical values
- **D.** Classification predicts discrete categorical labels, whereas regression predicts continuous numerical values

Question 29

SINGLE-ANSWER QUESTION • 16 Points

Which of the following statements about Support Vector Machines (SVM) is FALSE?

- **A.** Non-linear SVM uses kernel functions to project data into higher dimensions
- **B.** The support vectors determine the location and orientation of the optimal hyperplane
- **C.** SVM can only be used for unsupervised clustering without class labels
- **D.** The objective of linear SVM is to find a hyperplane that maximizes the margin between classes

Question 30

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT an unsupervised learning algorithm?

- **A.** Principal Component Analysis
- **B.** Decision Tree
- **C.** K-Means clustering
- **D.** Hierarchical clustering

Page Answers: **Q26**: A • **Q27**: B • **Q28**: D • **Q29**: C • **Q30**: B

Question 31

SINGLE-ANSWER QUESTION • 16 Points

In Tom Mitchell's well-known definition of machine learning, an algorithm learns from Experience (E) with respect to some Task (T) and Performance measure (P) if its performance on T, as measured by P, improves with E. What does 'E' represent in this context?

- **A.** Historical data and interactions
- **B.** CPU energy consumption
- **C.** Computational execution speed
- **D.** Validation loss variance

Question 32

SINGLE-ANSWER QUESTION • 16 Points

Which of the following metrics is best suited for evaluating a linear regression model?

- **A.** F1-Score
- **B.** Cross-Entropy Loss
- **C.** Precision and Recall
- **D.** Mean Squared Error

Question 33

SINGLE-ANSWER QUESTION • 16 Points

Which of the following feature selection approaches evaluates feature subsets by training an actual predictive model and measuring its validation score?

- **A.** Random projection method
- **B.** Filter method
- **C.** Wrapper method
- **D.** Constant variance threshold

Question 34

SINGLE-ANSWER QUESTION • 16 Points

When an overly complex machine learning model achieves 99.9% accuracy on the training set but only 52% accuracy on the test set, what problem is it experiencing?

- **A.** High Bias
- **B.** Vanishing Gradient
- **C.** Overfitting
- **D.** Underfitting

Question 35

SINGLE-ANSWER QUESTION • 16 Points

Which regularization technique adds the sum of absolute values of the coefficients (|w|) as a penalty term to the loss function?

- **A.** L1 Regularization
- **B.** L2 Regularization
- **C.** Dropout
- **D.** Batch Normalization

Page Answers: **Q31**: A • **Q32**: D • **Q33**: C • **Q34**: C • **Q35**: A

Question 36

SINGLE-ANSWER QUESTION • 16 Points

Which regularization technique adds the sum of squared values of the coefficients (w^2) as a penalty term to the loss function?

- **A.** Softmax Function
- **B.** L1 Regularization
- **C.** L2 Regularization
- **D.** Early Stopping

Question 37

SINGLE-ANSWER QUESTION • 16 Points

Which splitting criterion is used by the CART (Classification and Regression Trees) algorithm for classification trees?

- **A.** Information Gain Ratio
- **B.** Information Gain
- **C.** Mean Absolute Error
- **D.** Gini Impurity

Question 38

SINGLE-ANSWER QUESTION • 16 Points

Which splitting criterion is used by the C4.5 algorithm to reduce ID3's bias toward attributes with large numbers of distinct values?

- **A.** Mean Squared Error
- **B.** Information Gain Ratio
- **C.** Gini Index
- **D.** Information Gain

Question 39

SINGLE-ANSWER QUESTION • 16 Points

In a binary classification confusion matrix, what does 'Recall' (Sensitivity) measure?

- **A.** TP / (TP + FN) - Proportion of actual positives correctly identified
- **B.** TP / (TP + FP) - Proportion of predicted positives that are truly positive
- **C.** (TP + TN) / (TP + TN + FP + FN) - Overall proportion of correct predictions
- **D.** TN / (TN + FP) - Proportion of actual negatives correctly identified

Question 40

SINGLE-ANSWER QUESTION • 16 Points

In a binary classification confusion matrix, what does 'Precision' measure?

- **A.** (TP + TN) / (TP + TN + FP + FN) - Overall proportion of correct predictions
- **B.** TP / (TP + FN) - Proportion of actual positives correctly identified
- **C.** TN / (TN + FP) - Proportion of actual negatives correctly identified
- **D.** TP / (TP + FP) - Proportion of predicted positives that are truly positive

Page Answers: **Q36**: C • **Q37**: D • **Q38**: B • **Q39**: A • **Q40**: D

Question 41

SINGLE-ANSWER QUESTION • 16 Points

What is the harmonic mean of Precision and Recall commonly known as?

- **A.** R-Squared
- **B.** Accuracy
- **C.** F1-Score
- **D.** ROC Curve

Question 42

SINGLE-ANSWER QUESTION • 16 Points

In the k-nearest neighbors (k-NN) classification algorithm, how is the class of an unlabeled sample typically predicted?

- **A.** By calculating the gradient of the loss function
- **B.** By constructing a maximum-margin separating hyperplane
- **C.** By majority voting among the k nearest neighbors
- **D.** By minimizing within-cluster variance

Question 43

SINGLE-ANSWER QUESTION • 16 Points

In the k-nearest neighbors (k-NN) algorithm, what is the effect of selecting a very small value for k?

- **A.** The decision boundary becomes completely linear
- **B.** The model has a higher probability of underfitting
- **C.** The model has a higher probability of overfitting due to noise
- **D.** The model requires zero distance calculations

Question 44

SINGLE-ANSWER QUESTION • 16 Points

In K-Means clustering, what does 'K' represent?

- **A.** The number of clusters to be partitioned
- **B.** The number of nearest neighbors for voting
- **C.** The dimensionality of input features
- **D.** The number of iterations until convergence

Question 45

SINGLE-ANSWER QUESTION • 16 Points

How are cluster centroids updated in standard K-Means clustering during each iterative step?

- **A.** By projecting support vector coordinates onto the hyper-plane boundary
- **B.** By computing the arithmetic mean of all data points currently assigned to that cluster
- **C.** By re-initializing coordinates randomly from a uniform Gaussian probability distribution
- **D.** By selecting the extreme boundary outlier with the maximum Euclidean distance

Page Answers: **Q41**: C • **Q42**: C • **Q43**: C • **Q44**: A • **Q45**: B

Question 46

SINGLE-ANSWER QUESTION • 16 Points

What is the primary operational mechanism of Bagging (Bootstrap Aggregating) in ensemble learning?

- **A.** Training base models sequentially where each model fits the residuals of previous learners
- **B.** Pruning deep decision branches to enforce low variance across individual shallow trees
- **C.** Discarding half of the training feature dimensions randomly during each gradient update
- **D.** Training multiple base models in parallel on bootstrapped subsets and aggregating their outputs

Question 47

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is a classic Boosting algorithm?

- **A.** AdaBoost
- **B.** Principal Component Analysis
- **C.** Random Forest
- **D.** K-Means

Question 48

SINGLE-ANSWER QUESTION • 16 Points

In Naive Bayes classification, which principle provides the foundation for calculating posterior class probabilities?

- **A.** Bayes' Theorem
- **B.** Gradient Descent Optimization
- **C.** Decision Tree Pruning
- **D.** K-Means Centroid Allocation

Question 49

SINGLE-ANSWER QUESTION • 16 Points

What fundamental assumption does the Naive Bayes classification algorithm make regarding feature attributes?

- **A.** All features follow a uniform spherical Gaussian distribution
- **B.** All features are linearly dependent on the class label
- **C.** All features are conditionally independent of each other given the class label
- **D.** Features must be continuously normalized between 0 and 1

Question 50

SINGLE-ANSWER QUESTION • 16 Points

In gradient descent, what happens if the learning rate is configured to be excessively large?

- **A.** The loss value may oscillate and fail to converge, or even diverge completely
- **B.** The model parameters freeze permanently and cease executing mathematical updates
- **C.** The algorithm converges instantaneously to the exact global loss minimum on epoch 1
- **D.** The GPU physical RAM allocation drops to zero and clears all dataset tensors

Page Answers: **Q46**: D • **Q47**: A • **Q48**: A • **Q49**: C • **Q50**: A

Question 51

SINGLE-ANSWER QUESTION • 16 Points

Which gradient descent variant calculates gradients and updates parameters using a small subset (mini-batch) of training samples at each step?

- **A.** Full Grid Search
- **B.** Batch Gradient Descent
- **C.** Stochastic Gradient Descent
- **D.** Mini-batch Gradient Descent

Question 52

SINGLE-ANSWER QUESTION • 16 Points

In K-Fold Cross-Validation, if K = 5, what percentage of the dataset is used for validation in each fold?

- **A.** 80%
- **B.** 50%
- **C.** 10%
- **D.** 20%

Question 53

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of the validation set in the machine learning workflow?

- **A.** Storing redundant duplicate records
- **B.** Hyperparameter tuning and model architecture selection
- **C.** Final unbiased assessment before production deployment
- **D.** Updating the internal weights via backpropagation

Question 54

SINGLE-ANSWER QUESTION • 16 Points

In machine learning feature engineering, what is the primary purpose of normalizing and standardizing features?

- **A.** To ensure that different input variables of a model fall into the same value range
- **B.** To convert categorical labels into one-hot encoded binary vectors
- **C.** To artificially increase training data by generating synthetic noise
- **D.** To eliminate negative weight coefficients across neural connections

Question 55

SINGLE-ANSWER QUESTION • 16 Points

What type of data problem occurs when an age column contains values like -15 or 999?

- **A.** Underfitting
- **B.** Outlier data
- **C.** Generalization
- **D.** Regularization

Page Answers: **Q51**: D • **Q52**: D • **Q53**: B • **Q54**: A • **Q55**: B

Question 56

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is an effective technique to prevent decision trees from overfitting?

- **A.** Feature scaling
- **B.** Increasing tree depth
- **C.** Removing stopping criteria
- **D.** Tree pruning

Question 57

SINGLE-ANSWER QUESTION • 16 Points

What is the function of the Softmax layer in multi-class classification models?

- **A.** Normalizing raw model logits into a valid probability distribution summing to 1
- **B.** Computing pairwise Euclidean distance matrices between all feature sample vectors
- **C.** Eliminating all negative weight coefficients across dense hidden layer parameters
- **D.** Projecting multidimensional continuous inputs onto a binary step threshold output

Question 58

SINGLE-ANSWER QUESTION • 16 Points

In Support Vector Machines (SVM), what is the primary objective of constructing an optimal hyperplane?

- **A.** To minimize the number of input features used during training
- **B.** To maximize the margin (gap) between the two categories in the feature space
- **C.** To compute pairwise Euclidean distances to all training points
- **D.** To eliminate outliers by pruning decision tree leaf nodes

Question 59

SINGLE-ANSWER QUESTION • 16 Points

In Support Vector Machines (SVM), which kernel function is most frequently used to handle non-linearly separable data?

- **A.** Linear kernel
- **B.** Gaussian kernel
- **C.** Step kernel
- **D.** Constant kernel

Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

In the machine learning workflow, why is Feature Selection necessary? (Multiple Choice)

- **A.** Simplifies models for easier interpretation
- **B.** Shortens model training time
- **C.** Eliminates the need for any training datasets or labels
- **D.** Avoids the curse of dimensionality and improves generalization

Page Answers: **Q56**: D • **Q57**: A • **Q58**: B • **Q59**: B • **Q60**: ABD

Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

In linear regression models expressed as y = w^T x + b, which of the following statements are correct?

- **A.** 'w' represents the weight vector determining the slope of each feature
- **B.** 'b' represents the regularization hyperparameter controlling L1 sparsity penalties
- **C.** 'b' represents the bias term allowing the hyperplane to shift away from the origin
- **D.** The parameter vector 'w' must always be strictly perpendicular to the input feature vector 'x'

Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following ensemble learning algorithms are based on sequential boosting and iterative residual fitting?

- **A.** AdaBoost
- **B.** Gradient Boosted Decision Trees
- **C.** Random Forest
- **D.** Extra Trees

Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are typical application scenarios of Reinforcement Learning?

- **A.** Customer segmentation based on purchasing history
- **B.** Autonomous vehicle driving and traffic decisions
- **C.** Playing complex strategic games such as Chess and Go
- **D.** Industrial robot path planning and motion control

Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

When evaluating classification models on imbalanced datasets (e.g., fraud detection where positive cases are 0.1%), which of the following evaluation strategies are appropriate?

- **A.** Precision, Recall, and F1-Score
- **B.** Relying only on overall Accuracy without inspecting the Confusion Matrix
- **C.** Area Under the Precision-Recall Curve or ROC-AUC
- **D.** Evaluating Mean Absolute Error on categorical predictions

Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Semi-Supervised Learning are correct?

- **A.** It uses a small amount of labeled data combined with a large pool of unlabeled data
- **B.** It helps reduce the high cost and labor required for detailed manual data annotation
- **C.** It completely eliminates the need for any initial ground-truth supervision or seed labels
- **D.** It assumes that neighboring data points in feature space tend to share similar task labels

Page Answers: **Q61**: AC • **Q62**: AB • **Q63**: BCD • **Q64**: AC • **Q65**: ABD

Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following evaluation metrics are specifically designed for classification models rather than continuous regression?

- **A.** F1-Score
- **B.** Mean Squared Error
- **C.** Area Under the ROC Curve
- **D.** Mean Absolute Percentage Error

Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following hyperparameters may be defined and tuned during model training?

- **A.** Model weights and biases
- **B.** Number of training iterations
- **C.** Batch size
- **D.** Learning rate

Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about model parameters and hyperparameters are true?

- **A.** Weights and biases in neural networks are model parameters
- **B.** Hyperparameters are set manually by developers prior to model training
- **C.** Learning rate and tree maximum depth are hyperparameters
- **D.** Model parameters are learned and updated automatically by the algorithm from training data

Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about datasets in machine learning are true?

- **A.** The test set should be used to update gradient weights during training
- **B.** The training set is used to optimize model parameters
- **C.** The test set is used to evaluate final model generalization performance
- **D.** The validation set is used for hyperparameter tuning and model selection

Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following tasks belong to the data preprocessing stage in machine learning?

- **A.** Data deduplication
- **B.** Missing value imputation
- **C.** Feature normalization and standardization
- **D.** Outlier detection and handling

Page Answers: **Q66**: AC • **Q67**: BCD • **Q68**: ABCD • **Q69**: BCD • **Q70**: ABCD

Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following methods can help prevent model overfitting?

A. Applying cross-validation during hyperparameter selection
B. Expanding the training dataset size or applying data augmentation
C. Adding L1 or L2 regularization terms to the loss function
D. Adding millions of redundant parameters to a tiny dataset

Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are unsupervised machine learning algorithms?

A. Multiple Linear Regression
B. K-Means Clustering
C. Principal Component Analysis
D. Support Vector Machines

Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

When evaluating continuous regression models in the presence of extreme anomalous data points (outliers), which of the following statements regarding MSE, RMSE, and MAE are correct?

A. Mean Squared Error (MSE) squares prediction errors, causing large outliers to heavily distort the score
B. Root Mean Squared Error (RMSE) preserves the original unit scale of the target variable, making it more intuitive than MSE
C. Mean Squared Error (MSE) is completely immune to extreme outliers regardless of error magnitude
D. Mean Absolute Error (MAE) avoids squaring errors and is therefore the most robust regression metric when outliers exist

Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following ensemble learning algorithms are based on Boosting principles?

A. AdaBoost
B. Random Forest
C. XGBoost
D. Gradient Boosting Decision Tree fitting negative gradients

Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about K-Nearest Neighbors (KNN) are true?

A. KNN makes predictions based on the class labels of the k closest training examples
B. Feature scaling is essential for KNN to prevent high-magnitude features from dominating distance calculations
C. KNN requires weeks of intensive offline gradient descent training before it can make predictions
D. KNN is sensitive to the choice of distance metric

Page Answers: **Q71**: ABC • **Q72**: BC • **Q73**: ABD • **Q74**: ACD • **Q75**: ABD

Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Support Vector Machines (SVM) are correct?

A. Data points located on the margin boundaries are called support vectors

B. The optimal hyperplane maximizes the geometric margin between classes

C. Kernel functions allow SVM to construct non-linear decision boundaries

D. Standard linear SVM requires quadratic memory that decreases as feature dimensions grow

Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are commonly used kernel functions in Support Vector Machines?

A. Linear Kernel

B. Sigmoid Kernel

C. Radial Basis Function

D. Polynomial Kernel

Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following impurity and splitting criteria are commonly used in Decision Tree algorithms (ID3 and CART)?

A. Information Gain

B. Silhouette Separation Coefficient

C. Gini Impurity Index

D. Cosine Dissimilarity Metric

Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following metrics are used to evaluate the performance of continuous regression models?

A. Mean Squared Error

B. Mean Absolute Error

C. Classification Accuracy

D. Recall or Sensitivity

Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Gradient Descent variants are true?

A. Stochastic Gradient Descent updates parameters using a single random sample per step, making it fast but causing fluctuating loss curves

B. Mini-batch Gradient Descent balances computational stability and speed by using small subsets of data

C. Batch Gradient Descent uses all training samples to calculate gradient, guaranteeing smooth convergence but being computationally slow on massive data

D. All gradient descent methods update parameters without using derivatives

Page Answers: **Q76**: ABC • **Q77**: ABCD • **Q78**: AC • **Q79**: AB • **Q80**: ABC

Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are automated search strategies commonly used for model hyperparameter tuning?

- **A.** Grid Search
- **B.** Random Search
- **C.** Constant Value Imputation
- **D.** One-Hot Category Encoding

Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are core components in a Reinforcement Learning system?

- **A.** State and Action
- **B.** Agent
- **C.** Reward signal
- **D.** Environment

Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common statistical imputation strategies for handling missing values in continuous numeric features?

- **A.** Mean Value Imputation
- **B.** Random Label Flipping
- **C.** Median Value Imputation
- **D.** Softmax Probability Normalization

Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are benefits of Feature Scaling (Normalization / Standardization)?

- **A.** Improves numerical stability of optimization algorithms
- **B.** Speeds up gradient descent convergence
- **C.** Prevents features with large scales from dominating distance-based models
- **D.** Guarantees 100% test accuracy on any arbitrary model

Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Random Forest are correct?

- **A.** It requires base trees to be trained strictly in serial order
- **B.** It is generally more resistant to overfitting than an unpruned single decision tree
- **C.** It uses bootstrap aggregation to generate diverse training subsets
- **D.** It selects a random subset of features at each split node

Page Answers: **Q81**: AB • **Q82**: ABCD • **Q83**: AC • **Q84**: ABC • **Q85**: BCD

Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are geometric spatial distance metrics commonly used in KNN and K-Means?

- **A.** Euclidean Distance
- **B.** Manhattan Distance
- **C.** Categorical Cross-Entropy Loss
- **D.** Margin Hinge Loss Function

Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about K-Means clustering are true?

- **A.** The results can vary depending on the initial centroid positions
- **B.** It is sensitive to extreme outlier points
- **C.** It is a supervised classification technique that requires pre-annotated ground-truth labels
- **D.** It requires specifying the number of clusters K in advance

Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common methods of feature selection in machine learning?

- **A.** Filter method
- **B.** Wrapper method
- **C.** Embedded method
- **D.** Gradient descent optimization

Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Logistic Regression are true?

- **A.** Despite its name, Logistic Regression is primarily used for classification problems
- **B.** It optimizes model parameters using maximum likelihood estimation
- **C.** It uses the Sigmoid function to output probability values between 0 and 1
- **D.** It can only be used when there are at least 1,000 output classes

Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

What four basic outcomes are recorded in a binary confusion matrix?

- **A.** True Positive
- **B.** False Negative
- **C.** False Positive
- **D.** True Negative

Page Answers: **Q86**: AB • **Q87**: ABD • **Q88**: ABC • **Q89**: ABC • **Q90**: ABCD

Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding the K-Nearest Neighbors (KNN) algorithm are TRUE?

- **A.** KNN is an instance-based 'lazy learning' algorithm that performs no explicit training phase
- **B.** The decision boundary is determined locally by the majority class among the k closest neighbors
- **C.** The choice of distance metric directly impacts classification results
- **D.** KNN produces a compact mathematical parametric formula independent of the training dataset

Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding the Softmax activation function in multi-class classification are TRUE?

- **A.** It transforms unnormalized model logits into a valid probability distribution
- **B.** The sum of all output probabilities across all mutually exclusive classes equals exactly 1
- **C.** Every individual output probability value is strictly bounded between 0 and 1
- **D.** Larger input logit values correspond monotonically to higher output probability values

Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn, the core method called on an estimator instance to train the machine learning algorithm on feature matrix X and target labels y is model.____(X, y).

Answer: _____

Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn, after training an estimator, the method used to generate predicted target values for new unseen samples is model.____(X_test).

Answer: _____

Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn classification models (such as LogisticRegression), the method that returns calibrated class probability estimates for each sample is model.____(X_test).

Answer: _____

Page Answers: **Q91**: ABC • **Q92**: ABCD • **Q93**: fit • **Q94**: predict • **Q95**: predict_proba

Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn preprocessing (e.g., StandardScaler), the combined method that calculates statistics on training data and applies scaling in a single step is scaler.____(X_train).

Answer: _____

Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn, the function from sklearn.model_selection commonly used to randomly partition a dataset into training and testing subsets is ____.

Answer: _____

Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn's KMeans clustering estimator, the hyperparameter that specifies the desired number of clusters to form is ____.

Answer: _____

Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn's LinearRegression, the learned weight slope coefficients of the model are stored in the model.____ attribute.

Answer: _____

Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

In NumPy, the function used to initialize an array of a given shape populated entirely with zeros is np.____().

Answer: _____

Page Answers: **Q96**: fit_transform • **Q97**: train_test_split • **Q98**: n_clusters • **Q99**: coef_ • **Q100**: zeros

HCIA-AI V4.0 Certification Practice Exam • Chapter 3

Module 3: Deep Learning & Foundation Models (100 Questions & Answer Key)

Curated by: Umed Jamal Nouri • v1.1 (Nov 2026) • github.com/UmedJaMala

Question 1

TRUE OR FALSE • 16 Points

Typically, in a deep neural network that uses the Sigmoid activation function, the gradient may decrease close to 0 within five layers, causing severe vanishing gradients.

A. True
B. False

Question 2

TRUE OR FALSE • 16 Points

When training a deep neural network with the Softsign function, gradients never saturate regardless of how deep the network is.

A. True
B. False

Question 3

TRUE OR FALSE • 16 Points

The Tanh activation function can effectively eliminate and completely solve the vanishing gradient problem in arbitrarily deep neural networks.

A. True
B. False

Question 4

TRUE OR FALSE • 16 Points

Batch Normalization is a technique used in deep learning to standardize layer inputs, stabilizing internal covariate shift and accelerating model convergence.

A. True
B. False

Question 5

TRUE OR FALSE • 16 Points

A single-layer perceptron without hidden layers is capable of solving linearly non-separable problems such as the XOR logical operation.

A. True
B. False

Page Answers: **Q1**: True • **Q2**: False • **Q3**: False • **Q4**: True • **Q5**: False

Question 6

TRUE OR FALSE • 16 Points

The maximum derivative of the standard Sigmoid function is 1.0, which occurs when its input x equals 0.

A. True

B. False

Question 7

TRUE OR FALSE • 16 Points

The ReLU activation function outputs 0 for all negative inputs ($x < 0$) and has a derivative of 1 for all positive inputs ($x > 0$).

A. True

B. False

Question 8

TRUE OR FALSE • 16 Points

During neural network inference (testing), Dropout continues to randomly set 50% of the neurons' outputs to zero.

A. True

B. False

Question 9

TRUE OR FALSE • 16 Points

In Convolutional Neural Networks, the pooling layer introduces learnable weight parameters that are updated via backpropagation.

A. True

B. False

Question 10

TRUE OR FALSE • 16 Points

Parameter sharing in CNNs means that every single pixel in an image requires a completely unique, independent set of convolution kernel weights.

A. True

B. False

Page Answers: **Q6**: False • **Q7**: True • **Q8**: False • **Q9**: False • **Q10**: False

Question 11

TRUE OR FALSE • 16 Points

Compared with standard RNNs, CNNs are better suited for extracting 2D spatial grid features from images.

A. True
B. False

Question 12

TRUE OR FALSE • 16 Points

Transformer models discard recurrent loops completely and rely on Self-Attention mechanisms and Positional Encoding to capture token dependencies.

A. True
B. False

Question 13

TRUE OR FALSE • 16 Points

The Adam optimizer combines the principles of Momentum (first moment estimation) and RMSProp (second moment estimation).

A. True
B. False

Question 14

TRUE OR FALSE • 16 Points

LSTM networks contain only a single gate known as the Reset Gate.

A. True
B. False

Question 15

TRUE OR FALSE • 16 Points

The Gated Recurrent Unit (GRU) contains five separate gates, which makes it much more computationally complex than standard LSTM.

A. True
B. False

Page Answers: **Q11**: True • **Q12**: True • **Q13**: True • **Q14**: False • **Q15**: False

Question 16

TRUE OR FALSE • 16 Points

In ResNet, residual shortcut (skip) connections allow gradients to flow directly through identity paths, enabling effective training of networks with hundreds of layers.

A. True
B. False

Question 17

TRUE OR FALSE • 16 Points

Exploding gradients during deep network training can lead to numerical overflow and NaN loss values, which can be mitigated by Gradient Clipping.

A. True
B. False

Question 18

TRUE OR FALSE • 16 Points

The AdaGrad optimizer monotonically increases the learning rate during training, eventually causing infinite learning rates.

A. True
B. False

Question 19

TRUE OR FALSE • 16 Points

In the Transformer architecture, the Scaled Dot-Product Attention divides the dot products of Q and K by the square root of the key dimension (sqrt(d_k)) to prevent softmax gradients from becoming extremely small.

A. True
B. False

Question 20

TRUE OR FALSE • 16 Points

BERT is an autoregressive decoder-only language model trained primarily to predict the very next token in a causal sequence.

A. True
B. False

Page Answers: **Q16**: True • **Q17**: True • **Q18**: False • **Q19**: True • **Q20**: False

Question 21

TRUE OR FALSE • 16 Points

GPT models adopt a bidirectional encoder architecture where each token attends at the same time to all past and future tokens during causal generation.

A. True
B. False

Question 22

TRUE OR FALSE • 16 Points

Data augmentation techniques (e.g., random flipping, cropping, and color jittering) reduce training set size and increase model underfitting.

A. True
B. False

Question 23

TRUE OR FALSE • 16 Points

Cross-Entropy loss is widely used with the Softmax activation function for multi-class classification neural networks.

A. True
B. False

Question 24

TRUE OR FALSE • 16 Points

Backpropagation calculates the gradient of the loss function with respect to each network parameter using the mathematical Chain Rule of calculus.

A. True
B. False

Question 25

TRUE OR FALSE • 16 Points

In standard RNNs, Backpropagation Through Time (BPTT) unfolds the recurrent computational graph across time steps to compute weight gradients.

A. True
B. False

Page Answers: **Q21**: False • **Q22**: False • **Q23**: True • **Q24**: True • **Q25**: True

Question 26

SINGLE-ANSWER QUESTION • 16 Points

During neural network training, which of the following values are continuously updated by optimization algorithms using gradient descent?

- **A.** Sample labels
- **B.** Learning rate
- **C.** Model parameters
- **D.** Batch size

Question 27

SINGLE-ANSWER QUESTION • 16 Points

The ReLU function is commonly used in deep learning. What is the value range (output range) of the standard ReLU function?

- **A.** (-inf, +inf)
- **B.** (-1.0, +1.0)
- **C.** [0, +inf)
- **D.** (0.0, 1.0)

Question 28

SINGLE-ANSWER QUESTION • 16 Points

The Sigmoid activation function is monotonic and continuous, but when used in very deep networks, which problem is most likely to emerge?

- **A.** Vanishing gradient
- **B.** Gradient explosion
- **C.** Underfitting caused by high variance
- **D.** Loss function becoming negative

Question 29

SINGLE-ANSWER QUESTION • 16 Points

Which of the following activation functions can effectively reduce the vanishing gradient problem in positive activation regions?

- **A.** ReLU
- **B.** Tanh
- **C.** Softsign
- **D.** Sigmoid

Question 30

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT a recurrent neural network model?

- **A.** Long Short-Term Memory
- **B.** Logistic Regression Classifier
- **C.** Gated Recurrent Unit
- **D.** Bidirectional RNN

Page Answers: **Q26**: C • **Q27**: C • **Q28**: A • **Q29**: A • **Q30**: B

Question 31

SINGLE-ANSWER QUESTION • 16 Points

If a 32 x 32 feature map is processed through a 2 x 2 pooling layer with stride = 2 and no padding, what is the size of the output feature map?

- **A.** 30 x 30
- **B.** 32 x 32
- **C.** 64 x 64
- **D.** 16 x 16

Question 32

SINGLE-ANSWER QUESTION • 16 Points

If a 32 x 32 image is input and a 5 x 5 convolution kernel is applied with stride = 1 and padding = 0, what is the spatial dimension of the output feature map?

- **A.** 27 x 27
- **B.** 30 x 30
- **C.** 28 x 28
- **D.** 32 x 32

Question 33

SINGLE-ANSWER QUESTION • 16 Points

Which layer in a Convolutional Neural Network achieves translational invariance and reduces spatial dimensions without introducing trainable parameters?

- **A.** Batch Normalization Layer
- **B.** Pooling Layer
- **C.** Convolutional Layer
- **D.** Fully-Connected Layer

Question 34

SINGLE-ANSWER QUESTION • 16 Points

Which optimizer uses both the exponentially decaying average of past gradients (first moment) and the past squared gradients (second moment)?

- **A.** Standard SGD
- **B.** Adam
- **C.** Momentum SGD
- **D.** AdaGrad

Question 35

SINGLE-ANSWER QUESTION • 16 Points

In the LSTM cell, which gate is responsible for deciding what percentage of information from the previous cell state (C_{t-1}) should be discarded?

- **A.** Forget Gate
- **B.** Input Gate
- **C.** Reset Gate
- **D.** Output Gate

Page Answers: **Q31**: D • **Q32**: C • **Q33**: B • **Q34**: B • **Q35**: A

Question 36

SINGLE-ANSWER QUESTION • 16 Points

Which gate in the LSTM cell decides which newly computed candidate values should be stored in the updated cell state?

- **A.** Update Gate
- **B.** Forget Gate
- **C.** Output Gate
- **D.** Input Gate

Question 37

SINGLE-ANSWER QUESTION • 16 Points

What is the primary motivation behind the Residual Connection (shortcut connection) introduced in ResNet?

- **A.** Enabling identity mappings F(x) + x so gradients flow smoothly, preventing network degradation
- **B.** Replacing non-linear activation functions with direct linear identity pass-through layers
- **C.** Flattening two-dimensional spatial feature maps into single-dimension sequential vectors
- **D.** Doubling kernel weight parameter counts per layer to maximize model memorization capacity

Question 38

SINGLE-ANSWER QUESTION • 16 Points

In the Transformer Self-Attention formula Attention(Q, K, V) = softmax(Q K^T / sqrt(d_k)) V, what does 'd_k' represent?

- **A.** Total number of training samples
- **B.** Total vocabulary size
- **C.** Dimension of the key vector
- **D.** Number of layers in the encoder

Question 39

SINGLE-ANSWER QUESTION • 16 Points

Why is Positional Encoding essential in the Transformer architecture?

- **A.** To replace dense feedforward projection layers with fixed sinusoidal trigonometric operations
- **B.** To calculate cross-entropy loss gradients across multi-token target vocabulary dictionaries
- **C.** To provide the permutation-invariant self-attention mechanism with sequence order awareness
- **D.** To project sparse token index embeddings into low-dimensional normalized spatial coordinates

Question 40

SINGLE-ANSWER QUESTION • 16 Points

Which activation function is defined as f(x) = max(0.01x, x) to solve the 'Dying ReLU' problem?

- **A.** Exponential Linear Unit
- **B.** Scaled Exponential Unit
- **C.** Gaussian Error Linear Unit
- **D.** Leaky Rectified Linear Unit

Page Answers: **Q36**: D • **Q37**: A • **Q38**: C • **Q39**: C • **Q40**: D

Question 41

SINGLE-ANSWER QUESTION • 16 Points

What happens when a hidden neuron experiences the 'Dying ReLU' problem?

- **A.** Its weight values explode toward positive infinity, triggering numerical NaN overflow errors
- **B.** Its activation function spontaneously transitions into a continuous hyperbolic tangent curve
- **C.** Its input stays negative, causing 0 output and 0 gradient, permanently halting parameter updates
- **D.** Its internal bias coefficient inverts polarity, forcing the activation to saturate at positive one

Question 42

SINGLE-ANSWER QUESTION • 16 Points

Which loss function is mathematically defined as L = - sum(y_i * log(p_i)) for multi-class classification?

- **A.** Mean Squared Error
- **B.** Cross-Entropy Loss
- **C.** Hinge Loss
- **D.** Mean Absolute Error

Question 43

SINGLE-ANSWER QUESTION • 16 Points

In Convolutional Neural Networks, what is a 1x1 convolution commonly used for?

- **A.** Doubling the spatial height and width dimensions of feature maps through bilinear upsampling
- **B.** Cross-channel pooling to reduce or expand channel depth while injecting non-linear computation
- **C.** Extracting directional edge gradients across spatial diagonal pixel coordinates in input images
- **D.** Binarizing continuous feature activations into discrete zero-or-one boolean threshold outputs

Question 44

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of the Output Gate in an LSTM unit?

- **A.** Discarding obsolete historical context from the previous cell state vector during reset steps
- **B.** Resetting the internal forget gate bias to zero at the start of every sequence
- **C.** Controlling which filtered portion of the updated cell state is exposed as hidden state h_t
- **D.** Scaling the input embedding matrix by the inverse square root of hidden dimension

Question 45

SINGLE-ANSWER QUESTION • 16 Points

What is the key advantage of Multi-Head Attention over single-head attention in Transformers?

- **A.** It processes multiple independent batch sequences at the same time using a single shared linear matrix
- **B.** It projects inputs into diverse representation subspaces, allowing joint attention to different features
- **C.** It eliminates the requirement for positional encodings by computing temporal ordering dynamically
- **D.** It replaces floating-point matrix multiplications with low-precision integer bitwise operations

Page Answers: **Q41**: C • **Q42**: B • **Q43**: B • **Q44**: C • **Q45**: B

Question 46

SINGLE-ANSWER QUESTION • 16 Points

In deep learning, what does 'Weight Decay' directly correspond to in terms of loss regularization?

- **A.** L1 Regularization
- **B.** Dropout probability
- **C.** L2 Regularization
- **D.** Learning rate scheduling

Question 47

SINGLE-ANSWER QUESTION • 16 Points

What regularization method temporarily removes units (both hidden and visible) along with their incoming and outgoing edges with probability p during each training iteration?

- **A.** Dropout
- **B.** Gradient clipping
- **C.** Batch normalization
- **D.** Feature scaling

Question 48

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes the training procedure of the BERT foundation model?

- **A.** K-Means clustering of raw pixels
- **B.** Masked Language Modeling and Next Sentence Prediction
- **C.** Supervised reinforcement learning from human feedback
- **D.** Next token prediction only using causal masking

Question 49

SINGLE-ANSWER QUESTION • 16 Points

Which classical CNN architecture won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) in 2012, sparking the modern deep learning boom?

- **A.** ResNet-152
- **B.** Transformer
- **C.** LeNet-5
- **D.** AlexNet

Question 50

SINGLE-ANSWER QUESTION • 16 Points

What is the primary function of the 'C' (Cell State) vector in an LSTM network?

- **A.** Acting as an unobstructed conveyor belt that carries linear memory across steps with additive updates
- **B.** Calculating cross-entropy objective loss against ground truth label targets on every recurrent step
- **C.** Applying softmax normalization directly to the unscaled input gate logits at each step
- **D.** Multiplying hidden state vectors by a fixed independent rotation matrix at each step

Page Answers: **Q46**: C • **Q47**: A • **Q48**: B • **Q49**: D • **Q50**: A

Question 51

SINGLE-ANSWER QUESTION • 16 Points

In a deep neural network using the standard Sigmoid activation function sigma(x), what is the maximum value of its first derivative sigma'(x), and where does it occur?

- **A.** Maximum derivative is 0.25 (1/4), occurring at x = 0
- **B.** Maximum derivative is 1.0, occurring at x = 0
- **C.** Maximum derivative is 0.5, occurring as x approaches +infinity
- **D.** Maximum derivative is 2.0, occurring at x = 1

Question 52

SINGLE-ANSWER QUESTION • 16 Points

Why are non-linear activation functions mandatory between successive layers in deep neural networks?

- **A.** To initialize all neural weight parameters uniformly to constant values of positive one point zero
- **B.** To convert floating-point tensor activations into discrete binary booleans for hardware caching
- **C.** To prevent multiple stacked linear layers from mathematically collapsing into a single linear map
- **D.** To restrict operating system memory consumption and prevent CUDA kernel deadlocks during training

Question 53

SINGLE-ANSWER QUESTION • 16 Points

Which optimizer maintains an exponential moving average of past gradients to help accelerate in directions with persistent gradients and dampen oscillations?

- **A.** Nearest Neighbor Search
- **B.** Standard SGD without momentum
- **C.** Exhaustive Grid Search
- **D.** Momentum Optimizer

Question 54

SINGLE-ANSWER QUESTION • 16 Points

In an image with 3 color channels (RGB) of size 224 x 224, what are the dimensions of a single convolution kernel of spatial size 3 x 3?

- **A.** 1 x 1 x 3
- **B.** 3 x 3 x 3
- **C.** 224 x 224 x 3
- **D.** 3 x 3 x 1

Question 55

SINGLE-ANSWER QUESTION • 16 Points

What is the formula for calculating output spatial dimension O given input size W, kernel size K, padding P, and stride S in a CNN layer?

- **A.** O = floor((W - K + 2P) / S) - 1
- **B.** O = floor((W - K + 2P) / S)
- **C.** O = floor((W - K + 2P) / S) + 2
- **D.** O = floor((W - K + 2P) / S) + 1

Page Answers: **Q51**: A • **Q52**: C • **Q53**: D • **Q54**: B • **Q55**: D

Question 56

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is a key architectural characteristic of the Swish activation function f(x) = x * sigmoid(beta * x)?

- **A.** It restricts all activated output values strictly within the closed real interval of zero and one
- **B.** It eliminates backward gradient propagation by computing forward parameter updates analytically
- **C.** It truncates all non-zero floating-point numbers into eight-bit signed integer representations
- **D.** It is smooth and non-monotonic, consistently reducing gradient vanishing in deep architectures

Question 57

SINGLE-ANSWER QUESTION • 16 Points

Which layer normalization technique in deep learning normalizes activations across features for each individual sample, widely used in Transformer models?

- **A.** Batch Normalization
- **B.** Min-Max Normalization
- **C.** Layer Normalization
- **D.** Weight Normalization

Question 58

SINGLE-ANSWER QUESTION • 16 Points

What is 'Early Stopping' in neural network training?

- **A.** Halting model training when validation performance ceases improving, thereby preventing overfitting
- **B.** Deleting intermediate training checkpoints automatically whenever the loss curve plateaus on disk
- **C.** Terminating model optimization immediately after epoch one regardless of error convergence rates
- **D.** Continuing gradient updates until training loss reaches zero across all epochs

Question 59

SINGLE-ANSWER QUESTION • 16 Points

Which architecture is specifically designed for sequence-to-sequence tasks (e.g., machine translation) using two separate RNN or Transformer blocks?

- **A.** Single Decision Tree
- **B.** Encoder-Decoder Architecture
- **C.** Linear Hyperplane Sorter
- **D.** K-Means Clusterer

Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

In the Transformer architecture, which mechanisms are integrated into each sub-layer to stabilize gradient flow and accelerate deep network training? (Multiple Choice)

- **A.** Residual connection
- **B.** Max pooling
- **C.** Layer normalization
- **D.** Convolution kernel

Page Answers: **Q56**: D • **Q57**: C • **Q58**: A • **Q59**: B • **Q60**: AC

Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following factors contribute to the Vanishing or Exploding Gradient problem during backpropagation through time (BPTT) in standard RNNs?

- **A.** Repeated matrix multiplications of weight matrices across extended sequential time steps
- **B.** Use of saturating activation functions like Sigmoid whose derivatives are bounded
- **C.** Operating on fixed-width non-overlapping spatial convolutional sliding windows
- **D.** Long-range temporal dependencies that exponentially attenuate or escalate backpropagated error

Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following represent recognized methods of Parameter-Efficient Fine-Tuning (PEFT) for large foundation models?

- **A.** Full Model Parameter Retraining on entire multi-terabyte web crawl datasets
- **B.** LoRA injecting low-rank decomposition matrices
- **C.** Prefix Tuning and Prompt Tuning conditioning model activations with learnable vectors
- **D.** Re-initializing all transformer feedforward projection layers from random uniform distributions

Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding LoRA (Low-Rank Adaptation) fine-tuning are correct?

- **A.** It freezes the original pre-trained model weight matrices during fine-tuning
- **B.** It completely removes the multi-head attention mechanism from Transformer layers
- **C.** It introduces trainable low-rank decomposition matrices alongside attention projections
- **D.** It greatly reduces the number of trainable parameters and GPU memory consumption

Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following activation functions have strictly bounded output ranges that squash inputs into finite numerical intervals?

- **A.** Sigmoid function
- **B.** Tanh function
- **C.** Standard ReLU function
- **D.** Leaky ReLU function

Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

According to the HCIA-AI V4.0 curriculum, which of the following statements accurately describe the structure and properties of a Feedforward Neural Network (FNN)?

- **A.** Signals are transmitted unidirectionally from the input layer to the output layer with no feedback loops
- **B.** The complete network architecture can be represented mathematically as a Directed Acyclic Graph (DAG)
- **C.** The bias term 'b' acts as the threshold required for activating a neuron
- **D.** Feedback edges route output activations cyclically back to input nodes during forward inference

Page Answers: **Q61**: ABD • **Q62**: BC • **Q63**: ACD • **Q64**: AB • **Q65**: ABC

Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following activation functions are smooth, bounded S-shaped (sigmoidal) curves commonly used in neural networks?

- **A.** Sigmoid function
- **B.** Hyperbolic Tangent or Tanh function
- **C.** Standard Rectified Linear Unit
- **D.** Leaky ReLU function

Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

In the Transformer scaled dot-product attention formula, which vectors are multiplied together via dot-product to compute raw compatibility scores?

- **A.** Query vector
- **B.** Value vector
- **C.** Key vector
- **D.** Residual shortcut vector

Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following algorithms are commonly used first-order gradient-based optimizers in deep learning?

- **A.** RMSProp
- **B.** Momentum SGD
- **C.** K-Means
- **D.** Adam

Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following components are foundational architectural building blocks of Convolutional Neural Networks (CNNs)?

- **A.** Recurrent hidden-to-hidden sequential transition matrices
- **B.** Convolutional layers with learnable spatial filter kernels
- **C.** Pooling layers
- **D.** Non-linear activation functions

Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized regularization techniques used to prevent overfitting in deep neural networks?

- **A.** Early stopping based on validation loss monitoring
- **B.** L1 and L2 weight decay regularization
- **C.** Data augmentation expanding training sample diversity
- **D.** Dropout randomly deactivating neurons during training

Page Answers: **Q66**: AB • **Q67**: AC • **Q68**: ABD • **Q69**: BCD • **Q70**: ABCD

Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding the Adam optimizer are correct?

- **A.** It tracks an exponentially decaying average of past gradients
- **B.** It completely eliminates the necessity of error backpropagation gradient evaluation
- **C.** It tracks an exponentially decaying average of past squared gradients
- **D.** It includes bias-correction terms to counteract zero-bias initialization

Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are the primary pooling operations commonly used in Convolutional Neural Networks?

- **A.** Max Pooling
- **B.** Median Value Sorting Filter
- **C.** Average Pooling
- **D.** Cross-Entropy Loss Normalization

Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Long Short-Term Memory (LSTM) are true?

- **A.** LSTM introduces dedicated gate mechanisms to reduce vanishing gradients in standard RNNs
- **B.** The Cell State acts as a linear conveyor belt across long time sequences
- **C.** The Forget Gate controls information retention from previous cell state
- **D.** LSTM can only be used for image classification and cannot process text

Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are key architectural advantages of the Transformer over classical RNN models?

- **A.** Ability to scale efficiently to billions of parameters
- **B.** Direct global dependency modeling without distance-related attenuation
- **C.** Eliminating the need for any training data
- **D.** Full parallelization during training across all sequence tokens

Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are classic deep Convolutional Neural Network architectures?

- **A.** VGGNet
- **B.** LeNet-5
- **C.** AlexNet
- **D.** ResNet

Page Answers: **Q71**: ACD • **Q72**: AC • **Q73**: ABC • **Q74**: ABD • **Q75**: ABCD

Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following structural mechanisms are integrated inside a standard Transformer Encoder block?

- **A.** Multi-Head Self-Attention layer
- **B.** Depthwise spatial 2D max-pooling subsampling
- **C.** Layer Normalization and residual shortcut connections
- **D.** Position-wise Feedforward Neural Network

Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized mathematical limitations of the classic Sigmoid activation function?

- **A.** Vanishing gradient problem when inputs have large absolute values
- **B.** Outputs are not zero-centered, causing zigzag gradient updates during backpropagation
- **C.** Output values unbounded toward positive infinity
- **D.** Complete absence of first-order mathematical differentiability

Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are typical foundation model methods?

- **A.** Autoregressive language models
- **B.** Single-perceptron linear classifiers
- **C.** Encoder-decoder sequence models
- **D.** Masked autoencoder language models

Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Dropout are correct?

- **A.** During evaluation/testing, all neurons are kept active and outputs are scaled appropriately
- **B.** It randomly sets activations of selected neurons to zero with probability p during training
- **C.** It prevents complex co-adaptation of feature detectors
- **D.** It is an unsupervised clustering algorithm

Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are standard loss functions used for multi-class classification and continuous regression respectively?

- **A.** Triplet Contrastive Distance Metric
- **B.** Silhouette Cluster Separation Index
- **C.** Categorical Cross-Entropy Loss
- **D.** Mean Squared Error

Page Answers: **Q76**: ACD • **Q77**: AB • **Q78**: ACD • **Q79**: ABC • **Q80**: CD

Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Batch Normalization are true?

- **A.** It introduces learnable scale and shift parameters
- **B.** It normalizes each feature dimension across the mini-batch during training
- **C.** It allows higher learning rates and accelerates training convergence
- **D.** It can only be used on the final output classification layer

Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Backpropagation (BP) are correct?

- **A.** It generates random training labels for supervised datasets
- **B.** It computes parameter gradients backward from the output layer to the input layer
- **C.** It relies on the mathematical chain rule of differentiation
- **D.** Gradients computed by BP are passed to optimizers to update model weights

Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements correctly describe the standard ReLU activation function $f(x) = \max(0, x)$?

- **A.** Its output is strictly bounded within the closed real range of negative one and positive one
- **B.** It possesses a smooth, non-linear curve with continuous second-order derivatives everywhere
- **C.** For positive inputs, its derivative is constant 1, preventing gradient vanishing
- **D.** For negative inputs, its output is constant 0, inducing sparse network representation

Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are classic Parameter-Efficient Fine-Tuning (PEFT) methods for adapting large pre-trained models?

- **A.** LoRA
- **B.** Prompt Tuning
- **C.** Full retraining
- **D.** Random initialization

Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following problems can occur when training very deep neural networks without appropriate normalization or skip connections?

- **A.** Faster convergence
- **B.** Vanishing gradients
- **C.** Network degradation
- **D.** Exploding gradients

Page Answers: **Q81**: ABC • **Q82**: BCD • **Q83**: CD • **Q84**: AB • **Q85**: BCD

Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about the Softmax function are correct?

- **A.** Each output probability is strictly bounded between 0 and 1
- **B.** It converts real-valued logits into a probability distribution
- **C.** It can only be used when there are exactly two classes
- **D.** The sum of all output probabilities across classes equals exactly 1

Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

Unlike LSTM which contains three gates, the Gated Recurrent Unit (GRU) simplifies recurrent gating into which distinct gates?

- **A.** Forget Gate
- **B.** Reset Gate
- **C.** Output Gate
- **D.** Update Gate

Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are benefits of Data Augmentation in Computer Vision?

- **A.** Helps prevent model overfitting
- **B.** Deletes corrupted image files from cloud storage
- **C.** Increases the diversity of training samples without gathering new data
- **D.** Improves the generalization ability of neural networks

Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common types of Positional Encodings used to provide sequence order awareness in Transformers?

- **A.** Sinusoidal encodings
- **B.** Learned Positional Embeddings
- **C.** Stochastic Gaussian Noise Additive Perturbations
- **D.** Batch Normalization Running Variance Scaling

Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are hyperparameters of a Convolutional Layer?

- **A.** Padding size
- **B.** Kernel size
- **C.** Stride
- **D.** Number of output channels

Page Answers: **Q86**: ABD • **Q87**: BD • **Q88**: ACD • **Q89**: AB • **Q90**: ABCD

Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary structural layers commonly found in Convolutional Neural Networks (CNNs)?

- **A.** Convolutional Layers
- **B.** Pooling Layers
- **C.** Fully Connected Layers
- **D.** Recurrent Feedback Memory Units

Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

In deep neural network training, which of the following official techniques are used to mitigate Exploding Gradients and Vanishing Gradients?

- **A.** Applying Gradient Clipping to cap gradients exceeding a threshold, preventing exploding gradients
- **B.** Applying L1 and L2 weight regularization to constrain weight growth and resolve exploding gradients
- **C.** Using ReLU activations (derivative 1 for x > 0) and LSTM gating to alleviate vanishing gradients
- **D.** Initializing every weight tensor across all hidden layers strictly to constant zero

Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

A Feedforward Neural Network (FNN) transmits signals unidirectionally with no feedback, so its topology can be represented as a Directed Acyclic Graph (abbreviation: ____).

Answer: _____

Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

In a Feedforward Neural Network neuron, the bias parameter 'b' represents the activation ____ required to fire the neuron.

Answer: _____

Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

For the standard Sigmoid activation function, the maximum value of its derivative sigma'(x) (occurring at x = 0) is ____. (Enter a decimal number.)

Answer: _____

Page Answers: **Q91**: ABC • **Q92**: ABC • **Q93**: DAG • **Q94**: threshold • **Q95**: 0.25

Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

Alongside Gradient Clipping, applying L1 or L2 weight ____ is an effective deep learning technique to mitigate the Exploding Gradient problem.

Answer: _____

Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

Given a 32 x 32 input image, a 5 x 5 convolution kernel with stride 1 and padding 0, the spatial width (and height) of the output feature map is ____. (Enter a number.)

Answer: _____

Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

On a CNN layer with 128 filters of spatial size 3 x 3 processing an input feature map of size 28 x 28 x 64, the depth of each convolution kernel is ____. (Enter a number.)

Answer: _____

Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

In Transformer Scaled Dot-Product Attention Attention(Q, K, V) = softmax((Q K^T) / sqrt(d_k)) V, the scaling factor sqrt(d_k) uses d_k, which represents the dimension of the ____ vector.

Answer: _____

Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

In multi-class neural network classifiers, the sum of all output probabilities produced by the Softmax layer across every class is always ____. (Enter a number.)

Answer: _____

Page Answers: **Q96**: regularization • **Q97**: 28 • **Q98**: 64 • **Q99**: Key (or key) • **Q100**: 1

HCIA-AI V4.0 Certification Practice Exam • Chapter 4

Module 4: AI Development Framework (100 Questions & Answer Key)

Curated by: Umed Jamal Nouri • v1.1 (Nov 2026) • github.com/UmedJaMala

Question 1

TRUE OR FALSE • 16 Points

PyTorch adopts a dynamic computational graph (Define-by-Run), which constructs the graph on the fly as operators are executed, enabling intuitive debugging with standard Python tools.

A. True
B. False

Question 2

TRUE OR FALSE • 16 Points

In static computational graphs (Define-and-Run), the graph structure is dynamically rebuilt at every execution step, allowing arbitrary changes to layer shapes during runtime without recompilation.

A. True
B. False

Question 3

TRUE OR FALSE • 16 Points

In PyTorch, calling optimizer.zero_grad() is necessary before loss.backward() because gradients accumulate by default across successive backward calls.

A. True
B. False

Question 4

TRUE OR FALSE • 16 Points

In PyTorch, the nn.CrossEntropyLoss criterion requires users to manually pass data through an explicit nn.Softmax layer before feeding it into the loss function.

A. True
B. False

Question 5

TRUE OR FALSE • 16 Points

A Tensor in deep learning frameworks is strictly limited to 2-dimensional matrices and cannot represent data with 3 or more dimensions.

A. True
B. False

Page Answers: **Q1**: True • **Q2**: False • **Q3**: True • **Q4**: False • **Q5**: False

Question 6

TRUE OR FALSE • 16 Points

Eager execution executes operations immediately and returns concrete tensor values without needing an explicit Session.run() call.

A. True
B. False

Question 7

TRUE OR FALSE • 16 Points

In PyTorch, custom neural network modules do not need to inherit from torch.nn.Module and can execute without defining a forward() method.

A. True
B. False

Question 8

TRUE OR FALSE • 16 Points

The torch.utils.data.DataLoader class is used only for saving trained model weight checkpoints to the hard drive.

A. True
B. False

Question 9

TRUE OR FALSE • 16 Points

Model quantization converts low-bit INT8 representations into high-precision FP64 to significantly increase model file size and memory footprint.

A. True
B. False

Question 10

TRUE OR FALSE • 16 Points

In AI model persistence, the Checkpoint (.ckpt) file format uses Protocol Buffers to store training parameter values by default (and can optionally save model structure and training state) for resuming interrupted training or fine-tuning.

A. True
B. False

Page Answers: **Q6**: True • **Q7**: False • **Q8**: False • **Q9**: False • **Q10**: True

Question 11

TRUE OR FALSE • 16 Points

Model pruning increases model complexity and expands file size by randomly adding duplicate layers to the neural network.

A. True
B. False

Question 12

TRUE OR FALSE • 16 Points

In PyTorch, calling model.eval() turns on Dropout and updates running statistics in Batch Normalization layers.

A. True
B. False

Question 13

TRUE OR FALSE • 16 Points

ONNX (Open Neural Network Exchange) provides an open ecosystem and universal intermediate representation format allowing models to transfer across different AI frameworks.

A. True
B. False

Question 14

TRUE OR FALSE • 16 Points

MindSpore is Huawei's open-source all-scenario AI development framework designed to support unified APIs across cloud, edge, and device environments.

A. True
B. False

Question 15

TRUE OR FALSE • 16 Points

Automatic differentiation in deep learning frameworks automatically computes exact derivative values without requiring developers to manually write analytical gradient equations.

A. True
B. False

Page Answers: **Q11**: False • **Q12**: False • **Q13**: True • **Q14**: True • **Q15**: True

Question 16

TRUE OR FALSE • 16 Points

A PyTorch Tensor residing in CPU memory can perform direct hardware matrix multiplications with a Tensor residing in GPU (CUDA) memory without explicit device transfer.

A. True
B. False

Question 17

TRUE OR FALSE • 16 Points

In PyTorch, setting tensor.requires_grad = False signals to the autograd engine that gradients must be actively tracked and computed for that tensor.

A. True
B. False

Question 18

TRUE OR FALSE • 16 Points

TensorFlow 2.0 adopted Eager Execution as its default operating mode to significantly improve ease of development and debugging.

A. True
B. False

Question 19

TRUE OR FALSE • 16 Points

In deep learning frameworks, operator fusion combines multiple successive computational kernels into a single kernel to minimize memory access overhead and improve execution efficiency.

A. True
B. False

Question 20

TRUE OR FALSE • 16 Points

TFRecord is a binary file storage format designed in TensorFlow for efficient high-throughput sequential data reading.

A. True
B. False

Page Answers: **Q16**: False • **Q17**: False • **Q18**: True • **Q19**: True • **Q20**: True

Question 21

TRUE OR FALSE • 16 Points

When constructing a custom Dataset in PyTorch, developers must implement both the __len__() and __getitem__() methods.

A. True
B. False

Question 22

TRUE OR FALSE • 16 Points

The torch.no_grad() context manager forces PyTorch to track and calculate gradients for all tensor operations during evaluation.

A. True
B. False

Question 23

TRUE OR FALSE • 16 Points

In PyTorch, .pt and .pth are the default file formats for saving complete models (architecture and parameters) or parameter dictionaries, whereas converting a PyTorch model to a generic .bin format requires explicit format conversion.

A. True
B. False

Question 24

TRUE OR FALSE • 16 Points

Hardware acceleration in modern AI frameworks is achieved only through CPU multi-threading, while GPUs and NPUs cannot be used.

A. True
B. False

Question 25

TRUE OR FALSE • 16 Points

In Huawei's AI full-stack architecture, CANN (Compute Architecture for Neural Networks) provides high-performance operator libraries directly interfacing with Ascend AI chips.

A. True
B. False

Page Answers: **Q21**: True • **Q22**: False • **Q23**: True • **Q24**: False • **Q25**: True

Question 26

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT an AI deep learning framework?

- **A.** MySQL
- **B.** TensorFlow
- **C.** PyTorch
- **D.** MindSpore

Question 27

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT a core characteristic of PyTorch?

- **A.** Mandatory static compilation of entire code into C++ sessions before executing single operators
- **B.** Native Pythonic development experience supporting standard control flow and interactive debuggers
- **C.** Seamless acceleration across different hardware targets through unified CUDA and NPU backends
- **D.** Dynamic computational graph construction where graphs are built on the fly

Question 28

SINGLE-ANSWER QUESTION • 16 Points

What is used in computational graphs to represent operations (such as matrix multiplication and convolution)?

- **A.** Edges
- **B.** Nodes
- **C.** Weights
- **D.** File descriptors

Question 29

SINGLE-ANSWER QUESTION • 16 Points

In a computational graph, what do the directed edges connecting nodes represent?

- **A.** Operators
- **B.** Callbacks
- **C.** Tensors
- **D.** Optimizers

Question 30

SINGLE-ANSWER QUESTION • 16 Points

In PyTorch, which module is specifically responsible for automatic differentiation of tensors?

- **A.** torch.multiprocessing
- **B.** torch.distributed
- **C.** torch.autograd
- **D.** torch.optim.lr

Page Answers: **Q26**: A • **Q27**: A • **Q28**: B • **Q29**: C • **Q30**: C

Question 31

SINGLE-ANSWER QUESTION • 16 Points

Which method in PyTorch calculates the backward gradients of a loss scalar with respect to all tracked leaf tensors?

- **A.** model.forward()
- **B.** loss.backward()
- **C.** tensor.zero_grad()
- **D.** optimizer.step()

Question 32

SINGLE-ANSWER QUESTION • 16 Points

Which method in PyTorch updates the model parameters based on their computed gradients?

- **A.** loss.backward()
- **B.** model.eval()
- **C.** optimizer.step()
- **D.** torch.cuda.empty_cache()

Question 33

SINGLE-ANSWER QUESTION • 16 Points

What happens if a developer forgets to call optimizer.zero_grad() at the beginning of each training iteration in PyTorch?

- **A.** The interpreter raises a fatal runtime exception and halts execution immediately on the forward pass
- **B.** Gradients from the current batch accumulate onto historical gradients, corrupting parameter updates
- **C.** The model skips forward propagation completely and freezes all parameter tensors at zero
- **D.** PyTorch automatically halves the optimizer learning rate after every ten training batches

Question 34

SINGLE-ANSWER QUESTION • 16 Points

In PyTorch, how do you move a model and input tensors to GPU for accelerated processing?

- **A.** model.compile()
- **B.** model.transfer()
- **C.** model.serialize()
- **D.** model.cuda()

Question 35

SINGLE-ANSWER QUESTION • 16 Points

Which PyTorch package contains popular computer vision datasets, model architectures, and image transformation utilities?

- **A.** torchaudio
- **B.** torchvision
- **C.** torchrec
- **D.** torchtext

Page Answers: **Q31**: B • **Q32**: C • **Q33**: B • **Q34**: D • **Q35**: B

Question 36

SINGLE-ANSWER QUESTION • 16 Points

In the PyTorch torchvision.transforms module, which transformation converts a PIL Image or NumPy array into a floating-point Tensor normalized to [0.0, 1.0]?

- **A.** transforms.Grayscale()
- **B.** transforms.Resize()
- **C.** transforms.Normalize()
- **D.** transforms.ToTensor()

Question 37

SINGLE-ANSWER QUESTION • 16 Points

What is the primary function of the collate_fn argument in PyTorch DataLoader?

- **A.** Merging a list of individual sampled dataset elements into a formatted, stacked mini-batch tensor
- **B.** Compiling high-performance CUDA operator kernels into native machine instructions ahead of time
- **C.** Serializing and exporting trained model architectures into cross-platform ONNX runtime binary formats
- **D.** Deleting unreferenced and redundant raw media files from external network-attached storage disks

Question 38

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes the 'Define-by-Run' design pattern used in modern deep learning frameworks?

- **A.** The computational graph is dynamically constructed on the fly as each individual operator executes
- **B.** The model must be exported to static ONNX binary format before forward pass execution
- **C.** The execution graph is statically declared and frozen in advance before any data tensors can flow
- **D.** Operators are defined and orchestrated through external relational database schemas and SQL queries

Question 39

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is a major advantage of Static Computational Graphs?

- **A.** Direct integration with native Python if-else conditions and standard loop statements during runtime
- **B.** Total elimination of compilation phases, enabling instant execution of newly modified operator lines
- **C.** Native line-by-line interactive debugging support using standard Python debugging utilities like pdb
- **D.** Extensive ahead-of-time whole-graph optimizations, memory reuse planning, and easy embedded deployment

Question 40

SINGLE-ANSWER QUESTION • 16 Points

Which model compression method involves removing less important weight parameters (setting them to zero) based on magnitude or sensitivity?

- **A.** Model Quantization
- **B.** Knowledge Distillation
- **C.** Model Pruning
- **D.** Data Augmentation

Page Answers: **Q36**: D • **Q37**: A • **Q38**: A • **Q39**: D • **Q40**: C

Question 41

SINGLE-ANSWER QUESTION • 16 Points

In Knowledge Distillation, what does the compact Student model learn from during training?

- **A.** Soft target probabilities from the larger Teacher model alongside hard ground-truth target labels
- **B.** Low-level operating system kernel diagnostic logs recorded during hardware boot-up sequences
- **C.** Zeroing out random activation channels during inference passes
- **D.** Raw Python source code comments and docstrings parsed directly from external code repositories

Question 42

SINGLE-ANSWER QUESTION • 16 Points

What type of quantization converts 32-bit floating-point weights (FP32) to 8-bit integers (INT8)?

- **A.** Binarized Weight Pruning
- **B.** Unstructured Distillation
- **C.** INT8 Quantization
- **D.** FP64 Double Precision

Question 43

SINGLE-ANSWER QUESTION • 16 Points

Which PyTorch module implements a two-dimensional convolutional layer?

- **A.** torch.nn.Conv2d
- **B.** torch.nn.BatchNorm1d
- **C.** torch.nn.Linear
- **D.** torch.nn.Embedding

Question 44

SINGLE-ANSWER QUESTION • 16 Points

In PyTorch, which module is used to define a fully connected (dense) linear layer?

- **A.** torch.nn.Linear
- **B.** torch.nn.Dropout2d
- **C.** torch.nn.Conv1d
- **D.** torch.nn.RNNCell

Question 45

SINGLE-ANSWER QUESTION • 16 Points

What is the recommended PyTorch method for saving only the learned parameters (weights and biases) of a model?

- **A.** torch.save(model, PATH)
- **B.** json.dump(model)
- **C.** pickle.dump(model)
- **D.** torch.save(model.state_dict(), PATH)

Page Answers: **Q41**: A • **Q42**: C • **Q43**: A • **Q44**: A • **Q45**: D

Question 46

SINGLE-ANSWER QUESTION • 16 Points

What is the recommended PyTorch method for restoring learned parameters into an instantiated model architecture?

- **A.** model.restore_all_parameters
- **B.** model = torch.deserialize_weights
- **C.** model.load_state_dict
- **D.** torch.compile_and_mount_weights

Question 47

SINGLE-ANSWER QUESTION • 16 Points

Why is model.eval() called prior to running validation or testing on a PyTorch neural network?

- **A.** It sets all gradient accumulators to zero and clears temporary intermediate autograd computation buffers
- **B.** It recompiles the Python model into optimized native C++ assembly binaries for rapid standalone execution
- **C.** It unmounts the training dataset partitions from physical storage to maximize available operating system RAM
- **D.** It switches Dropout and BatchNorm to evaluation mode, disabling random dropping and freezing statistics

Question 48

SINGLE-ANSWER QUESTION • 16 Points

In Huawei's MindSpore framework, what is the unified programming format that supports both dynamic debugging and static acceleration?

- **A.** C-only mode and Fortran mode
- **B.** TCP mode and UDP mode
- **C.** SQL mode and NoSQL mode
- **D.** PyNative mode and Graph mode

Question 49

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is an open-source framework developed by Google that introduced XLA (Accelerated Linear Algebra) and automatic differentiation for Python/NumPy programs?

- **A.** Theano
- **B.** PyTorch
- **C.** Caffe
- **D.** JAX

Question 50

SINGLE-ANSWER QUESTION • 16 Points

In PyTorch, what tensor operation changes the tensor view/shape without altering the underlying data in memory?

- **A.** tensor.view() or tensor.reshape()
- **B.** tensor.backward_propagation()
- **C.** tensor.to_cuda_device()
- **D.** tensor.item_to_scalar()

Page Answers: **Q46**: C • **Q47**: D • **Q48**: D • **Q49**: D • **Q50**: A

Question 51

SINGLE-ANSWER QUESTION • 16 Points

In PyTorch, which function concatenates a sequence of tensors along a specified existing dimension?

A. torch.chunk()
B. torch.squeeze()
C. torch.cat()
D. torch.split()

Question 52

SINGLE-ANSWER QUESTION • 16 Points

What is the purpose of the torch.squeeze() function?

A. Normalizing tensor values between 0 and 1
B. Compressing the tensor with ZIP compression
C. Adding a new dimension of size 1 at a specified position
D. Removing all dimensions of size 1 from the tensor

Question 53

SINGLE-ANSWER QUESTION • 16 Points

What is the purpose of torch.unsqueeze(dim) in tensor manipulation?

A. Inserting a new singleton dimension at the specified tensor index position
B. Deleting all negative numerical values across the tensor buffer and clipping to zero
C. Transferring tensor memory buffers from CPU host RAM to GPU device memory partitions
D. Computing the inverse matrix transformation across two-dimensional square tensor slices

Question 54

SINGLE-ANSWER QUESTION • 16 Points

Which of the following represents the correct sequence in a typical PyTorch training step?

A. output = model(input) -> optimizer.step() -> loss.backward() -> optimizer.zero_grad() -> log_batch_metrics()
B. loss.backward() -> optimizer.step() -> optimizer.zero_grad() -> output = model(input) -> loss = criterion(output, target)
C. optimizer.step() -> loss.backward() -> output = model(input) -> optimizer.zero_grad() -> loss = criterion(output, target)
D. optimizer.zero_grad() -> output = model(input) -> loss = criterion(output, target) -> loss.backward() -> optimizer.step()

Question 55

SINGLE-ANSWER QUESTION • 16 Points

In PyTorch, which module is commonly used to adjust the learning rate during training according to predefined epoch schedules?

A. torch.autograd.grad
B. torchvision.models
C. torch.optim.lr_scheduler
D. torch.utils.data.Sampler

Page Answers: **Q51**: C • **Q52**: D • **Q53**: A • **Q54**: D • **Q55**: C

Question 56

SINGLE-ANSWER QUESTION • 16 Points

Which tool in PyTorch allows serializing dynamic models into static TorchScript graphs for production deployment without Python dependencies?

- **A.** torch.nn.parallel or torch.cuda.amp
- **B.** torch.jit.trace or torch.jit.script
- **C.** torch.save_weights or torch.load_bin
- **D.** torch.optim.sgd or torch.optim.adam

Question 57

SINGLE-ANSWER QUESTION • 16 Points

What role does CANN play in Huawei's Ascend AI software stack?

- **A.** A cloud-based web browser designed for visualizing distributed multi-node training metrics and logs
- **B.** The computing architecture and chip enablement layer, exposing optimized operator APIs
- **C.** A mobile operating system kernel tailored specifically for consumer wearable smart devices and phones
- **D.** A relational database management system storing user authentication tokens and private access keys

Question 58

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is Huawei's lightweight AI engine designed for ultra-fast, low-latency model inference on mobile and IoT edge devices?

- **A.** MindSpore Lite
- **B.** Kunpeng OS
- **C.** ModelArts Cloud
- **D.** GaussDB

Question 59

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is a common high-performance deployment engine developed by NVIDIA for deep learning inference on GPUs?

- **A.** PyTorch Eager
- **B.** Matplotlib
- **C.** TensorRT
- **D.** Pandas

Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Model Compression and Pruning techniques are correct? (Multiple Choice)

- **A.** Structured pruning removes entire contiguous channels or layers to achieve direct hardware acceleration
- **B.** Pruning permanently doubles the precision of weight tensors to 64-bit double precision
- **C.** Model pruning reduces parameter count and inference memory footprint while striving to preserve accuracy
- **D.** Pruning can only be performed before training and cannot be applied to pre-trained weights

Page Answers: **Q56**: B • **Q57**: B • **Q58**: A • **Q59**: C • **Q60**: AC

Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

In PyTorch data loading pipelines using DataLoader, which parameters directly optimize throughput and CPU-GPU transfer efficiency?

- **A.** num_workers
- **B.** pin_memory
- **C.** batch_size
- **D.** learning_rate

Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following techniques improve data transfer speeds and pipeline throughput in deep learning training?

- **A.** Using page-locked pinned host memory
- **B.** Asynchronous multi-threaded data pre-fetching and batch queuing
- **C.** Writing intermediate mini-batches to uncompressed text CSV files between training steps
- **D.** Storing training data in optimized binary container formats

Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following loss functions are commonly implemented in modern deep learning frameworks (MindSpore, PyTorch) for continuous regression tasks?

- **A.** Categorical Cross-Entropy Loss
- **B.** Mean Squared Error Loss
- **C.** Binary Cross-Entropy with Logits
- **D.** Smooth L1 Loss

Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

In AI compiler software stacks (such as Huawei CANN and TVM), which benefits are achieved by Operator Fusion?

- **A.** Significant reduction in intermediate tensor memory read/write access overhead
- **B.** Converting non-linear neural networks into closed-form linear equation systems
- **C.** Minimization of hardware kernel launch overhead and execution scheduling latency
- **D.** Doubling the physical floating-point register count on the processor silicon core

Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following services and capabilities are provided by Huawei Cloud ModelArts AI development platform?

- **A.** Bare-metal FPGA bitstream synthesis and hardware register-transfer level logic routing
- **B.** Automated data labeling, dataset management, and feature store integration
- **C.** Distributed large-scale model training with automated hyperparameter optimization
- **D.** One-click model deployment, real-time online inference API hosting, and batch serving

Page Answers: **Q61**: ABC • **Q62**: ABD • **Q63**: BD • **Q64**: AC • **Q65**: BCD

Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary functions provided by an AI Development Framework?

- **A.** High-level front-end programming APIs for defining neural network layers
- **B.** Automatic differentiation and gradient computation engine
- **C.** Model optimization, serialization, and deployment tools
- **D.** Hardware abstraction and computation scheduling

Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are defining characteristics of Eager Execution (Define-by-Run) mode in deep learning frameworks?

- **A.** Immediate execution of operations with direct numerical feedback
- **B.** Mandatory ahead-of-time compilation of the complete computational graph
- **C.** Seamless line-by-line debugging using standard Python debugging tools like pdb
- **D.** Immutability of network layer topologies during runtime execution

Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following phases are included in the end-to-end deep learning engineering lifecycle?

- **A.** Model validation, evaluation, and hyperparameter tuning
- **B.** Data preparation, cleansing, and preprocessing pipelines
- **C.** Model architecture construction and training loop execution
- **D.** Manual hexadecimal byte rewriting of pre-trained binary weights before serving

Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary attributes of a PyTorch Tensor?

- **A.** requires_grad
- **B.** shape
- **C.** dtype
- **D.** device

Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following hardware accelerators are primarily targeted by modern deep learning frameworks for high-throughput matrix computation?

- **A.** Legacy Optical Disc Drives
- **B.** Serial RS-232 Port Controllers
- **C.** Graphics Processing Units
- **D.** Neural Processing Units

Page Answers: **Q66**: ABCD • **Q67**: AC • **Q68**: ABC • **Q69**: ABCD • **Q70**: CD

Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Static Computational Graphs are true?

A. It enables extensive compiler-level global graph optimizations and operator fusions
B. It allows native interactive debugging using standard Python print statements inside the graph execution
C. It is typically easier to export to standalone embedded C++ runtimes
D. The graph topology is built once and executed repeatedly

Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Dynamic Computational Graphs are true?

A. The computational graph is built dynamically in memory during each forward pass
B. Debugging is very straightforward because inspection is synchronous
C. Dynamic control flow is fully supported
D. It completely eliminates the need for GPU drivers

Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

When implementing, training, and saving the LeNet CNN architecture in PyTorch (as presented in the HCIA-AI V4.0 training material and lab guide), which of the following statements are correct?

A. Convolutional feature maps are flattened via x = x.view(-1, 16*5*5) before entering nn.Linear(16*5*5, 120)
B. Pooling operations such as nn.AvgPool2d(kernel_size=2, stride=2) or torch.max_pool2d(x, 2) downsample feature maps without learnable weights
C. Saving only learned weights with torch.save(net.state_dict(), path) requires instantiating net = LeNet() before calling net.load_state_dict(...)
D. Calling net.eval() automatically overwrites saved .pth checkpoint files with random uninitialized weights

Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common loss functions available in PyTorch (torch.nn)?

A. nn.BCELoss
B. nn.CrossEntropyLoss
C. nn.MSELoss
D. nn.L1Loss

Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following optimization algorithms are implemented in PyTorch's torch.optim module?

A. torch.optim.SGD
B. torch.optim.Adam
C. torch.optim.Softmax
D. torch.optim.RMSprop

Page Answers: **Q71**: ACD • **Q72**: ABC • **Q73**: ABC • **Q74**: ABCD • **Q75**: ABD

Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

When creating a custom Dataset class in PyTorch, which magic methods MUST be implemented?

A. __len__() (returning the total size of the dataset)
B. __init_weights__() (initializing random network parameters)
C. __backward__() (computing analytic loss gradients)
D. __getitem__() (fetching an individual data sample by index)

Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

According to the HCIA-AI V4.0 Model File specification, which of the following statements accurately describe standard model file formats?

A. Checkpoint (.ckpt) uses Protocol Buffers to store parameters by default and is used to resume interrupted training or fine-tune
B. ONNX (.onnx) provides a general model expression for model migration between frameworks or execution on TensorRT
C. Generic binary (.bin) is a common binary format requiring explicit conversion from PyTorch models
D. PyTorch (.pt or .pth) is PyTorch's default model file format for saving and loading complete models or weights

Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

Which execution modes are natively supported by Huawei's MindSpore AI development framework?

A. Graph Mode
B. PyNative Mode
C. Assembly Mode
D. Flash Emulation Mode

Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are valid operations for altering tensor shapes in PyTorch?

A. tensor.transpose()
B. tensor.squeeze()
C. tensor.reshape()
D. tensor.view()

Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary benefits of wrapping inference execution inside the with torch.no_grad(): context block?

A. Automatically pruning 50% of the neural network weights
B. Disabling autograd tracking to greatly reduce GPU memory consumption
C. Accelerating forward pass execution by skipping computation graph generation
D. Eliminating the necessity of loading pretrained model weights into RAM

Page Answers: **Q76**: AD • **Q77**: ABCD • **Q78**: AB • **Q79**: ABCD • **Q80**: BC

Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following operations occur during model backpropagation and optimization?

- **A.** Storing gradients into the .grad attribute of parameter tensors
- **B.** Erasing the original dataset files from storage
- **C.** Updating weight parameters in the negative gradient direction using the learning rate
- **D.** Computing partial derivatives of the loss function via the chain rule

Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are image transformations available in torchvision.transforms?

- **A.** transforms.RandomHorizontalFlip
- **B.** transforms.ColorJitter
- **C.** transforms.Normalize
- **D.** transforms.RandomCrop

Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are major advantages of Post-Training Quantization (e.g., FP32 to INT8)?

- **A.** Greatly reduced model memory and storage footprint on disk
- **B.** Guaranteed zero accuracy loss across all complex downstream tasks
- **C.** Doubling the quantity of trainable floating-point parameters
- **D.** Accelerated inference throughput and reduced latency on edge hardware

Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common distributed training methods supported by modern AI frameworks?

- **A.** Standalone single-core CPU execution without distributed communication
- **B.** Data Parallelism
- **C.** Model or Tensor Parallelism
- **D.** Pipeline Parallelism

Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following components are part of Huawei's full-stack AI development ecosystem?

- **A.** Ascend AI processors
- **B.** ModelArts development platform
- **C.** MindSpore AI framework
- **D.** CANN compute architecture

Page Answers: **Q81**: ACD • **Q82**: ABCD • **Q83**: AD • **Q84**: BCD • **Q85**: ABCD

Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are reasons why AI development frameworks are necessary for deep learning?

- **A.** Hand-calculating and hardcoding analytic Jacobian matrices for every layer
- **B.** Frameworks automatically eliminate all need for domain training data collection
- **C.** Frameworks provide unified multi-hardware acceleration
- **D.** Frameworks provide automated memory management and memory reuse across execution graphs

Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about PyTorch DataLoader are correct?

- **A.** It supports multi-process data loading via num_workers
- **B.** It automatically designs neural network architectures
- **C.** It allows shuffling of samples before each epoch
- **D.** It supports automatic batching of data samples

Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are widely adopted open standard formats/runtimes for cross-framework model deployment and inference?

- **A.** Open Neural Network Exchange
- **B.** HTML5 Web Document Format
- **C.** TensorRT or MindSpore Lite runtime formats
- **D.** Comma-Separated Values Tabular Format

Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following layers in torch.nn require input dimension configurations during initialization?

- **A.** nn.Flatten
- **B.** nn.Conv2d
- **C.** nn.Linear
- **D.** nn.ReLU

Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are advantages of Knowledge Distillation?

- **A.** Guarantees that the student model has zero weight parameters
- **B.** Does not require altering the hardware architecture of edge deployment devices
- **C.** Enables lightweight models to achieve accuracy close to large teacher models
- **D.** Significantly reduces inference latency on resource-constrained devices

Page Answers: **Q86**: CD • **Q87**: ACD • **Q88**: AC • **Q89**: BC • **Q90**: BCD

Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

In deep learning computational graphs, which of the following statements regarding nodes and edges are TRUE?

A. Nodes represent mathematical operations or mathematical functions
B. Edges represent the flow of multidimensional arrays between computational operations
C. Automatic differentiation engines traverse the graph in reverse topological order during backpropagation
D. Computational graphs can only execute sequential linear pipelines and prohibit branching

Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Static and Dynamic Computational Graphs are correct?

A. Dynamic computational graphs construct the execution graph on-the-fly during runtime
B. Static computational graphs build the full symbolic graph structure prior to execution
C. Static computational graphs help compiler-level optimizations such as operator fusion and memory reuse
D. Modern AI frameworks can convert dynamic functions into optimized static graphs

Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the core package responsible for automatic differentiation and gradient tracking is torch.____.

Answer: _____

Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, all custom neural network models and layers must inherit from the base class torch.nn.____.

Answer: _____

Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

According to the HCIA-AI V4.0 Model File specification, the Checkpoint (____) file format uses Protocol Buffers to store network parameters for resuming interrupted training or fine-tuning.

Answer: _____

Page Answers: **Q91**: ABC • **Q92**: ABCD • **Q93**: autograd • **Q94**: Module • **Q95**: .ckpt (or ckpt)

Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

In Huawei MindSpore, the forward computational execution of a neural network must be defined inside the def ____(self, x): method of nn.Cell.

Answer: _____

Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

In Huawei MindSpore, all neural network models and functional layers must inherit from the base class mindspore.nn.____.

Answer: _____

Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

In Huawei MindSpore, learned network weights are saved to a checkpoint file on disk using the function mindspore.____(network, path).

Answer: _____

Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

To migrate a trained PyTorch (.pt/.pth) model between different AI frameworks or deploy it on an inference engine like TensorRT, developers convert it into the open ____ format.

Answer: _____

Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the class from torch.utils.data used to combine a dataset with mini-batching, shuffling, and multi-threaded loading is ____.

Answer: _____

Page Answers: **Q96**: construct • **Q97**: Cell • **Q98**: save_checkpoint • **Q99**: ONNX (or .onnx) • **Q100**: DataLoader

HCIA-AI V4.0 Certification Practice Exam • Chapter 5

Module 5: AI Business Process & Prompt Engineering (100 Questions & Answer Key)

Curated by: Umed Jamal Nouri • v1.1 (Nov 2026) • github.com/UmedJaMala

Question 1

TRUE OR FALSE • 16 Points

In Data Parallelism, the model architecture is partitioned layer by layer across devices while the entire dataset remains on a single GPU.

A. True
B. False

Question 2

TRUE OR FALSE • 16 Points

In Pipeline Parallelism, the weight parameters within each individual layer are partitioned across multiple GPUs for matrix multiplication.

A. True
B. False

Question 3

TRUE OR FALSE • 16 Points

Model fine-tuning primarily adjusts the number of neural network layers and changes the overall network topology rather than tuning existing weight parameters.

A. True
B. False

Question 4

TRUE OR FALSE • 16 Points

The core idea of Adapter Tuning is to insert small, trainable adapter sub-modules into fixed layers of a pre-trained model while freezing the original backbone weights.

A. True
B. False

Question 5

TRUE OR FALSE • 16 Points

In traditional AI business development, Requirement Analysis is the starting phase where business objectives are translated into clear machine learning problem formulations.

A. True
B. False

Page Answers: **Q1**: False • **Q2**: False • **Q3**: False • **Q4**: True • **Q5**: True

Question 6

TRUE OR FALSE • 16 Points

Concept Drift refers to the physical hardware failure of hard drives and GPUs during model serving.

A. True

B. False

Question 7

TRUE OR FALSE • 16 Points

Tensor Parallelism splits individual weight matrices (such as attention projection matrices or MLP layers) across multiple accelerators to reduce per-device memory usage.

A. True

B. False

Question 8

TRUE OR FALSE • 16 Points

Retrieval-Augmented Generation (RAG) completely eliminates the need for an external knowledge base by embedding all world facts directly into model activation weights.

A. True

B. False

Question 9

TRUE OR FALSE • 16 Points

In Prompt Engineering, Zero-Shot prompting provides multiple input-output demonstration examples before presenting the actual query.

A. True

B. False

Question 10

TRUE OR FALSE • 16 Points

In the AI business development pipeline, manual Feature Engineering (Optional) is strictly mandatory for deep learning Computer Vision (CV) and NLP tasks, rather than for Recommendation Systems.

A. True

B. False

Page Answers: **Q6**: False • **Q7**: True • **Q8**: False • **Q9**: False • **Q10**: False

Question 11

TRUE OR FALSE • 16 Points

Chain of Thought (CoT) prompting encourages language models to generate explicit intermediate reasoning steps, improving performance on multi-step logic and math problems.

A. True
B. False

Question 12

TRUE OR FALSE • 16 Points

Zero-Shot-CoT requires developers to manually author dozens of detailed step-by-step reasoning exemplars for every prompt.

A. True
B. False

Question 13

TRUE OR FALSE • 16 Points

Reinforcement Learning from Human Feedback (RLHF) aligns large model outputs with human intent, helpfulness, and safety guidelines.

A. True
B. False

Question 14

TRUE OR FALSE • 16 Points

In the RLHF pipeline, the Reward Model (RM) is trained on human pairwise preference rankings comparing different model responses to the same prompt.

A. True
B. False

Question 15

TRUE OR FALSE • 16 Points

LoRA (Low-Rank Adaptation) updates 100% of all pre-trained model weights, requiring complete retraining of all attention and feedforward layers.

A. True
B. False

Page Answers: **Q11**: True • **Q12**: False • **Q13**: True • **Q14**: True • **Q15**: False

Question 16

TRUE OR FALSE • 16 Points

A/B testing is an offline mathematical proof technique where models are tested without ever exposing them to real user traffic.

A. True
B. False

Question 17

TRUE OR FALSE • 16 Points

Prompt Injection is a security vulnerability where malicious user inputs trick a language model into ignoring its developer system instructions and executing unauthorized actions.

A. True
B. False

Question 18

TRUE OR FALSE • 16 Points

Data Contamination refers to physical dust and humidity damaging server hard drives in data centers.

A. True
B. False

Question 19

TRUE OR FALSE • 16 Points

Prompt Tuning prepends continuous learnable virtual token embeddings to the input sequence while freezing the underlying language model weights.

A. True
B. False

Question 20

TRUE OR FALSE • 16 Points

Prefix Tuning attaches learnable prefix vectors to the keys and values of attention layers in each Transformer block.

A. True
B. False

Page Answers: **Q16**: False • **Q17**: True • **Q18**: False • **Q19**: True • **Q20**: True

Question 21

TRUE OR FALSE • 16 Points

Supervised Fine-Tuning (SFT) transforms raw pre-trained models into conversational instruction-following assistants using paired (instruction, output) data.

A. True
B. False

Question 22

TRUE OR FALSE • 16 Points

Large language models deployed in enterprise production do not require monitoring because foundation models are permanently immune to factual decay.

A. True
B. False

Question 23

TRUE OR FALSE • 16 Points

The ReAct framework combines reasoning traces and task-specific external tool actions in an interleaved loop to enable autonomous agent problem-solving.

A. True
B. False

Question 24

TRUE OR FALSE • 16 Points

In large model training, 3D Parallelism refers to at the same time combining Data Parallelism, Tensor Parallelism, and Pipeline Parallelism across thousands of accelerators.

A. True
B. False

Question 25

TRUE OR FALSE • 16 Points

Evaluating large models using an LLM-as-a-Judge approach is completely immune to self-enhancement bias or position bias.

A. True
B. False

Page Answers: **Q21**: True • **Q22**: False • **Q23**: True • **Q24**: True • **Q25**: False

Question 26

SINGLE-ANSWER QUESTION • 16 Points

Within the broad landscape of large language models, what is the primary goal of Supervised Fine-Tuning (SFT)?

- **A.** Substituting internal attention layers with traditional rule-based scripts
- **B.** Compressing the total parameter capacity of the model by ninety percent using pruning
- **C.** Adapting the pre-trained model to follow specific user instructions and conversational formats
- **D.** Re-training all weights from scratch without pre-trained checkpoints

Question 27

SINGLE-ANSWER QUESTION • 16 Points

Which of the following statements is TRUE regarding parallel training strategies for large models?

- **A.** Data parallelism divides model layers across devices, whereas tensor parallelism splits batches
- **B.** Tensor parallelism splits single layers, whereas pipeline parallelism partitions sequential layers
- **C.** Large models can only be trained on standalone workstation GPUs without distributed clustering
- **D.** Pipeline parallelism executes matrix multiplications synchronously on single-core microcontrollers

Question 28

SINGLE-ANSWER QUESTION • 16 Points

Which distributed training strategy partitions the sequential layers of a deep neural network across multiple accelerators like stages in an assembly line?

- **A.** Pipeline Parallelism
- **B.** K-Means Clustering
- **C.** Data Parallelism
- **D.** Zero-Shot Prompting

Question 29

SINGLE-ANSWER QUESTION • 16 Points

Which PEFT method freezes original weights W_0 and injects trainable rank decomposition matrices B and A such that W = W_0 + B * A (where rank r << min(d, k))?

- **A.** Full fine-tuning
- **B.** Gradient descent
- **C.** LoRA
- **D.** Grid search

Question 30

SINGLE-ANSWER QUESTION • 16 Points

What is the primary advantage of Parameter-Efficient Fine-Tuning (PEFT) compared to Full Fine-Tuning?

- **A.** Greatly lowers GPU memory and trainable parameters while preventing catastrophic forgetting
- **B.** Completely eliminates the requirement for labeled training datasets through unsupervised clustering
- **C.** Enforces simultaneous end-to-end retraining of all backbone layers across distributed cloud clusters
- **D.** Scales up total backbone parameter count tenfold by stacking additional multi-head attention layers

Page Answers: **Q26**: C • **Q27**: B • **Q28**: A • **Q29**: C • **Q30**: A

Question 31

SINGLE-ANSWER QUESTION • 16 Points

In the traditional AI project lifecycle, what phase immediately follows Model Training before production deployment?

- **A.** Problem Requirement Analysis
- **B.** Hyperparameter Sensitivity Profiling
- **C.** Model Evaluation and Validation
- **D.** Data Collection

Question 32

SINGLE-ANSWER QUESTION • 16 Points

Which prompting technique enables multi-step deductive reasoning by simply appending 'Let us think step by step'?

- **A.** Post-Training Vector Quantization
- **B.** Zero-Shot Chain-of-Thought
- **C.** In-Context Few-Shot Feature Alignment
- **D.** Differential Data Anonymization

Question 33

SINGLE-ANSWER QUESTION • 16 Points

What is the primary function of Retrieval-Augmented Generation (RAG) in enterprise AI applications?

- **A.** Rewriting native operating system kernel scheduling algorithms using interpreted Python code
- **B.** Querying external knowledge stores to inject real-time, domain context and reduce hallucination
- **C.** Deleting outdated customer database records automatically based on temporal retention rules
- **D.** Training foundational transformer base models from scratch using random initialization weights

Question 34

SINGLE-ANSWER QUESTION • 16 Points

In the RLHF (Reinforcement Learning from Human Feedback) workflow, which algorithm optimizes the policy model?

- **A.** Ordinary Least Squares Linear Regression
- **B.** K-Means Centroid Coordinate Clustering
- **C.** Information Gain Ratio Decision Tree
- **D.** Proximal Policy Optimization

Question 35

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes the 'Reward Model' in the RLHF process?

- **A.** A model trained on human preference rankings to output a scalar quality score for responses
- **B.** An ACID relational transactional accounting database
- **C.** A server billing script calculating enterprise electricity consumption during training runs
- **D.** A hyperparameter grid search module tuning learning rates across validation folds

Page Answers: **Q31**: C • **Q32**: B • **Q33**: B • **Q34**: D • **Q35**: A

Question 36

SINGLE-ANSWER QUESTION • 16 Points

What security threat occurs when an adversary crafts adversarial prompts to bypass safety guardrails and elicit restricted or dangerous content?

- **A.** Batch Normalization
- **B.** Cross-Validation
- **C.** Jailbreaking
- **D.** Weight Decay

Question 37

SINGLE-ANSWER QUESTION • 16 Points

What are the four core structural components of a well-designed professional prompt?

- **A.** HTML, CSS, JavaScript, and PHP
- **B.** IP Address, Port Number, Gateway, and Subnet Mask
- **C.** Instruction, Context, Input Data, and Output Indicator
- **D.** Loss Function, Optimizer, Batch Size, and Epoch Count

Question 38

SINGLE-ANSWER QUESTION • 16 Points

What is the primary operational mechanism of Adapter Tuning?

- **A.** Rebuilding the entire tokenizer vocabulary dictionary by parsing multi-language web archives
- **B.** Inserting small bottleneck feedforward layers into the transformer while keeping backbone weights frozen
- **C.** Pruning all attention heads from the transformer to force pure sequential recurrent processing
- **D.** Multiplying base model learning rates by one thousand to accelerate gradient convergence speed

Question 39

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT an advantage of Zero-Shot-CoT compared to Few-Shot-CoT?

- **A.** It offers better task scalability and flexibility
- **B.** It always guarantees strictly superior accuracy over hand-crafted Few-Shot-CoT in every benchmark
- **C.** It requires no manual engineering of multi-step problem exemplars
- **D.** It is simple to implement across diverse tasks using a standardized prompt suffix

Question 40

SINGLE-ANSWER QUESTION • 16 Points

What type of fine-tuning updates all model parameters at the same time across the entire network architecture?

- **A.** Low-Rank Adaptation Fine-Tuning
- **B.** Prefix Tuning Parameter Adaptation
- **C.** Full Model Fine-Tuning
- **D.** Prompt Tuning Soft Embedding Strategy

Page Answers: **Q36**: C • **Q37**: C • **Q38**: B • **Q39**: B • **Q40**: C

Question 41

SINGLE-ANSWER QUESTION • 16 Points

In enterprise AI production monitoring, what does 'Data Drift' refer to?

- **A.** Physical signal attenuation occurring along undersea fiber-optic telecommunication cables
- **B.** Changes in input feature distribution P(X) over time compared to the original training distribution
- **C.** Floating-point numerical precision rounding errors across different CPU registers
- **D.** Accidental deletion of validation split partitions from cloud object storage repositories

Question 42

SINGLE-ANSWER QUESTION • 16 Points

Which evaluation approach uses state-of-the-art foundation models (such as GPT-4) to grade and evaluate responses generated by candidate models?

- **A.** Keyword matching
- **B.** Grid search
- **C.** LLM-as-a-Judge
- **D.** Unit testing

Question 43

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of the 'Instruction' component in a prompt?

- **A.** Supplying pseudorandom seed integers to initialize GPU CUDA calculation kernels
- **B.** Specifying the exact task, operational behavior, or objective the model is required to perform
- **C.** Allocating static memory blocks within dedicated hardware video RAM buffers
- **D.** Defining firewall subnet routing policies for outbound API transmission endpoints

Question 44

SINGLE-ANSWER QUESTION • 16 Points

Which vector database operation is used in RAG to find the most relevant context documents?

- **A.** Dense vector semantic similarity search
- **B.** Exact lexical keyword matching using inverted boolean indices
- **C.** Exact regular expression wildcard string matching
- **D.** Uniform random sampling across database records

Question 45

SINGLE-ANSWER QUESTION • 16 Points

Why is 'Human-in-the-Loop' (HITL) crucial in high-risk AI business processes?

- **A.** To provide human expert oversight, safety validation, and ethical accountability for critical model decisions
- **B.** To automatically retrain neural network weights in real time without human evaluation
- **C.** To execute scalar arithmetic calculations on host CPU registers
- **D.** To guarantee 100% rule-based elimination of all software bugs in database schemas

Page Answers: **Q41**: B • **Q42**: C • **Q43**: B • **Q44**: A • **Q45**: A

Question 46

SINGLE-ANSWER QUESTION • 16 Points

What is the primary goal of the Pre-training stage in foundation model development?

- **A.** Learning universal language representations and world knowledge from vast unlabeled datasets
- **B.** Fitting an isolated decision tree classifier to predict categorical churn on tabular datasets
- **C.** Restricting model output syntax strictly to RFC-compliant JSON dictionary structures
- **D.** Memorizing fixed polite greeting phrases tailored for enterprise customer service chatbots

Question 47

SINGLE-ANSWER QUESTION • 16 Points

In LoRA, what is the hyperparameter 'rank' (r)?

- **A.** The inner low dimension of decomposition matrices A and B
- **B.** The total count of physical GPU acceleration nodes allocated in the cluster rack
- **C.** The relative performance ranking score achieved by the model on public benchmarks
- **D.** The batch size threshold configured for distributed asynchronous gradient updates

Question 48

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes 'Direct Preference Optimization' (DPO)?

- **A.** A sorting algorithm optimizing B-tree indices in relational database management systems
- **B.** An analytical closed-form solver calculating regression weights via ordinary least squares
- **C.** A graph search algorithm finding minimum spanning trees across distributed networks
- **D.** An alignment algorithm directly optimizing policy weights on human preferences without a separate reward model

Question 49

SINGLE-ANSWER QUESTION • 16 Points

What does MMLU (Massive Multitask Language Understanding) evaluate in large language models?

- **A.** Cross-domain knowledge and problem-solving across 57 academic subjects
- **B.** Network packet routing hop counts between routers
- **C.** Graphics processing unit rendering framerates and polygon rasterization speeds in video games
- **D.** Input/output operations per second across high-throughput solid-state storage arrays

Question 50

SINGLE-ANSWER QUESTION • 16 Points

In a RAG pipeline, what component is responsible for transforming raw text chunks into dense vector embeddings?

- **A.** Reranker
- **B.** Tokenizer
- **C.** Vector database
- **D.** Embedding model

Page Answers: **Q46**: A • **Q47**: A • **Q48**: D • **Q49**: A • **Q50**: D

Question 51

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is an example of an Output Indicator in prompt engineering?

- **A.** 'Translate the following business email into fluent Spanish.'
- **B.** 'Provide the response in valid JSON format containing keys: summary and action_items.'
- **C.** 'You are a certified senior cloud architect with twenty years of experience.'
- **D.** 'Input context: The quick brown fox jumps over the lazy dog in the meadow.'

Question 52

SINGLE-ANSWER QUESTION • 16 Points

What is the primary risk of 'Over-Refusal' in safety-aligned foundation models?

- **A.** The model exhaustively consumes all available host memory during batch token generation
- **B.** The model emits vulgar phrases and unmoderated offensive tokens to end users
- **C.** The model triggers unhandled kernel panics across underlying distributed server runtimes
- **D.** The model rejects benign, completely safe queries due to overly sensitive safety guardrails

Question 53

SINGLE-ANSWER QUESTION • 16 Points

In distributed large model training, which collective communication operation gathers and sums gradients across all worker nodes in Data Parallelism?

- **A.** Broadcast
- **B.** Scatter
- **C.** Gather
- **D.** AllReduce

Question 54

SINGLE-ANSWER QUESTION • 16 Points

What is 'Catastrophic Forgetting' in neural networks and foundation models?

- **A.** Loss of saved model checkpoint files caused by unrecoverable physical disk sectors
- **B.** Omission of standard library import statements in application deployment scripts
- **C.** System authentication failure resulting from forgotten user database passwords
- **D.** The tendency of a model to lose previously learned capabilities upon learning new tasks

Question 55

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is a classic Python framework for building LLM and RAG application workflows with composable chains and agents?

- **A.** LangChain Framework
- **B.** Matplotlib Library
- **C.** LightGBM Framework
- **D.** Scikit-Learn Toolkit

Page Answers: **Q51**: B • **Q52**: D • **Q53**: D • **Q54**: D • **Q55**: A

Question 56

SINGLE-ANSWER QUESTION • 16 Points

What is the primary metric for evaluating code generation capabilities in models like DeepSeek-Coder?

- **A.** Perplexity
- **B.** Mean squared error
- **C.** Pass@k
- **D.** BLEU score

Question 57

SINGLE-ANSWER QUESTION • 16 Points

In traditional machine learning project deployment, what is 'Canary Deployment'?

- **A.** Replacing software inference models with fixed-function application-specific integrated circuits
- **B.** Terminating all active web production servers at the same time during peak commercial traffic hours
- **C.** Replacing the entire production cluster at the same time without staged traffic routing
- **D.** Directing a small percentage of live traffic to the new model to verify stability before full rollout

Question 58

SINGLE-ANSWER QUESTION • 16 Points

Which prompt engineering technique prompts the model to assign itself a specific expert identity (e.g., 'Act as an expert cybersecurity auditor')?

- **A.** Token Pruning
- **B.** Data Imputation
- **C.** Temperature 0 Sampling
- **D.** Role Prompting

Question 59

SINGLE-ANSWER QUESTION • 16 Points

What is the impact of setting LLM inference 'Temperature' to 0?

- **A.** The model generates highly divergent, chaotic text with maximum Shannon entropy values
- **B.** Token generation velocity drops to zero because computation threads enter permanent sleep
- **C.** The model produces deterministic, greedy responses by always selecting the highest-probability token
- **D.** Model checkpoint weights fail cryptographic hash integrity checks

Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

In Large Language Model text generation, which decoding hyperparameters directly regulate the diversity and randomness of sampled tokens? (Multiple Choice)

- **A.** Temperature
- **B.** Top-P or Nucleus Sampling
- **C.** Top-K Sampling
- **D.** Weight Decay Coefficient

Page Answers: **Q56**: C • **Q57**: D • **Q58**: D • **Q59**: C • **Q60**: ABC

Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are dedicated vector database technologies commonly used for high-dimensional semantic similarity search in RAG?

- **A.** Milvus
- **B.** SQLite
- **C.** Chroma
- **D.** MySQL

Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following data sources are commonly aggregated and curated for Large Language Model self-supervised pre-training?

- **A.** Filtered, deduplicated, and toxicity-cleansed web text crawl datasets
- **B.** Curated open-source code repositories, technical documentation, and commit histories
- **C.** Encrypted private cryptographic SSL handshake session payloads
- **D.** High-quality digitized books, peer-reviewed academic papers, and encyclopedic reference collections

Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

In Large Language Model architecture and application design, why is the Context Window size a critical technical specification?

- **A.** It establishes the maximum combined token capacity for input prompts, retrieved external context, and generated responses
- **B.** Standard full self-attention memory and compute complexity scales quadratically with respect to context sequence length
- **C.** It directly bounds the volume of long-document grounding context that can be injected in RAG workflows
- **D.** It fixes the physical silicon substrate surface area and transistor die packaging thickness

Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

In an enterprise AI lifecycle and MLOps architecture, which components manage system monitoring and production reliability?

- **A.** Real-time logging of inference latency, token throughput, and cost consumption
- **B.** Periodic physical power-cycling of isolated database compute nodes
- **C.** Data distribution drift detection and model output degradation monitoring
- **D.** Automated fallback routing and circuit-breaker mechanisms for high-availability serving

Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

In modern Mixture of Experts (MoE) architectures (such as DeepSeekMoE), which benefits are achieved by fine-grained expert segmentation?

- **A.** Increased specialized expert diversity and higher representation capacity per parameter budget
- **B.** Finer granularity in routing decisions, enabling more precise factual and linguistic specialization
- **C.** Complete elimination of multi-head attention mechanisms from the transformer backbone
- **D.** Restricting inference execution strictly to single-threaded CPU instruction sets

Page Answers: **Q61**: AC • **Q62**: ABD • **Q63**: ABC • **Q64**: ACD • **Q65**: AB

Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about large foundation models are true?

- **A.** They possess significantly larger parameter counts than traditional deep learning models
- **B.** They eliminate the need for token embeddings and positional encodings
- **C.** They exhibit emergent capabilities at massive scale
- **D.** They demonstrate strong cross-task generalization through pre-training

Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following domains can foundation model technologies be applied to?

- **A.** Multimodal creative content creation
- **B.** Intelligent customer service and conversational assistants
- **C.** Automated software programming and code generation
- **D.** Scientific literature analysis and research hypothesis generation

Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements accurately characterize Parameter-Efficient Fine-Tuning (PEFT) methods covered in HCIA-AI V4.0?

- **A.** Adapter Tuning inserts small trainable modules into pre-trained model layers while freezing the main backbone
- **B.** Prefix Tuning and Prompt Tuning avoid altering internal Transformer block structure by adding trainable virtual/prompt tokens
- **C.** In LoRA, matrix A starts with random Gaussian values and matrix B starts as a zero matrix before merging BA with frozen weights
- **D.** Full-parameter fine-tuning updates strictly fewer weights than Prompt Tuning while freezing backbone layers

Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following Parameter-Efficient Fine-Tuning (PEFT) methods freeze the base model weights and introduce trainable low-rank or prompt matrices?

- **A.** LoRA
- **B.** Full-Parameter End-to-End Retraining
- **C.** Prompt Tuning
- **D.** Random Gaussian Weight Re-initialization

Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

The development and enterprise operation of large models requires which of the following lifecycle stages?

- **A.** Pre-training, instruction fine-tuning, and alignment
- **B.** Model serving, monitoring, and prompt engineering
- **C.** Data collection, cleaning, and filtering
- **D.** Safety testing, model evaluation, and monitoring

Page Answers: **Q66**: ACD • **Q67**: ABCD • **Q68**: ABC • **Q69**: AC • **Q70**: ABCD

Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following distributed parallelism strategies partition model layers across pipeline stages and tensor operations across multiple accelerators respectively?

- **A.** Unconstrained Greedy Decoding
- **B.** Data Anonymization and Token Masking
- **C.** Pipeline Parallelism
- **D.** Tensor Parallelism

Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

In the standard RLHF (Reinforcement Learning from Human Feedback) process, which training stages follow Supervised Fine-Tuning (SFT)?

- **A.** Raw Web Data Scraping and Filtering
- **B.** Reward Model Training on human preference comparisons
- **C.** Static C++ Kernel Compilation
- **D.** Policy Optimization using PPO

Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

A standard enterprise RAG (Retrieval-Augmented Generation) pipeline mainly consists of which core functional modules?

- **A.** Pre-training module
- **B.** Retrieval module
- **C.** Generation module
- **D.** Fine-tuning module

Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are advantages of Retrieval-Augmented Generation (RAG) over full model fine-tuning?

- **A.** Enables verifiable source attribution and citations in answers
- **B.** Greatly lowers compute costs compared to continually retraining large models
- **C.** Allows updating knowledge in real time by simply adding documents to the vector database
- **D.** Eliminates the need for a generative language model completely

Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are key dimensions evaluated when assessing large language models?

- **A.** Helpfulness
- **B.** Honesty
- **C.** Efficiency
- **D.** Harmlessness

Page Answers: **Q71**: CD • **Q72**: BD • **Q73**: BC • **Q74**: ABC • **Q75**: ABD

Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common Prompt Engineering techniques?

- **A.** Role-Based Prompting
- **B.** Few-Shot In-Context Learning
- **C.** Zero-Shot Prompting
- **D.** Chain-of-Thought Prompting

Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following represent common forms of Prompt Injection security threats in LLM applications?

- **A.** Direct Prompt Injection
- **B.** Indirect Prompt Injection
- **C.** Database Schema Migration Lockout
- **D.** Ethernet Network Cable Disconnection

Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Zero-Shot-CoT are true?

- **A.** It has high generalization across diverse reasoning tasks
- **B.** It requires no manually authored input-output reasoning exemplars
- **C.** It triggers step-by-step reasoning using prompt phrases like 'Let's think step by step'
- **D.** It completely eliminates all reasoning capability

Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following tasks are performed during the Data Preparation phase of a large language model project?

- **A.** Text extraction from diverse file formats
- **B.** Quality filtering and removal of toxic or low-quality text
- **C.** Tokenization using algorithms like BPE
- **D.** Deduplication of identical and near-duplicate documents

Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common challenges encountered in Large Model Evaluation?

- **A.** High cost and subjectivity of human expert evaluation
- **B.** Inability to count the number of test questions
- **C.** Benchmark saturation on popular public test sets
- **D.** Test set contamination in pre-training datasets

Page Answers: **Q76**: ABCD • **Q77**: AB • **Q78**: ABC • **Q79**: ABCD • **Q80**: ACD

Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are defining mathematical properties of LoRA (Low-Rank Adaptation)?

- **A.** Multiplies total trainable parameters tenfold by adding layers
- **B.** Freezes the original pre-trained weight matrix W_0
- **C.** Decomposes weight updates delta W into the product of two low-rank matrices B * A
- **D.** Requires training all backbone parameters from epoch zero

Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are key features of the ReAct (Reason + Act) prompting framework?

- **A.** Observing tool outputs to update the reasoning trajectory
- **B.** Disabling all model attention layers
- **C.** Executing external tools and APIs to retrieve real-time data
- **D.** Generating reasoning thoughts before selecting actions

Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common online deployment strategies for safely rolling out new AI model versions to production users?

- **A.** Direct overwriting
- **B.** Manual cutover
- **C.** Canary deployment
- **D.** Blue-green deployment

Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common evaluation benchmarks for large language models?

- **A.** HumanEval
- **B.** Massive Multitask Language Understanding
- **C.** Cinebench
- **D.** Grade School Math 8K

Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following techniques can reduce prompt injection attacks?

- **A.** Directly pasting raw user input as system instructions
- **B.** Output content filtering and safety checks
- **C.** Using separate system instruction channels that cannot be overridden by user inputs
- **D.** Input validation and prompt delimiters

Page Answers: **Q81**: BC • **Q82**: ACD • **Q83**: CD • **Q84**: ABD • **Q85**: BCD

Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following automated n-gram overlap metrics are commonly used to evaluate machine translation and text summarization quality?

- **A.** Bilingual Evaluation Understudy
- **B.** Recall-Oriented Understudy for Gisting Evaluation
- **C.** Silhouette Cluster Separation Index
- **D.** K-Means Elbow Inertia

Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Vector Databases in RAG systems are true?

- **A.** They enable fast semantic retrieval over millions of enterprise documents
- **B.** They support Approximate Nearest Neighbor search algorithms
- **C.** They index high-dimensional embeddings generated by neural networks
- **D.** They can only store 10-word plain text files

Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following factors should be considered during model selection in an enterprise AI project?

- **A.** Target business accuracy and latency requirements
- **B.** Open-source licensing and commercial compliance
- **C.** Favorite color of the software engineer
- **D.** Available computational budget for training and inference

Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary operational advantages of Adapter Tuning?

- **A.** Keeps the base model backbone frozen, preventing catastrophic forgetting
- **B.** Multiplies total model parameters by ten to maximize capacity
- **C.** Guarantees zero floating-point computation latency
- **D.** Significantly reduces storage and training compute requirements

Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common causes of Concept Drift in production machine learning systems?

- **A.** Changes in consumer shopping behavior over holiday seasons
- **B.** Evolution of user vocabulary and slang on social media
- **C.** Macroeconomic fluctuations and policy changes
- **D.** The screen resolution of client laptops

Page Answers: **Q86**: AB • **Q87**: ABC • **Q88**: ABD • **Q89**: AD • **Q90**: ABC

Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following represent core sequential training stages in the development of modern Large Language Models?

- **A.** Pre-training
- **B.** Supervised Fine-Tuning
- **C.** Data deduplication
- **D.** RLHF alignment

Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following methodologies are established techniques for aligning Large Language Models with human values and helpfulness?

- **A.** Reinforcement Learning from Human Feedback
- **B.** Direct Preference Optimization
- **C.** Reinforcement Learning from AI Feedback
- **D.** Rejection Sampling and Best-of-N Policy Optimization

Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

The parameter-efficient fine-tuning technique that decomposes weight updates into two low-rank matrices is called ____.

Answer: _____

Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

The enterprise technique that retrieves relevant factual documents from an external knowledge base to ground model generation is called ____ (Retrieval-Augmented Generation).

Answer: _____

Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

In distributed model training, when a single layer's matrix operations are split across multiple GPUs, this is called ____ Parallelism.

Answer: _____

Page Answers: **Q91**: ABD • **Q92**: ABCD • **Q93**: LoRA (or Low-Rank Adaptation) • **Q94**: RAG • **Q95**: Tensor

Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

In distributed model training, when sequential layers of a neural network are partitioned across sequential devices like an assembly line, this is called ____ Parallelism.

Answer: _____

Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

Implemented in C/C++, the standalone deployment tool ____ runs Large Language Model inference locally or in the cloud without requiring an AI development framework.

Answer: _____

Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

The security vulnerability where an attacker uses crafted prompts to bypass guardrails and violate system rules is called ____ (or Prompt Injection).

Answer: _____

Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

In prompt engineering, the strategy of providing a few solved demonstration examples inside the prompt is known as ____-Shot Prompting.

Answer: _____

Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

In the traditional AI business process, after model training and offline validation, models are often evaluated on live traffic using ____ testing.

Answer: _____

Page Answers: **Q96**: Pipeline • **Q97**: llama.cpp • **Q98**: Jailbreaking (or Jailbreak) • **Q99**: Few • **Q100**: A/B

HCIA-AI V4.0 Certification Practice Exam • Chapter 6

Module 6: Cutting-Edge AI Applications (100 Questions & Answer Key)

Curated by: Umed Jamal Nouri • v1.1 (Nov 2026) • github.com/UmedJaMala

Question 1

TRUE OR FALSE • 16 Points

Huawei's smart voice assistant Celia (Xiaoyi) integrates Automatic Speech Recognition (ASR), Natural Language Understanding (NLU), and Text-to-Speech (TTS) technologies.

A. True
B. False

Question 2

TRUE OR FALSE • 16 Points

In autonomous driving systems, LiDAR sensors rely completely on capturing passive ambient visible sunlight and do not emit any laser light.

A. True
B. False

Question 3

TRUE OR FALSE • 16 Points

Collaborative Filtering in recommendation systems relies only on the textual metadata keywords of items and completely ignores historical user behavior logs.

A. True
B. False

Question 4

TRUE OR FALSE • 16 Points

Huawei's Qiankun Intelligent Driving solution provides full-stack autonomous driving capabilities covering urban scenarios, highways, and automated valet parking.

A. True
B. False

Question 5

TRUE OR FALSE • 16 Points

The Cold Start problem in recommendation systems refers to server processors running at low temperatures during winter.

A. True
B. False

Page Answers: **Q1**: True • **Q2**: False • **Q3**: False • **Q4**: True • **Q5**: False

Question 6

TRUE OR FALSE • 16 Points

In intelligent vehicles, HarmonyOS Cockpit enables seamless cross-device interconnection between smartphones, smart watches, and vehicle infotainment displays.

A. True
B. False

Question 7

TRUE OR FALSE • 16 Points

Driver Monitoring Systems (DMS) are designed to monitor tire pressure and engine oil levels, having nothing to do with driver fatigue or attention.

A. True
B. False

Question 8

TRUE OR FALSE • 16 Points

Embodied Artificial Intelligence focuses on developing intelligent agents that perceive physical environments through multimodal sensors and physically interact using robotic actuators.

A. True
B. False

Question 9

TRUE OR FALSE • 16 Points

In AI4Science, Huawei's Pangu-Weather model relies only on manual handwritten differential equations without using any deep neural networks.

A. True
B. False

Question 10

TRUE OR FALSE • 16 Points

In smart homes, AI-enabled cameras with acoustic detection can recognize abnormal events such as glass breaking or infant crying to trigger security alarms.

A. True
B. False

Page Answers: **Q6**: True • **Q7**: False • **Q8**: True • **Q9**: False • **Q10**: True

Question 11

TRUE OR FALSE • 16 Points

Simultaneous Localization and Mapping (SLAM) is a technology used only for chemical synthesis and cannot be applied to robot navigation.

A. True
B. False

Question 12

TRUE OR FALSE • 16 Points

Item-based collaborative filtering recommends items to a user based purely on the user's demographic age and gender without analyzing item-item interaction history.

A. True
B. False

Question 13

TRUE OR FALSE • 16 Points

End-to-End (E2E) autonomous driving directly maps raw sensor inputs (e.g., multi-camera video streams) into vehicle trajectory and control outputs using a unified deep neural network.

A. True
B. False

Question 14

TRUE OR FALSE • 16 Points

In AI + Molecular Dynamics, neural network potentials enable atomic simulations over long biological timescales with quantum-mechanical level accuracy.

A. True
B. False

Question 15

TRUE OR FALSE • 16 Points

Smart home climate control systems only support hard-coded manual thermostat switches and cannot dynamically adapt to occupant presence or outdoor weather patterns.

A. True
B. False

Page Answers: **Q11**: False • **Q12**: False • **Q13**: True • **Q14**: True • **Q15**: False

Question 16

TRUE OR FALSE • 16 Points

Keyword Spotting (KWS) in voice assistants requires continuous cloud-based training of 100-billion parameter models for every millisecond of background audio.

A. True
B. False

Question 17

TRUE OR FALSE • 16 Points

Sensor fusion in intelligent driving combines data from multiple complementary sensors (cameras, millimeter-wave radars, LiDAR) to overcome the environmental limitations of any individual sensor.

A. True
B. False

Question 18

TRUE OR FALSE • 16 Points

User-based collaborative filtering assumes that users with similar historical ratings will have completely opposite preferences for future items.

A. True
B. False

Question 19

TRUE OR FALSE • 16 Points

AI-assisted materials science can screen millions of virtual crystal compositions to identify candidate compounds for novel solid-state batteries.

A. True
B. False

Question 20

TRUE OR FALSE • 16 Points

Autonomous mobile robots (AMR) in smart warehouses rely only on physical railway tracks and cannot perform dynamic obstacle avoidance.

A. True
B. False

Page Answers: **Q16**: False • **Q17**: True • **Q18**: False • **Q19**: True • **Q20**: False

Question 21

TRUE OR FALSE • 16 Points

In natural language dialogue management, Dialogue State Tracking (DST) maintains the current state and contextual slot values across multiple turns of user conversation.

A. True
B. False

Question 22

TRUE OR FALSE • 16 Points

Millimeter-wave (mmWave) radars are severely degraded by light rain or fog and cannot detect vehicle velocities.

A. True
B. False

Question 23

TRUE OR FALSE • 16 Points

In AI + Quantum Chemistry, machine learning models can predict molecular electronic properties and energy states significantly faster than traditional Density Functional Theory (DFT).

A. True
B. False

Question 24

TRUE OR FALSE • 16 Points

Content-based recommendation algorithms recommend items only based on popularity rankings across the general population without considering item attributes.

A. True
B. False

Question 25

TRUE OR FALSE • 16 Points

Modern intelligent robots use Large Multimodal Models (LMMs) to ground abstract high-level linguistic commands into sequence-level physical robotic action primitives.

A. True
B. False

Page Answers: **Q21**: True • **Q22**: False • **Q23**: True • **Q24**: False • **Q25**: True

Question 26

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT an advanced AI application domain?

- **A.** Autonomous vehicular path trajectory planning
- **B.** Medical radiological diagnostic segmentation
- **C.** Traditional relational database B-tree indexing
- **D.** Intelligent power grid dispatch optimization

Question 27

SINGLE-ANSWER QUESTION • 16 Points

In Huawei's smart voice assistant architecture, which component is responsible for understanding the user's underlying intent and extracting key entity slots from transcribed text?

- **A.** Automatic Speech Recognition System
- **B.** Natural Language Understanding Engine
- **C.** Text-to-Speech Acoustic Synthesizer
- **D.** Dialog State Management Controller

Question 28

SINGLE-ANSWER QUESTION • 16 Points

Which sensor in an autonomous driving vehicle is most effective at directly capturing high-resolution color, texture, and text details of traffic signs and signal lights?

- **A.** Millimeter-wave Radar
- **B.** Optical HD Camera
- **C.** Ultrasonic Sensor
- **D.** Wheel Speed Encoder

Question 29

SINGLE-ANSWER QUESTION • 16 Points

Which sensor emits radio waves and uses the Doppler effect to measure the precise relative radial velocity of moving objects, even in adverse weather like fog or heavy rain?

- **A.** Monocular RGB Optical Camera Sensor
- **B.** Ultrasonic Close-Range Proximity Sensor
- **C.** Millimeter-wave Radar Unit
- **D.** Micro-Electro-Mechanical Gyroscope

Question 30

SINGLE-ANSWER QUESTION • 16 Points

Which recommendation approach recommends products to User A by finding other users whose past rating profiles are mathematically closest to User A?

- **A.** Random Permutation Generator
- **B.** User-Based Collaborative Filtering
- **C.** Item-Based Collaborative Filtering
- **D.** Content-Based Metadata Matching

Page Answers: **Q26**: C • **Q27**: B • **Q28**: B • **Q29**: C • **Q30**: B

Question 31

SINGLE-ANSWER QUESTION • 16 Points

When an e-commerce platform launches a brand-new product with zero customer reviews, clicks, or purchase records, which challenge does it face?

- **A.** Catastrophic Forgetting
- **B.** Pipeline Parallelism Stall
- **C.** Vanishing Gradient Problem
- **D.** Item Cold Start Problem

Question 32

SINGLE-ANSWER QUESTION • 16 Points

Huawei's flagship commercial smart voice assistant available across its HarmonyOS device ecosystem is known as:

- **A.** Celia
- **B.** Siri
- **C.** Alexa
- **D.** Bixby

Question 33

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of the Planning & Decision-Making system in autonomous driving?

- **A.** Generating collision-free trajectories and behavioral actions based on fused perception data
- **B.** Extracting 3D bounding boxes and semantic lane markings from raw camera and LiDAR sensors
- **C.** Estimating centimeter-level vehicle pose relative to high-definition prior spatial maps
- **D.** Converting target trajectory commands into low-level steering angle and braking pressure signals

Question 34

SINGLE-ANSWER QUESTION • 16 Points

What is the core function of SLAM (Simultaneous Localization and Mapping) in autonomous robotics?

- **A.** Executing global route navigation across pre-computed static highway road networks
- **B.** Estimating robot spatial pose while building a consistent map of an unknown surrounding environment
- **C.** Regulating closed-loop PID servo motor torque across robotic manipulator joints
- **D.** Classifying static road traffic signs using pre-trained 2D convolutional networks

Question 35

SINGLE-ANSWER QUESTION • 16 Points

In smart home ecosystems, what is the primary benefit of Edge AI processing on smart hubs?

- **A.** Ensuring low-latency inference and local user privacy protection without constant cloud offloading
- **B.** Offloading all raw audiovisual streams to remote public cloud data centers
- **C.** Requiring continuous gigabit internet bandwidth for routine offline tasks
- **D.** Replacing local neural inference chips with remote cloud virtual machines

Page Answers: **Q31**: D • **Q32**: A • **Q33**: A • **Q34**: B • **Q35**: A

Question 36

SINGLE-ANSWER QUESTION • 16 Points

Which Huawei foundation model has demonstrated groundbreaking achievements in high-speed, accurate global medium-range numerical weather prediction?

- **A.** Pangu-Weather
- **B.** Pangu-Mining
- **C.** Pangu-Code
- **D.** Pangu-Finance

Question 37

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of Text-to-Speech (TTS) in an intelligent voice assistant?

- **A.** Segmenting audio files into 1-second chunks
- **B.** Encrypting Wi-Fi passwords
- **C.** Converting structured response text into natural, fluent human-like spoken audio
- **D.** Listening to spoken microphone input and transcribing it into text

Question 38

SINGLE-ANSWER QUESTION • 16 Points

Which technology enables modern voice assistants to listen and recognize user voice commands across noisy living rooms?

- **A.** Far-field microphone arrays combined with acoustic beamforming, dereverberation, and noise suppression
- **B.** Optical infrared cameras detecting micro-vibrations across human vocal cords through window glass
- **C.** Rule-based symbolic syntax parsing using context-free grammar trees
- **D.** Mechanical diaphragm springs that physically amplify sound waves using resonance chamber cones

Question 39

SINGLE-ANSWER QUESTION • 16 Points

What level of autonomous driving (according to SAE J3016) represents 'High Driving Automation' where the vehicle handles all driving tasks under specific operational design domains (ODD) without expecting driver intervention?

- **A.** Level 4
- **B.** Level 1
- **C.** Level 0
- **D.** Level 2

Question 40

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes the 'Smart Cockpit' in intelligent vehicles?

- **A.** The physical aerodynamic rear spoiler adjusting vehicle downforce at high highway cruising speeds
- **B.** Heavy-duty cast-iron engine mounting brackets absorbing rotational powertrain vibrations
- **C.** In-cabin multimodal infotainment displays, HUD, voice control, and driver monitoring systems
- **D.** Hydraulic brake caliper fluid pressure reservoirs regulating emergency stopping distances

Page Answers: **Q36**: A • **Q37**: C • **Q38**: A • **Q39**: A • **Q40**: C

Question 41

SINGLE-ANSWER QUESTION • 16 Points

How does AI accelerate drug discovery in pharmaceutical science?

- **A.** Automating plastic mold injection machines manufacturing pill bottles in fulfillment warehouses
- **B.** Replacing living human clinical trial volunteers with basic mathematical spreadsheet formulas
- **C.** Scheduling cargo airplane delivery flight paths for commercial wholesale pharmaceutical shipments
- **D.** Predicting protein 3D structures, molecular docking affinity, and optimizing candidate chemical molecules

Question 42

SINGLE-ANSWER QUESTION • 16 Points

Which technology allows an autonomous vehicle to maintain accurate sub-meter positioning in urban canyons?

- **A.** Dead-reckoning wheel odometry without external spatial map correction
- **B.** Single-frequency civil GPS positioning without inertial sensor fusion
- **C.** Monocular optical flow estimation without metric depth calibration
- **D.** High-Definition Map matching fused with multi-sensor GNSS-RTK and LiDAR visual odometry

Question 43

SINGLE-ANSWER QUESTION • 16 Points

What does 'Embodied AI' mainly require that distinguishes it from purely virtual AI software?

- **A.** Training only on unlabeled text datasets
- **B.** Replacing self-attention layers with static 1D max-pooling operations
- **C.** Registered corporate developer accounts on public social media web platforms and message boards
- **D.** A physical agent interacting with the real physical world through sensors and physical actuators

Question 44

SINGLE-ANSWER QUESTION • 16 Points

In a smart home, what type of AI sensor can detect human occupancy and breathing without optical cameras?

- **A.** Millimeter-wave Radar Micro-Doppler Motion Sensors
- **B.** Atmospheric barometric altitude pressure measurement transducers
- **C.** Passive photoresistor cadmium-sulfide ambient illuminance sensors
- **D.** Passive monocular RGB cameras without active illumination

Question 45

SINGLE-ANSWER QUESTION • 16 Points

Which deep learning recommendation model introduced by Google combines a linear generalized model (for memorization) with a deep feedforward network (for generalization)?

- **A.** Pure K-Means
- **B.** LeNet-5
- **C.** Wide & Deep Model
- **D.** Single Decision Tree

Page Answers: **Q41**: D • **Q42**: D • **Q43**: D • **Q44**: A • **Q45**: C

Question 46

SINGLE-ANSWER QUESTION • 16 Points

In voice assistants, what is the role of Keyword Spotting (KWS)?

- **A.** Continuous low-power acoustic monitoring to detect pre-configured wake words
- **B.** Translating multi-hour foreign language cinema dialogue scripts into localized subtitle files
- **C.** Synthesizing full audiobooks from raw plain-text electronic manuscripts using neural vocoders
- **D.** Measuring indoor air quality and room humidity levels using solid-state chemical sensors

Question 47

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of Ultrasonic Sensors in modern production cars?

- **A.** Short-range obstacle distance measurement during low-speed automated parking maneuvers
- **B.** Offline batch sentiment classification of customer review datasets
- **C.** Recognizing highway speed limit traffic signs and overhead optical electronic toll gantries
- **D.** Tracking high-speed oncoming vehicles at three hundred meters in dense highway fog

Question 48

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is a classic technique used to solve the User Cold-Start problem in recommendation systems?

- **A.** Blanking the consumer screen completely until the user performs ten physical purchase transactions
- **B.** Requiring new account registrants to solve five complex multi-variable calculus equations
- **C.** Presenting an interactive onboarding interest selection questionnaire to capture initial user preferences
- **D.** Automatically terminating inactive account sessions after thirty seconds of scrolling dormancy

Question 49

SINGLE-ANSWER QUESTION • 16 Points

In robotics, what is the computational problem of calculating joint motor angles required to position an end-effector gripper at a target coordinate?

- **A.** Gradient Descent
- **B.** Inverse Kinematics
- **C.** Data Augmentation
- **D.** Batch Normalization

Question 50

SINGLE-ANSWER QUESTION • 16 Points

How does AI + Quantum Mechanics assist in materials science?

- **A.** Exact analytical closed-form solution of multi-electron Schrödinger equations without approximation
- **B.** Substituting physical experimental verification with static database lookup tables
- **C.** Applying macroscopic Newtonian rigid-body mechanics to quantum wavepacket states
- **D.** Predicting electronic ground-state energies and molecular wavefunctions via neural approximations

Page Answers: **Q46**: A • **Q47**: A • **Q48**: C • **Q49**: B • **Q50**: D

Question 51

SINGLE-ANSWER QUESTION • 16 Points

Which component in dialogue systems decides what action the system should take next?

- **A.** Dialogue Policy
- **B.** Speech Recognition
- **C.** Intent Classifier
- **D.** Speech Synthesis

Question 52

SINGLE-ANSWER QUESTION • 16 Points

What is 'V2X' (Vehicle-to-Everything) communication in intelligent transportation?

- **A.** Eliminating electronic brake sensor communication buses
- **B.** Wireless data exchange between vehicles, roadside units, pedestrians, and network infrastructure
- **C.** Executing offline k-means clustering on static vehicle registration tables
- **D.** Retrofitting internal combustion engines with commercial compressed liquid natural gas cylinders

Question 53

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes 'Matrix Factorization' in collaborative filtering recommendation systems?

- **A.** Inverting matrices to solve linear regression equations
- **B.** Joining tables across relational databases using primary keys
- **C.** Applying convolutional filters to rotate image feature maps
- **D.** Decomposing the user-item interaction matrix into lower-dimensional user and item feature vectors

Question 54

SINGLE-ANSWER QUESTION • 16 Points

Which Huawei intelligent vehicle solution provides high-precision navigation and autonomous driving capabilities?

- **A.** HiSilicon Multimedia Video Decoder Service
- **B.** MindSpore Model Zoo Hub Repository
- **C.** Huawei Qiankun ADS
- **D.** ModelArts Lite Edge Container Daemon

Question 55

SINGLE-ANSWER QUESTION • 16 Points

In intelligent robotics, what is the role of 'Tactile Sensors' integrated into robotic grippers?

- **A.** Measuring long-range obstacle distances using time-of-flight laser pulses
- **B.** Estimating rotational angular velocity and linear acceleration of the robot chassis
- **C.** Capturing wide-angle stereoscopic RGB depth maps of the surrounding workspace
- **D.** Measuring contact forces, shear slip, and pressure distributions during delicate object manipulation

Page Answers: **Q51**: A • **Q52**: B • **Q53**: D • **Q54**: C • **Q55**: D

Question 56

SINGLE-ANSWER QUESTION • 16 Points

What is the primary objective of Occupancy Grid Mapping in autonomous navigation?

- **A.** Organizing unstructured text log records chronologically inside cloud object storage buckets
- **B.** Offline batch backpropagation across historical server log archives
- **C.** Assigning numbered reserved seating tickets across municipal public transportation buses
- **D.** Discretizing surrounding spatial terrain into grid cells representing the probability of obstacles

Question 57

SINGLE-ANSWER QUESTION • 16 Points

In smart buildings, what does AI-driven HVAC optimization achieve?

- **A.** Replacing physical metal ventilation ductwork with wireless digital microwave communication links
- **B.** Dynamically adjusting thermal heating and cooling based on real-time occupancy and weather forecasts
- **C.** Manually rotating mechanical window blind louvers based on printed solar equinox almanac charts
- **D.** Overclocking building electricity consumption by running redundant industrial refrigeration units

Question 58

SINGLE-ANSWER QUESTION • 16 Points

What enables modern humanoid robots to understand complex unstructured natural language commands and act?

- **A.** Hard-coded sequential conditional if-else statements executed by programmable logic controllers
- **B.** Vision-Language-Action foundation models translating multimodal perception into robotic actions
- **C.** Purely reactive open-loop motor commands without visual or semantic scene grounding
- **D.** Classical PID single-variable feedback controllers operating without language interpretation

Question 59

SINGLE-ANSWER QUESTION • 16 Points

In recommendation systems, what does 'Popularity Bias' refer to?

- **A.** Ranking all catalog items strictly by reverse chronological upload timestamp
- **B.** Displaying discount promotional items only to verified enterprise corporate account holders
- **C.** Suppressing item recommendations completely for users who exhibit high daily browsing activity
- **D.** The algorithmic tendency to over-recommend viral head items while ignoring the diverse long tail

Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Huawei Pangu-Weather AI model are correct? (Multiple Choice)

- **A.** It employs a 3D Earth-Specific Transformer architecture to process 3D atmospheric variables
- **B.** It generates global weather forecasts at hour-level precision in just seconds on Ascend NPUs
- **C.** Its forecasting precision surpasses traditional Numerical Weather Prediction systems across key metrics
- **D.** It relies completely on 2D convolutional edge-detection kernels without temporal modeling

Page Answers: **Q56**: D • **Q57**: B • **Q58**: B • **Q59**: D • **Q60**: ABC

Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common application scenarios and tasks of Natural Language Generation (NLG)?

- **A.** Automated multi-document summarization and executive report drafting
- **B.** Syntactic part-of-speech tagging and dependency tree parsing
- **C.** Conversational dialogue agent response generation in customer support chatbots
- **D.** Lexical named entity boundary extraction and classification

Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

In Autonomous Driving perception systems, which spatial parameters are estimated for detected 3D bounding boxes?

- **A.** 3D spatial center coordinates relative to vehicle sensor coordinate origin
- **B.** Physical object dimensional extents
- **C.** Camera intrinsic focal length and optical lens radial distortion calibration matrix
- **D.** Object orientation yaw angle and estimated 3D velocity vector

Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following domain applications naturally benefit from Graph Neural Networks (GNNs) due to their non-Euclidean relational structure?

- **A.** Uniform 2D pixel grid convolution on standard grayscale bitmap images
- **B.** Molecular chemical property prediction and drug-target interaction modeling
- **C.** Financial fraud ring detection and anti-money laundering in transaction graphs
- **D.** Social network community clustering, user influence modeling, and citation analysis

Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

In intelligent vehicles, which vehicular sub-systems are commonly actuated through Drive-by-Wire (DbW) electronic control signals?

- **A.** Steer-by-Wire
- **B.** Passive suspension
- **C.** Brake-by-Wire
- **D.** Throttle-by-Wire

Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following represent recognized, high-impact clinical and medical applications of Artificial Intelligence in Healthcare?

- **A.** Radiological lesion detection, organ segmentation, and anomaly screening in CT and MRI scans
- **B.** Genomic variant pathogenicity prediction, mutation risk scoring, and personalized oncology analysis
- **C.** Automated digital pathology slide whole-image tumor tissue grading and classification
- **D.** AI-assisted molecular screening for de novo drug discovery and target-protein binding affinity prediction

Page Answers: **Q61**: AC • **Q62**: ABD • **Q63**: BCD • **Q64**: ACD • **Q65**: ABCD

Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following core functional modules constitute a traditional modular voice assistant system (such as Huawei Celia)?

- **A.** Automatic Speech Recognition
- **B.** Text-to-Speech
- **C.** Dialogue Management and Natural Language Understanding
- **D.** Unsupervised optical flow motion vector estimation

Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary sensors used in autonomous driving perception systems?

- **A.** Light Detection and Ranging
- **B.** Millimeter-Wave Radar
- **C.** Ultrasonic Sensors
- **D.** Optical HD Cameras

Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are typical AI-powered capabilities deployed in Smart Home environments?

- **A.** Intelligent voice-activated appliance control and conversational queries
- **B.** Smart climate control and energy optimization based on occupancy patterns
- **C.** Central bank interbank financial settlement transaction clearing
- **D.** Acoustic anomaly detection

Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following represent classic main types of recommendation algorithms?

- **A.** Random Permutation Shuffling
- **B.** Collaborative Filtering
- **C.** Content-Based Filtering
- **D.** Reverse Lexical Alphabetical Sorting

Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Sensor Fusion in autonomous driving are correct?

- **A.** Millimeter-wave radar directly measures target velocity and operates reliably in fog, rain, and snow
- **B.** Combining multiple sensors overcomes the physical limitations of individual sensors
- **C.** LiDAR provides direct, highly accurate 3D spatial coordinate measurements regardless of ambient lighting
- **D.** Cameras provide rich visual color, semantic text, and high angular resolution

Page Answers: **Q66**: ABC • **Q67**: ABCD • **Q68**: ABD • **Q69**: BC • **Q70**: ABCD

Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following scientific fields are actively transformed by AI for Science (AI4Science)?

- **A.** Quantum mechanical electronic energy and molecular wavefunction approximations
- **B.** Novel materials discovery and crystalline structure simulation
- **C.** Global medium-range numerical weather and atmospheric climate prediction
- **D.** High-frequency relational database transaction concurrency locking

Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are capabilities supported by Huawei's HarmonyOS Smart Cockpit?

- **A.** Driver fatigue and distraction monitoring
- **B.** Direct physical wireless override of municipal traffic intersection signal controllers
- **C.** Multi-screen sharing and phone-to-car seamless navigation continuation
- **D.** Intelligent voice dialogue and in-cabin multi-zone sound pickup

Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following strategies can effectively solve the Cold Start problem in recommendation systems?

- **A.** Completely barring newly registered users from accessing platform content
- **B.** Using demographic onboarding preference questionnaires during user registration
- **C.** Using item metadata, attributes, and tags for content-based recommendations
- **D.** Recommending globally popular, highly-rated, or trending items to new users

Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following tasks belong to the Perception layer rather than the Planning or Control layers in an autonomous vehicle?

- **A.** 3D Object Detection
- **B.** Lane Boundary Semantic Segmentation
- **C.** Collision-Free Trajectory Optimization
- **D.** Steering Actuator Torque Regulation

Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Intelligent Robots and Embodied AI are correct?

- **A.** They rely on multimodal foundation models to interpret real-world commands
- **B.** They integrate perception, high-level reasoning, and physical manipulation
- **C.** They process static tabular datasets without real-world sensor-actuator interaction
- **D.** They require spatial reasoning and dynamics control to interact with physical objects

Page Answers: **Q71**: ABC • **Q72**: ACD • **Q73**: BCD • **Q74**: AB • **Q75**: ABD

Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized technical advantages of Huawei's Pangu-Weather model over traditional Numerical Weather Prediction (NWP)?

- **A.** Eliminates all physical radar and satellite observation sensors
- **B.** 10,000x faster inference speed on Ascend AI hardware
- **C.** Superior forecast accuracy on extreme weather events like tropical cyclone tracks
- **D.** Operates with zero historical meteorological observation data

Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

In robotics, SLAM mainly solves which two tasks at the same time in an unknown environment?

- **A.** Localization
- **B.** Path planning
- **C.** Mapping
- **D.** Motion control

Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common interaction modalities supported in modern intelligent cockpits?

- **A.** Touchscreen gesture control
- **B.** Vision-based contactless air gestures
- **C.** Natural voice interaction
- **D.** Driver gaze and eye-tracking interaction

Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are benefits of deploying AI in Industrial Manufacturing and Smart Logistics?

- **A.** Autonomous mobile robots for automated material handling and palletizing
- **B.** Predictive equipment maintenance based on sensor vibration monitoring
- **C.** Guaranteeing zero inventory warehouse management
- **D.** Automated optical inspection for high-speed product defect detection

Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are metrics commonly used to evaluate recommendation systems?

- **A.** Normalized Discounted Cumulative Gain
- **B.** Click-Through Rate and Conversion Rate
- **C.** Mean Absolute Error in rating prediction
- **D.** Precision@K and Recall@K

Page Answers: **Q76**: BC • **Q77**: AC • **Q78**: ABCD • **Q79**: ABD • **Q80**: ABCD

Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are key challenges in Autonomous Driving?

- **A.** Handling edge cases and unexpected road hazards
- **B.** Making vehicles invisible to human vision
- **C.** Operating reliably in extreme weather conditions
- **D.** Ensuring high safety redundancy across hardware, sensors, and computing compute units

Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

Collaborative Filtering algorithms are mainly divided into which classic memory-based approaches?

- **A.** Quantum Tunneling Filtering
- **B.** User-Based Collaborative Filtering
- **C.** Item-Based Collaborative Filtering
- **D.** Solid-State Cache Sector Filtering

Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are safety features enabled by AI in Smart Home security?

- **A.** Visual perimeter intrusion detection with human-vs-pet classification
- **B.** Facial recognition on smart doorbells to identify family members vs strangers
- **C.** Acoustic detection of glass breakage or smoke detector alarms
- **D.** Elderly fall detection and automated emergency alert dispatch

Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are typical autonomous driving scenarios supported by Huawei Qiankun ADS?

- **A.** Urban Navigation Cruise Assist
- **B.** Automated Valet Parking and Remote Parking
- **C.** Offline Tabular Data Warehouse ETL Indexing
- **D.** Highway Navigation Cruise Assist

Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

In voice assistant systems, Natural Language Understanding (NLU) primarily performs which semantic extraction tasks?

- **A.** Intent classification
- **B.** Slot filling
- **C.** Speech synthesis
- **D.** Echo cancellation

Page Answers: **Q81**: ACD • **Q82**: BC • **Q83**: ABCD • **Q84**: ABD • **Q85**: AB

Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements accurately characterize End-to-End Autonomous Driving compared to traditional modular pipelines?

- **A.** Directly maps raw sensor inputs to trajectory planning/control via unified neural networks
- **B.** Reduces compounding error accumulation across hand-engineered intermediate interfaces
- **C.** Eliminates the requirement for any sensor training data
- **D.** Uses large-scale multi-task driving demonstrations to learn complex driving behaviors

Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following represent high-impact application scenarios of Artificial Intelligence in environmental protection and biodiversity conservation?

- **A.** Acoustic detection of illegal chainsaw logging and poaching vehicles in protected rainforests
- **B.** Satellite imagery analysis to monitor deforestation rates, illegal mining, and glacial retreat
- **C.** Autonomous underwater vehicle visual surveying of coral reef bleaching and marine biodiversity
- **D.** Camera-trap computer vision recognition to track and monitor endangered wildlife populations

Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

In industrial and enterprise recommendation systems, which stages constitute the standard two-stage funnel architecture?

- **A.** Retrieval stage
- **B.** Ranking stage
- **C.** Data collection stage
- **D.** Feature labeling stage

Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following represent core sensory-motor capabilities required for Embodied AI humanoid robots?

- **A.** Multimodal Environmental Perception
- **B.** Physical Motor Actuation and Coordinated Bipedal Locomotion
- **C.** Formal Axiomatic Mathematical Theorem Proving
- **D.** Whole-Body Dynamic Balance and Real-Time Trajectory Control

Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are major components of Huawei's All-Scenario AI Strategy?

- **A.** MindSpore full-scenario AI computing framework
- **B.** Ascend series AI processors providing scalable compute
- **C.** ModelArts platform enabling end-to-end AI lifecycle management
- **D.** Unified architecture across consumer devices, enterprise edge, and public/private clouds

Page Answers: **Q86**: ABD • **Q87**: ABCD • **Q88**: AB • **Q89**: ABD • **Q90**: ABCD

Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

In intelligent voice interaction systems (AI voice assistants), which two speech processing technologies are mainly required?

- **A.** Automatic Speech Recognition
- **B.** Text-to-Speech synthesis
- **C.** Semantic Image Segmentation
- **D.** Machine Translation Alignment

Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized functional sub-disciplines of Speech and Acoustic Signal Processing?

- **A.** Speech Recognition
- **B.** Speaker Identification and Voiceprint Verification
- **C.** Speech Enhancement and Acoustic Noise Suppression
- **D.** Spoken Language Understanding and Acoustic Wake-Word Spotting

Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's official smart voice assistant across the HarmonyOS ecosystem is named ____.

Answer: _____

Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

In autonomous driving, the optical sensor that uses pulsed lasers to measure distances and generate 3D point clouds is called ____.

Answer: _____

Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

In robotics and autonomous vehicles, SLAM stands for Simultaneous Localization and ____.

Answer: _____

Page Answers: **Q91**: AB • **Q92**: ABCD • **Q93**: Celia (or Xiaoyi) • **Q94**: LiDAR • **Q95**: Mapping

Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

The challenge of providing accurate recommendations for new users or items with zero historical interaction is called the ____ Start problem.

Answer: _____

Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's intelligent driving solution for high-level autonomous driving across highways, urban roads, and parking is called Huawei ____ (or ADS).

Answer: _____

Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's large meteorological model for high-precision global weather forecasting is named Pangu-____.

Answer: _____

Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

The field of artificial intelligence that integrates physical bodies, sensors, and actuators to interact with physical environments is known as ____ AI.

Answer: _____

Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

In vehicle cockpits, DMS stands for Driver Monitoring System, which primarily detects driver ____ and distraction.

Answer: _____

Page Answers: **Q96**: Cold • **Q97**: Qiankun • **Q98**: Weather • **Q99**: Embodied • **Q100**: fatigue

HCIA-AI V4.0 Certification Practice Exam • Final Mock Exam

Module 7: Comprehensive Final Mock Exam (100 Questions & Answer Key)

Curated by: Umed Jamal Nouri • v1.1 (Nov 2026) • github.com/UmedJaMala

Question 1

TRUE OR FALSE • 16 Points

The emergence of foundation models represents a shift from developing isolated single-task models to pre-training a general model that adapts across diverse downstream tasks.

A. True
B. False

Question 2

TRUE OR FALSE • 16 Points

Speech processing is a computer vision technology that only processes visual photographs of lips without analyzing audio soundwaves.

A. True
B. False

Question 3

TRUE OR FALSE • 16 Points

DeepSeek models use Multi-Head Latent Attention (MLA) architecture primarily to compress KV cache memory overhead during high-concurrency inference.

A. True
B. False

Question 4

TRUE OR FALSE • 16 Points

In supervised machine learning, the training dataset contains input feature vectors without any corresponding target label values.

A. True
B. False

Question 5

TRUE OR FALSE • 16 Points

L1 regularization drives certain weight coefficients strictly to zero, effectively performing embedded feature selection.

A. True
B. False

Page Answers: **Q1**: True • **Q2**: False • **Q3**: True • **Q4**: False • **Q5**: True

Question 6

TRUE OR FALSE • 16 Points

In hyperparameter optimization, exhaustive grid search is always much faster and computationally cheaper than random search when exploring high-dimensional spaces.

A. True
B. False

Question 7

TRUE OR FALSE • 16 Points

In deep networks using the Sigmoid activation function, gradients may diminish exponentially towards zero within five layers, causing severe vanishing gradients.

A. True
B. False

Question 8

TRUE OR FALSE • 16 Points

The Tanh activation function outputs values in the range (-1, 1) and completely eliminates the vanishing gradient problem in arbitrarily deep networks.

A. True
B. False

Question 9

TRUE OR FALSE • 16 Points

Batch Normalization standardizes intermediate layer inputs across mini-batches, stabilizing internal covariate shift and accelerating convergence.

A. True
B. False

Question 10

TRUE OR FALSE • 16 Points

During testing/evaluation mode, Dropout continues to randomly set 50% of neuron outputs to zero.

A. True
B. False

Page Answers: **Q6**: False • **Q7**: True • **Q8**: False • **Q9**: True • **Q10**: False

Question 11

TRUE OR FALSE • 16 Points

In Convolutional Neural Networks, the pooling layer introduces learnable weight parameters that are updated via backpropagation.

A. True
B. False

Question 12

TRUE OR FALSE • 16 Points

In LSTM networks, the Forget Gate determines the learning rate of the gradient descent optimizer.

A. True
B. False

Question 13

TRUE OR FALSE • 16 Points

PyTorch adopts a dynamic computational graph (Define-by-Run) where the graph is constructed on the fly as operations execute.

A. True
B. False

Question 14

TRUE OR FALSE • 16 Points

In PyTorch, calling optimizer.zero_grad() is necessary before loss.backward() because gradients accumulate by default.

A. True
B. False

Question 15

TRUE OR FALSE • 16 Points

In PyTorch, nn.CrossEntropyLoss requires the model output to be passed through an explicit torch.sigmoid() function first when performing multi-class classification.

A. True
B. False

Page Answers: **Q11**: False • **Q12**: False • **Q13**: True • **Q14**: True • **Q15**: False

Question 16

TRUE OR FALSE • 16 Points

In Data Parallelism, the entire training dataset is kept on one single device while neural network weights are replicated across all devices without batch partitioning.

A. True
B. False

Question 17

TRUE OR FALSE • 16 Points

In Pipeline Parallelism, the weight parameters within each individual layer are partitioned across multiple GPUs.

A. True
B. False

Question 18

TRUE OR FALSE • 16 Points

The core idea of Adapter Tuning is to insert small trainable adapter sub-modules into fixed layers of a pre-trained model while freezing backbone weights.

A. True
B. False

Question 19

TRUE OR FALSE • 16 Points

Huawei's Qiankun Intelligent Driving solution provides full-stack autonomous driving capabilities covering urban roads, highways, and automated valet parking.

A. True
B. False

Question 20

TRUE OR FALSE • 16 Points

In AI4Science, Huawei's Pangu-Weather model achieves weather forecasting speeds thousands of times faster than traditional numerical weather prediction (NWP).

A. True
B. False

Page Answers: **Q16**: False • **Q17**: False • **Q18**: True • **Q19**: True • **Q20**: True

Question 21

TRUE OR FALSE • 16 Points

In the hierarchical relationship among AI disciplines, Deep Learning is a subset of Machine Learning, and Machine Learning is a subset of Artificial Intelligence.

A. True
B. False

Question 22

TRUE OR FALSE • 16 Points

Connectionism is an AI school of thought inspired by biological neural networks and synaptic connection weights.

A. True
B. False

Question 23

TRUE OR FALSE • 16 Points

The three essential driving elements for modern Artificial Intelligence development are data, algorithms, and computing power.

A. True
B. False

Question 24

TRUE OR FALSE • 16 Points

Image classification tasks output both the bounding box spatial coordinates and the category label for each detected item.

A. True
B. False

Question 25

TRUE OR FALSE • 16 Points

Without considering regularization terms, the support vectors of a Support Vector Machine (SVM) are composed of the points closest to the separating hyperplane.

A. True
B. False

Page Answers: **Q21**: True • **Q22**: True • **Q23**: True • **Q24**: False • **Q25**: True

Question 26

SINGLE-ANSWER QUESTION • 16 Points

A computer uses labeled images to learn and distinguish between cats and dogs. What type of machine learning does this scenario represent?

- **A.** Clustering
- **B.** Reinforcement learning
- **C.** Unsupervised learning
- **D.** Supervised learning

Question 27

SINGLE-ANSWER QUESTION • 16 Points

Which of the following statements is TRUE regarding classification and regression in supervised learning?

- **A.** Both classification and regression predict discrete categorical labels
- **B.** Classification predicts discrete categorical labels, whereas regression predicts continuous numerical values
- **C.** Classification predicts continuous numerical values, whereas regression predicts discrete categories
- **D.** Both classification and regression predict continuous numerical values

Question 28

SINGLE-ANSWER QUESTION • 16 Points

According to the HCIA-AI V4.0 Model File specification, which model format uses Protocol Buffers to save parameter values by default so interrupted training can be resumed or fine-tuned?

- **A.** Uncompressed Bitmap (.bmp)
- **B.** Checkpoint (.ckpt)
- **C.** Comma-Separated Values (.csv)
- **D.** Plain Text Log (.txt)

Question 29

SINGLE-ANSWER QUESTION • 16 Points

When an overly complex model achieves 99.8% training accuracy but only 55% testing accuracy, it is suffering from:

- **A.** Vanishing Gradient
- **B.** High Bias
- **C.** Underfitting
- **D.** Overfitting

Question 30

SINGLE-ANSWER QUESTION • 16 Points

What is the harmonic mean of Precision and Recall commonly known as?

- **A.** R-Squared
- **B.** Accuracy
- **C.** F1-Score
- **D.** ROC Curve

Page Answers: **Q26**: D • **Q27**: B • **Q28**: B • **Q29**: D • **Q30**: C

Question 31

SINGLE-ANSWER QUESTION • 16 Points

Which distance metric measures the straight-line distance between two points in spatial geometry?

- **A.** Euclidean Distance
- **B.** Manhattan Distance
- **C.** Cosine Distance
- **D.** Hamming Distance

Question 32

SINGLE-ANSWER QUESTION • 16 Points

Which splitting criterion is used by the CART algorithm for classification trees?

- **A.** Information Gain Ratio
- **B.** Information Gain
- **C.** Gini Impurity
- **D.** Mean Squared Error

Question 33

SINGLE-ANSWER QUESTION • 16 Points

During neural network training, which of the following values are continuously updated by optimization algorithms using gradient descent?

- **A.** Model parameters
- **B.** Sample labels
- **C.** Learning rate
- **D.** Batch size

Question 34

SINGLE-ANSWER QUESTION • 16 Points

What is the value range of the standard ReLU activation function?

- **A.** [0, +inf)
- **B.** (-1, +1)
- **C.** (0, +1.0)
- **D.** (-inf, 0)

Question 35

SINGLE-ANSWER QUESTION • 16 Points

Which activation function is defined as f(x) = max(0, x) and helps reduce vanishing gradients in positive regions?

- **A.** Tanh
- **B.** Sigmoid
- **C.** ReLU
- **D.** Softsign

Page Answers: **Q31**: A • **Q32**: C • **Q33**: A • **Q34**: A • **Q35**: C

Question 36

SINGLE-ANSWER QUESTION • 16 Points

If a 32 x 32 feature map is processed through a 2 x 2 pooling layer with stride = 2 and no padding, what is the output size?

- **A.** 30 x 30
- **B.** 64 x 64
- **C.** 32 x 32
- **D.** 16 x 16

Question 37

SINGLE-ANSWER QUESTION • 16 Points

If a 32 x 32 image is convolved with a 5 x 5 kernel using stride = 1 and padding = 0, what is the output spatial dimension?

- **A.** 30 x 30
- **B.** 28 x 28
- **C.** 27 x 27
- **D.** 32 x 32

Question 38

SINGLE-ANSWER QUESTION • 16 Points

Which optimizer uses both the first moment (momentum) and the second moment (RMSProp) with bias corrections?

- **A.** Adam
- **B.** Newton-Raphson
- **C.** AdaGrad
- **D.** Standard SGD

Question 39

SINGLE-ANSWER QUESTION • 16 Points

What is the primary motivation behind the Residual Connection (skip connection) introduced in ResNet?

- **A.** Enabling identity mappings F(x) + x so gradients flow smoothly, preventing network degradation
- **B.** Replacing non-linear activation functions with direct linear identity pass-through layers
- **C.** Doubling total parameter counts across all hidden layers to maximize model capacity
- **D.** Eliminating the requirement for dedicated GPU accelerators during forward inference passes

Question 40

SINGLE-ANSWER QUESTION • 16 Points

Why is Positional Encoding essential in the Transformer architecture?

- **A.** To project sparse token index embeddings into low-dimensional normalized spatial coordinates
- **B.** To replace dense feedforward projection layers with fixed sinusoidal trigonometric operations
- **C.** To calculate cross-entropy loss gradients across multi-token target vocabulary dictionaries
- **D.** To provide the permutation-invariant self-attention mechanism with sequential order awareness

Page Answers: **Q36**: D • **Q37**: B • **Q38**: A • **Q39**: A • **Q40**: D

Question 41

SINGLE-ANSWER QUESTION • 16 Points

In the LSTM cell, which gate is responsible for deciding what percentage of information from the previous cell state should be discarded?

- **A.** Input Gate
- **B.** Reset Gate
- **C.** Forget Gate
- **D.** Output Gate

Question 42

SINGLE-ANSWER QUESTION • 16 Points

What does PEFT stand for in foundation model fine-tuning?

- **A.** Parameter-Efficient Fine-Tuning
- **B.** Pre-trained Exact Feature Transfer
- **C.** Parallel-Executed Forward Training
- **D.** Performance-Enhanced Formula Tool

Question 43

SINGLE-ANSWER QUESTION • 16 Points

Which PEFT method freezes base weights and injects trainable low-rank decomposition matrices into attention layers?

- **A.** Gradient descent
- **B.** Full retraining
- **C.** LoRA
- **D.** Cross validation

Question 44

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT an AI deep learning framework?

- **A.** MindSpore
- **B.** MySQL
- **C.** PyTorch
- **D.** TensorFlow

Question 45

SINGLE-ANSWER QUESTION • 16 Points

In PyTorch, which module is specifically responsible for automatic differentiation?

- **A.** torch.distributed
- **B.** torch.cuda.stream
- **C.** torch.autograd
- **D.** torch.multiprocess

Page Answers: **Q41**: C • **Q42**: A • **Q43**: C • **Q44**: B • **Q45**: C

Question 46

SINGLE-ANSWER QUESTION • 16 Points

Which method in PyTorch calculates the backward gradients of a loss scalar with respect to leaf tensors?

- **A.** loss.backward()
- **B.** optimizer.step()
- **C.** tensor.zero_grad()
- **D.** model.forward()

Question 47

SINGLE-ANSWER QUESTION • 16 Points

Which method in PyTorch updates model parameters based on their computed gradients?

- **A.** optimizer.step()
- **B.** loss.backward()
- **C.** torch.cuda.empty_cache()
- **D.** model.eval()

Question 48

SINGLE-ANSWER QUESTION • 16 Points

What is the recommended PyTorch method for saving only the learned parameters (weights and biases) of a model?

- **A.** json.dump(model)
- **B.** pickle.dump(model)
- **C.** torch.save(model.state_dict(), PATH)
- **D.** torch.save(model, PATH)

Question 49

SINGLE-ANSWER QUESTION • 16 Points

Which model compression method converts 32-bit floating-point weights into 8-bit integers to reduce memory and accelerate inference?

- **A.** Knowledge Distillation
- **B.** Data Augmentation
- **C.** Model Quantization
- **D.** Model Pruning

Question 50

SINGLE-ANSWER QUESTION • 16 Points

What role does CANN play in Huawei's Ascend AI software stack?

- **A.** A mobile operating system kernel tailored specifically for consumer wearable smart devices
- **B.** The computing architecture and chip enablement layer interfacing with Ascend NPUs
- **C.** A relational database engine managing structured transactional business intelligence queries
- **D.** A cloud-based web dashboard designed for visualizing multi-node distributed training metrics

Page Answers: **Q46**: A • **Q47**: A • **Q48**: C • **Q49**: C • **Q50**: B

Question 51

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is Huawei's lightweight AI engine for ultra-fast, low-latency inference on mobile and IoT edge devices?

- **A.** Kunpeng OS
- **B.** MindSpore Lite
- **C.** GaussDB
- **D.** ModelArts Cloud

Question 52

SINGLE-ANSWER QUESTION • 16 Points

Within the broad landscape of foundation models, what is the primary purpose of Supervised Fine-Tuning (SFT)?

- **A.** Initializing model weights from scratch using uncurated random Gaussian noise distributions
- **B.** Adapting pre-trained models to follow specific user instructions and conversational formats
- **C.** Replacing deep neural attention layers with traditional rule-based scripts
- **D.** Pruning ninety percent of all weight parameters to minimize embedded deployment memory footprint

Question 53

SINGLE-ANSWER QUESTION • 16 Points

Which distributed training strategy partitions sequential layers of a deep neural network across multiple accelerators like stages in an assembly line?

- **A.** Zero-Shot Prompting
- **B.** Data Parallelism
- **C.** Pipeline Parallelism
- **D.** K-Means Clustering

Question 54

SINGLE-ANSWER QUESTION • 16 Points

What is the primary function of Retrieval-Augmented Generation (RAG)?

- **A.** Querying external knowledge stores to inject factual context into prompts, reducing hallucination
- **B.** Compiling high-performance native device driver modules for modern operating system kernels
- **C.** Deleting outdated customer database records automatically based on temporal retention rules
- **D.** Training foundational transformer base models from scratch using random initialization weights

Question 55

SINGLE-ANSWER QUESTION • 16 Points

In the RLHF workflow, which reinforcement learning algorithm is commonly used to optimize the language policy?

- **A.** Proximal Policy Optimization
- **B.** K-Means Centroid Clustering Algorithm
- **C.** Information Gain Ratio Decision Tree
- **D.** Ordinary Least Squares Linear Regression

Page Answers: **Q51**: B • **Q52**: B • **Q53**: C • **Q54**: A • **Q55**: A

Question 56

SINGLE-ANSWER QUESTION • 16 Points

What security threat occurs when an adversary crafts adversarial prompts to bypass safety guardrails and elicit dangerous content?

- **A.** Batch Normalization
- **B.** Weight Decay
- **C.** Cross-Validation
- **D.** Jailbreaking

Question 57

SINGLE-ANSWER QUESTION • 16 Points

Huawei's smart voice assistant across the HarmonyOS ecosystem is named:

- **A.** Siri
- **B.** Celia
- **C.** Alexa
- **D.** Bixby

Question 58

SINGLE-ANSWER QUESTION • 16 Points

Which sensor in autonomous driving emits laser pulses to construct high-precision 3D point cloud maps of the environment?

- **A.** Ultrasonic sensor
- **B.** Optical camera
- **C.** Millimeter-wave radar
- **D.** LiDAR

Question 59

SINGLE-ANSWER QUESTION • 16 Points

When an e-commerce platform launches a new product with zero user ratings or interactions, which problem does it encounter?

- **A.** Pipeline Stall
- **B.** Cold Start Problem
- **C.** Catastrophic Forgetting
- **D.** Vanishing Gradient

Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following core technologies allow mobile robots and autonomous vehicles to map unknown environments and localize themselves? (Multiple Choice)

- **A.** Simultaneous Localization and Mapping
- **B.** Visual Odometry using camera sensor fusion
- **C.** LiDAR point cloud scan matching algorithms
- **D.** Static geometric CAD blueprint querying without sensor feedback

Page Answers: **Q56**: D • **Q57**: B • **Q58**: D • **Q59**: B • **Q60**: ABC

Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following AI schools of thought originated from bionics and environmental cybernetics respectively?

A. Symbolism
B. Connectionism
C. Positivism
D. Behaviorism

Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common Computer Vision tasks?

A. Object Detection
B. Optical Character Recognition
C. Image Classification
D. Semantic Segmentation

Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are core compute innovations introduced in DeepSeek-V3?

A. Multi-Head Latent Attention for efficient KV caching
B. Static Global Average Pooling Layers
C. DeepSeekMoE architecture with shared experts
D. Recurrent Hebbian Plasticity Optimization

Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

In the HCIA-AI V4.0 AI business evaluation workflow, which of the following statements regarding Data Pre-Evaluation, Model Post-Evaluation, and Regression Metrics are TRUE?

A. Pre-Evaluation inspects dataset Quality, Diversity, and Balance/Fairness before model training to prevent bias
B. Post-Evaluation assesses the trained model on the test set using classification metrics or regression metrics
C. When extreme outliers are present in regression data, MAE is more robust than MSE and RMSE because it does not square errors
D. Manual Feature Engineering is mandatory for Computer Vision and NLP models rather than Recommendation Systems

Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following hyperparameters may be defined and tuned during machine learning training?

A. Learning rate
B. Number of epochs
C. Model weights and biases
D. Batch size

Page Answers: **Q61**: BD • **Q62**: ABCD • **Q63**: AC • **Q64**: ABC • **Q65**: ABD

Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements accurately reflect Feedforward Neural Network (FNN) properties and gradient stability mechanisms in HCIA-AI V4.0?

A. An FNN transmits signals unidirectionally with no feedback and can be represented as a Directed Acyclic Graph (DAG)

B. In an FNN neuron, the bias parameter 'b' represents the activation threshold

C. The Sigmoid activation function has a maximum derivative of 0.25 (1/4), contributing to vanishing gradients in deep networks

D. L1 and L2 weight regularization cause exploding gradients and must always be avoided during deep network training

Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following tasks belong to the data preprocessing stage?

A. Data deduplication

B. Feature normalization and standardization

C. Outlier handling

D. Missing value imputation

Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are supervised machine learning algorithms?

A. Support Vector Machines

B. K-Means Clustering

C. Multiple Linear Regression

D. Principal Component Analysis

Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following algorithms belong to Ensemble Learning?

A. Random Forest

B. AdaBoost

C. Single Perceptron

D. Gradient Boosted Decision Tree

Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following activation functions are smooth, bounded S-shaped (sigmoidal) curves commonly used in neural networks?

A. Sigmoid function

B. Hyperbolic Tangent or Tanh function

C. Standard Rectified Linear Unit

D. Leaky ReLU function

Page Answers: **Q66**: ABC • **Q67**: ABCD • **Q68**: AC • **Q69**: ABD • **Q70**: AB

Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

In the Transformer attention mechanism, which vectors are multiplied via dot-product to compute raw compatibility attention weights?

- **A.** Value vector
- **B.** Query vector
- **C.** Key vector
- **D.** Residual shortcut vector

Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following algorithms are commonly used first-order gradient-based optimizers in deep learning?

- **A.** RMSProp
- **B.** Momentum SGD
- **C.** Adam
- **D.** K-Means

Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are foundational components of a Convolutional Neural Network (CNN)?

- **A.** Convolutional layer with learnable filter kernels
- **B.** Pooling layer
- **C.** Learnable positional encodings and multi-head cross-attention projections
- **D.** Non-linear activation functions

Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary techniques for preventing overfitting in deep neural networks?

- **A.** Dropout
- **B.** Early stopping
- **C.** L1 and L2 weight regularization
- **D.** Data augmentation

Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are key architectural advantages of the Transformer over classical RNN models?

- **A.** Full parallelization during training across sequence tokens
- **B.** Ability to scale efficiently to billions of parameters
- **C.** Eliminating the need for any training data
- **D.** Direct global dependency modeling without distance decay

Page Answers: **Q71**: BC • **Q72**: ABC • **Q73**: ABD • **Q74**: ABCD • **Q75**: ABD

Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about the Softmax activation function are correct?

- **A.** It maps arbitrary real-valued logits into a valid probability distribution
- **B.** The sum of all output probabilities across all classes equals exactly 1
- **C.** Every individual output probability value is strictly bounded between 0 and 1
- **D.** It guarantees that all output probabilities are strictly negative numbers

Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary functions provided by an AI Development Framework?

- **A.** Model optimization, serialization, and deployment tools
- **B.** High-level front-end programming APIs
- **C.** Hardware abstraction and computation scheduling
- **D.** Automatic differentiation engine

Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are key characteristics of Eager Execution (Define-by-Run) mode?

- **A.** Immediate execution of operations with direct numerical feedback
- **B.** Line-by-line interactive debugging using standard Python tools like pdb
- **C.** Whole-graph ahead-of-time memory reuse planning and optimization
- **D.** Immutability of execution topologies during runtime

Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following techniques directly compress numerical weight precision or model footprint in deep neural networks?

- **A.** Quantization
- **B.** Pruning
- **C.** Knowledge distillation
- **D.** Data augmentation

Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about PyTorch DataLoader are correct?

- **A.** It supports parallel multi-process data loading via the num_workers parameter
- **B.** It automatically batches individual data samples into mini-batch tensors
- **C.** It automatically searches and compiles optimal neural network architectures without user code
- **D.** It supports stochastic shuffling of dataset samples at the beginning of each training epoch

Page Answers: **Q76**: ABC • **Q77**: ABCD • **Q78**: AB • **Q79**: ABC • **Q80**: ABD

Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are major distributed parallelism modes for training large foundation models?

- **A.** Model pruning
- **B.** Tensor parallelism
- **C.** Data parallelism
- **D.** Pipeline parallelism

Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

In the standard RLHF training workflow, which training stages follow Supervised Fine-Tuning (SFT)?

- **A.** Raw Web Data Scraping and Pre-tokenization
- **B.** Reward Model Training on human preference comparisons
- **C.** Policy Optimization using PPO
- **D.** Static C++ Kernel Compilation

Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

A standard enterprise RAG system mainly consists of which core functional modules?

- **A.** Retrieval module
- **B.** Pre-training module
- **C.** Generation module
- **D.** Fine-tuning module

Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common Prompt Engineering techniques?

- **A.** Chain-of-Thought Prompting
- **B.** Zero-Shot Prompting
- **C.** Role-Based Prompting
- **D.** Few-Shot In-Context Learning

Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are main safety risks associated with large model deployments?

- **A.** Guaranteed convergence to zero generalization error
- **B.** Sensitive data leakage from training memorization
- **C.** Hallucination of harmful or false advice
- **D.** Prompt Injection and Jailbreaking attacks

Page Answers: **Q81**: BCD • **Q82**: BC • **Q83**: AC • **Q84**: ABCD • **Q85**: BCD

Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following core technologies are integrated into an end-to-end Voice Assistant (e.g., Celia)?

- **A.** Dialogue Management
- **B.** Text-to-Speech
- **C.** Natural Language Understanding
- **D.** Photometric stereo surface normal reconstruction

Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary sensors used in autonomous driving perception systems?

- **A.** Optical HD Cameras
- **B.** Ultrasonic Sensors
- **C.** Millimeter-Wave Radar
- **D.** Light Detection and Ranging

Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are typical applications of AI in Smart Homes?

- **A.** Intelligent voice-activated appliance control
- **B.** Smart climate control and energy optimization
- **C.** Anomaly detection
- **D.** High-frequency automated stock market equity arbitrage

Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following represent the classic main types of recommendation algorithms?

- **A.** Collaborative Filtering
- **B.** Content-Based Filtering
- **C.** Random Permutation Shuffling
- **D.** Lexical Alphabetical Sorting

Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following components are part of Huawei's full-stack AI development ecosystem?

- **A.** ModelArts development platform
- **B.** Ascend AI processors
- **C.** CANN compute architecture
- **D.** MindSpore AI framework

Page Answers: **Q86**: ABC • **Q87**: ABCD • **Q88**: ABC • **Q89**: AB • **Q90**: ABCD

Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized as the three foundational pillars powering the rapid advancement of modern AI?

- **A.** Massive high-quality training Data
- **B.** High-performance AI Computing Power
- **C.** Advanced Deep Learning and Machine Learning Algorithms
- **D.** Highly structured relational database tables with fixed SQL schemas

Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Support Vector Machines (SVM) are correct?

- **A.** The objective of SVM is to find an optimal hyperplane that maximizes the margin
- **B.** Support vectors are the subset of training samples closest to the decision boundary
- **C.** Kernel functions allow SVM to map input vectors into higher-dimensional feature spaces
- **D.** In high-dimensional feature spaces, an optimal hyperplane is constructed to separate classes

Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the method called on the optimizer to reset previously calculated gradients before computing new gradients is optimizer.____().

Answer: _____

Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the method called on the scalar loss tensor to calculate gradients via backpropagation is loss.____().

Answer: _____

Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the optimizer method that updates model parameters based on computed gradients is optimizer.____().

Answer: _____

Page Answers: **Q91**: ABC • **Q92**: ABCD • **Q93**: zero_grad • **Q94**: backward • **Q95**: step

Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

In Huawei MindSpore, the forward execution logic of a neural network module is implemented inside the def ____(self, x): method.

Answer: _____

Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the method implemented inside an nn.Module subclass to define the forward computational graph is def ____(self, x):.

Answer: _____

Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

Implemented in C/C++, the standalone deployment tool ____ enables high-performance LLM inference locally or in the cloud without needing an AI framework.

Answer: _____

Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the method that returns a dictionary containing all learnable model weights for saving to disk is model.____().

Answer: _____

Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, to disable gradient tracking during model evaluation and save memory, code is executed inside the context with torch.____():.

Answer: _____

Page Answers: **Q96**: construct • **Q97**: forward • **Q98**: llama.cpp • **Q99**: state_dict • **Q100**: no_grad

HCIA-AI V4.0 Certification Practice Exam • Practice Set 1

Module 8: Practice Exam Set 1 (100 High-Yield Exam-Style Questions)

Curated by: Umed Jamal Nouri • v1.1 (Nov 2026) • github.com/UmedJaMala

Question 1

TRUE OR FALSE • 16 Points

In MindSpore, when constructing a custom neural network, developers define forward computation inside the backward() method and must not override construct().

A. True

B. False

Question 2

TRUE OR FALSE • 16 Points

In a neural network based on Connectionism, each individual neuron node strictly expresses a specific, isolated human semantic meaning.

A. True

B. False

Question 3

TRUE OR FALSE • 16 Points

Voice recognition (Automatic Speech Recognition) refers to the computational process of converting acoustic voice audio data into text data.

A. True

B. False

Question 4

TRUE OR FALSE • 16 Points

Huawei's Ascend 310 processor is designed only for massive cloud-scale training clusters, whereas Ascend 910 is an ultra-low-power chip designed for IoT edge watches.

A. True

B. False

Question 5

TRUE OR FALSE • 16 Points

'Batch inference is an offline batch job that performs model inference on bulk datasets. Therefore, there is no need for model training before executing batch inference.' This statement is correct.

A. True

B. False

Page Answers: **Q1**: False • **Q2**: False • **Q3**: True • **Q4**: False • **Q5**: False

Question 6

TRUE OR FALSE • 16 Points

In PyTorch, tensor calculations and neural network models are executed on the GPU by default without calling the .to(device) method.

A. True
B. False

Question 7

TRUE OR FALSE • 16 Points

In Python, tuples are mutable list-like objects whose elements can be freely appended, removed, and reassigned in-place after creation.

A. True
B. False

Question 8

TRUE OR FALSE • 16 Points

In Huawei's full-stack AI strategy, MindSpore is designed to support device, edge, and cloud collaborative AI development.

A. True
B. False

Question 9

TRUE OR FALSE • 16 Points

Overfitting can occur in both classification and regression machine learning problems.

A. True
B. False

Question 10

TRUE OR FALSE • 16 Points

In Huawei's Atlas computing platform, Atlas 500 is an AI cluster of supercomputers designed for deep underground nuclear simulation.

A. True
B. False

Page Answers: **Q6**: False • **Q7**: False • **Q8**: True • **Q9**: True • **Q10**: False

Question 11

TRUE OR FALSE • 16 Points

In MindSpore, the PyNative execution mode executes operations line by line like standard Python, providing convenient interactive debugging.

A. True
B. False

Question 12

TRUE OR FALSE • 16 Points

In NumPy, broadcasting allows arithmetic operations between arrays of differing shapes as long as their trailing dimensions are compatible.

A. True
B. False

Question 13

TRUE OR FALSE • 16 Points

In deep learning optimization, increasing the learning rate to an extremely high value guarantees that the model will always converge faster to the minimum loss.

A. True
B. False

Question 14

TRUE OR FALSE • 16 Points

In Huawei CANN architecture, TBE (Tensor Boost Engine) allows developers to write custom high-performance neural network operators for Ascend NPUs.

A. True
B. False

Question 15

TRUE OR FALSE • 16 Points

Huawei Atlas 900 is an AI cluster consisting of thousands of Ascend 910 processors, designed for scientific research and massive deep learning supercomputing.

A. True
B. False

Page Answers: **Q11**: True • **Q12**: True • **Q13**: False • **Q14**: True • **Q15**: True

Question 16

TRUE OR FALSE • 16 Points

In Python dictionary structures, mutable objects like lists can be safely used as dictionary keys.

A. True

B. False

Question 17

TRUE OR FALSE • 16 Points

Huawei HiLens is an end-cloud collaborative AI development platform providing multi-modal development tools and device-cloud synergy for vision applications.

A. True

B. False

Question 18

TRUE OR FALSE • 16 Points

In modern foundation models, Reinforcement Learning from Human Feedback (RLHF) completely replaces all pre-training data requirements.

A. True

B. False

Question 19

TRUE OR FALSE • 16 Points

In gradient descent optimization, saddle points are critical points where the first derivative (gradient) is zero, but which are neither local minima nor local maxima.

A. True

B. False

Question 20

TRUE OR FALSE • 16 Points

In Python, the list.append() method modifies the list in place and returns a new list containing the appended element.

A. True

B. False

Page Answers: **Q16**: False • **Q17**: True • **Q18**: False • **Q19**: True • **Q20**: False

Question 21

TRUE OR FALSE • 16 Points

Huawei Cloud ModelArts includes automated machine learning (AutoML) capabilities that automatically perform data preprocessing, feature engineering, and model selection.

A. True
B. False

Question 22

TRUE OR FALSE • 16 Points

The Sigmoid activation function is symmetric around the origin (0, 0) and has zero-centered outputs.

A. True
B. False

Question 23

TRUE OR FALSE • 16 Points

In Python, lambda expressions can contain multiple complex statements, loops, and exception handling blocks.

A. True
B. False

Question 24

TRUE OR FALSE • 16 Points

In MindSpore, Automatic Parallelism can automatically search for optimal parallel strategies (data, operator, and pipeline parallelism) without manual tensor slicing.

A. True
B. False

Question 25

TRUE OR FALSE • 16 Points

In convolutional neural networks (CNNs), pooling layers are used to reduce feature map spatial dimensions and help prevent overfitting.

A. True
B. False

Page Answers: **Q21**: True • **Q22**: False • **Q23**: False • **Q24**: True • **Q25**: True

Question 26

SINGLE-ANSWER QUESTION • 16 Points

Which of the following layers of Huawei's full-stack AI solution provides chip enablement and computing operator libraries for Ascend processors?

- **A.** Compute Architecture for Neural Networks chip enablement layer
- **B.** ModelArts one-stop AI application development cloud platform
- **C.** MindSpore unified training and inference deep learning framework
- **D.** MindX SDK application enablement layer

Question 27

SINGLE-ANSWER QUESTION • 16 Points

Consider a scenario where a machine learning algorithm is trained to filter spam emails. What constitutes the 'Experience (E)' in Mitchell's ML definition?

- **A.** The static network IP address assigned to the corporate email gateway server
- **B.** The raw execution throughput latency measured by the server operating system
- **C.** The percentage of spam messages correctly flagged by the classification model
- **D.** The historical database archive of labeled spam and legitimate email messages

Question 28

SINGLE-ANSWER QUESTION • 16 Points

A computer uses labeled images to learn and determine which pictures contain apples and which contain pears. Which type of machine learning does this represent?

- **A.** Reinforcement learning
- **B.** Unsupervised learning
- **C.** Supervised learning
- **D.** Semi-supervised learning

Question 29

SINGLE-ANSWER QUESTION • 16 Points

In the traditional AI business development process, which phase immediately follows Data Engineering and Feature Processing?

- **A.** Model Training and Validation
- **B.** Requirement Clarification
- **C.** Hardware Silicon Fabrication
- **D.** System Decommissioning

Question 30

SINGLE-ANSWER QUESTION • 16 Points

In Huawei Cloud, which service is paired with ModelArts to easily deploy trained vision AI models directly to edge devices?

- **A.** Object Storage Service
- **B.** HiLens
- **C.** Cloud Eye
- **D.** Elastic Cloud Server

Page Answers: **Q26**: A • **Q27**: D • **Q28**: C • **Q29**: A • **Q30**: B

Question 31

SINGLE-ANSWER QUESTION • 16 Points

Which of the following Atlas series hardware products is an AI accelerator card used in data center servers for high-throughput inference and video decoding?

- **A.** Atlas 500
- **B.** Atlas 200
- **C.** Atlas 300
- **D.** Atlas 900

Question 32

SINGLE-ANSWER QUESTION • 16 Points

Which of the following Atlas series hardware products is an embedded AI accelerator module designed for smart cameras, drones, and robotics?

- **A.** Atlas 300
- **B.** Atlas 800
- **C.** Atlas 900
- **D.** Atlas 200

Question 33

SINGLE-ANSWER QUESTION • 16 Points

In Python, which function is used to determine the number of items in a sequence (such as a string, list, or dictionary)?

- **A.** count()
- **B.** length()
- **C.** len()
- **D.** size()

Question 34

SINGLE-ANSWER QUESTION • 16 Points

Which of the following statements about Logistic Regression is correct?

- **A.** Logistic regression is an unsupervised clustering algorithm optimizing intra-cluster variance
- **B.** Logistic regression cannot solve binary classification problems without multi-layer perceptrons
- **C.** Logistic regression requires Mercer kernel functions to compute gradients during backpropagation
- **D.** Logistic regression applies the Sigmoid function to map linear combinations to probabilities

Question 35

SINGLE-ANSWER QUESTION • 16 Points

In neural networks, what is the vector of partial derivatives of a scalar loss function with respect to all parameters called?

- **A.** Learning Rate
- **B.** Momentum Factor
- **C.** Gradient
- **D.** Batch Size

Page Answers: **Q31**: C • **Q32**: D • **Q33**: C • **Q34**: D • **Q35**: C

Question 36

SINGLE-ANSWER QUESTION • 16 Points

In deep learning activation functions, which function maps any real value into a range between 0 and 1, commonly used in binary classification output layers?

- **A.** ReLU
- **B.** Tanh
- **C.** Leaky ReLU
- **D.** Sigmoid

Question 37

SINGLE-ANSWER QUESTION • 16 Points

In Python NumPy, what attribute gives the dimensional tuple of an array?

- **A.** ndarray.size
- **B.** ndarray.shape
- **C.** ndarray.dtype
- **D.** ndarray.ndim

Question 38

SINGLE-ANSWER QUESTION • 16 Points

In Python, what is the output of the slice expression a = [10, 20, 30, 40, 50]; print(a[1:4])?

- **A.** [20, 30, 40, 50]
- **B.** [10, 30, 50]
- **C.** [20, 30, 40]
- **D.** [10, 20, 30]

Question 39

SINGLE-ANSWER QUESTION • 16 Points

Which Huawei tool in ModelArts provides device-edge-cloud joint application development with hardware affinity and operator optimization?

- **A.** MindX DL
- **B.** MindSpore
- **C.** ModelBox
- **D.** AscendCL

Question 40

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of the DaVinci architecture inside Huawei Ascend AI processors?

- **A.** Managing cloud virtual machine billing and multi-tenant user authentication
- **B.** Providing front-end web visualization dashboards for data annotation teams
- **C.** Compiling high-level Python scripts directly into x86 desktop executables
- **D.** Providing dedicated 3D Cube matrix computing engines tailored for deep neural network operations

Page Answers: **Q36**: D • **Q37**: B • **Q38**: C • **Q39**: C • **Q40**: D

Question 41

SINGLE-ANSWER QUESTION • 16 Points

In MindSpore, which execution mode compiles the computation into a static computational graph for maximized execution performance?

- **A.** PyNative Mode
- **B.** Eager Script Mode
- **C.** Dynamic Trace Mode
- **D.** Graph Mode

Question 42

SINGLE-ANSWER QUESTION • 16 Points

Which of the following algorithms is based on Bayes' Theorem and assumes that feature attributes are conditionally independent of each other?

- **A.** Support Vector Machine
- **B.** K-Means
- **C.** Linear Regression
- **D.** Naive Bayes

Question 43

SINGLE-ANSWER QUESTION • 16 Points

In Python, which statement is used to handle exceptions and prevent program crashes upon errors?

- **A.** def ... return
- **B.** if ... else
- **C.** for ... in
- **D.** try ... except

Question 44

SINGLE-ANSWER QUESTION • 16 Points

Which of the following loss functions is most appropriate for training a multi-class neural network classifier?

- **A.** Mean Absolute Percentage Error
- **B.** Binary Hinge Loss Support Classifier
- **C.** Categorical Cross-Entropy Loss Model
- **D.** Mean Squared Error Continuous Loss

Question 45

SINGLE-ANSWER QUESTION • 16 Points

In Python, what is the data type of the variable x after executing x = (1, 2, 3)?

- **A.** dict
- **B.** tuple
- **C.** set
- **D.** list

Page Answers: **Q41**: D • **Q42**: D • **Q43**: D • **Q44**: C • **Q45**: B

Question 46

SINGLE-ANSWER QUESTION • 16 Points

Which Huawei development suite provides the Ascend deep learning development kit, compiler, and low-level runtime drivers?

- **A.** CANN Toolkit
- **B.** MindSpore Hub
- **C.** ModelArts Studio
- **D.** HarmonyOS SDK

Question 47

SINGLE-ANSWER QUESTION • 16 Points

In decision trees, what does Information Gain measure?

- **A.** The total quantity of leaf nodes generated after pruning cross-validation partitions
- **B.** The reduction in entropy achieved by partitioning data according to a specific feature
- **C.** The physical file storage footprint occupied by the serialized model checkpoint on disk
- **D.** The wall-clock execution duration required by the CPU compiler to finish tree training

Question 48

SINGLE-ANSWER QUESTION • 16 Points

In Python, which keyword is used to define a user-defined function?

- **A.** function
- **B.** sub
- **C.** def
- **D.** func

Question 49

SINGLE-ANSWER QUESTION • 16 Points

Which of the following operations in CNN reduces the spatial dimensions of feature maps while retaining dominant features?

- **A.** Padding
- **B.** Pooling
- **C.** Stride
- **D.** Convolution

Question 50

SINGLE-ANSWER QUESTION • 16 Points

In Python, what is the output of bool([])?

- **A.** False
- **B.** True
- **C.** Error
- **D.** None

Page Answers: **Q46**: A • **Q47**: B • **Q48**: C • **Q49**: B • **Q50**: A

Question 51

SINGLE-ANSWER QUESTION • 16 Points

What is the primary difference between a parameter and a hyperparameter in machine learning?

- **A.** Hyperparameters are learned automatically; parameters are configured manually
- **B.** Parameters are only used in regression; hyperparameters are only used in classification
- **C.** There is no conceptual difference between them
- **D.** Parameters are learned automatically from data; hyperparameters are configured manually before training

Question 52

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is an example of an unsupervised learning task?

- **A.** Grouping customer transaction behaviors into clusters without ground-truth labels
- **B.** Classifying radiology images as tumor or healthy based on expert diagnoses
- **C.** Transcribing speech recordings into text
- **D.** Predicting tomorrow's temperature from historical records

Question 53

SINGLE-ANSWER QUESTION • 16 Points

What is the purpose of Early Stopping in deep learning?

- **A.** Increasing the optimizer learning rate exponentially at every training epoch
- **B.** Pruning low-magnitude convolutional filters during post-training quantization
- **C.** Terminating training when validation loss stops improving, thereby preventing overfitting
- **D.** Doubling the number of hidden layers dynamically whenever training loss plateaus

Question 54

SINGLE-ANSWER QUESTION • 16 Points

In Python, which module is part of the standard library and used for parsing regular expressions?

- **A.** re
- **B.** regex_lib
- **C.** text_match
- **D.** string_parser

Question 55

SINGLE-ANSWER QUESTION • 16 Points

Which of the following techniques converts categorical labels into binary vectors where only one index is 1 and all others are 0?

- **A.** One-Hot Encoding
- **B.** Min-Max Normalization
- **C.** Z-Score Standardization
- **D.** PCA Projection

Page Answers: **Q51**: D • **Q52**: A • **Q53**: C • **Q54**: A • **Q55**: A

Question 56

SINGLE-ANSWER QUESTION • 16 Points

In deep learning, what does the term 'Batch Size' refer to?

- **A.** The cumulative file count of raw image files stored inside the uncompressed dataset directory
- **B.** The number of training samples processed in one forward and backward gradient computation pass
- **C.** The maximum storage capacity of physical server solid-state storage expressed in gigabytes
- **D.** The total number of convolutional feature extraction layers stacked inside the neural model

Question 57

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes the learning rate in gradient descent?

- **A.** A hyperparameter controlling the step size taken along the negative gradient to update weights
- **B.** The clock frequency of the processor execution cores measured in gigahertz during training
- **C.** The total quantity of cross-validation fold partitions allocated for model evaluation
- **D.** The cumulative duration taken by the cluster to finish training expressed in clock hours

Question 58

SINGLE-ANSWER QUESTION • 16 Points

In Python, which operator is used to check if two references point to the exact same object in memory?

- **A.** equals
- **B.** is
- **C.** in
- **D.** ==

Question 59

SINGLE-ANSWER QUESTION • 16 Points

What is the primary function of the Activation Function in an artificial neural network?

- **A.** Calculating objective cross-entropy loss gradients against ground truth label targets
- **B.** Re-initializing all internal weight parameters uniformly to zero at every training epoch
- **C.** Introducing non-linearity into the network, enabling it to learn complex non-linear mappings
- **D.** Rescaling input image pixel dimensions to match standardized square aspect ratio templates

Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Support Vector Machines (SVM) are correct? (Multiple Choice)

- **A.** The objective of SVM is to find a hyperplane that maximizes the geometric margin between classes
- **B.** Support vectors are the closest training sample points to the separating hyperplane
- **C.** SVM cannot be applied to binary classification problems under any configuration
- **D.** Kernel functions allow SVM to handle non-linearly separable datasets

Page Answers: **Q56**: B • **Q57**: A • **Q58**: B • **Q59**: C • **Q60**: ABD

Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

In Python NumPy library, which of the following functions are commonly used to initialize new multidimensional arrays?

- **A.** np.zeros() (creating an array populated completely with zeros)
- **B.** np.ones() (creating an array populated completely with ones)
- **C.** np.arange() (creating an array with evenly spaced values within a given interval)
- **D.** np.linalg.inv() (computes the multiplicative inverse of a square matrix)

Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

In the Python Pandas data science library, which core data structures represent one-dimensional and two-dimensional labeled data?

- **A.** Series
- **B.** DataFrame
- **C.** Tensor
- **D.** Tuple

Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

In Support Vector Machine optimization, what are the effects of adjusting the regularization penalty parameter C?

- **A.** A large C value assigns a high penalty to misclassified samples, resulting in narrower margins
- **B.** A small C value leads to higher model variance and an increased risk of overfitting
- **C.** A small C value tolerates more training classification errors in exchange for a wider margin
- **D.** Parameter C controls the trade-off between margin maximization and classification error minimization

Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements correctly describe training iterations and batches in mini-batch SGD?

- **A.** An Epoch is completed after evaluating a single mini-batch of training samples
- **B.** A Batch represents a subset of training data processed together in a single forward-backward pass
- **C.** An Iteration requires evaluating the entire training dataset from beginning to end
- **D.** The number of iterations per epoch equals total dataset size divided by batch size

Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding the Random Forest ensemble algorithm are TRUE?

- **A.** It constructs an ensemble of diverse decision trees trained on bootstrap samples of the training data
- **B.** At each node split, it selects the best split from a randomly chosen subset of features
- **C.** It trains decision trees sequentially, with each tree fitting residual errors of previous trees
- **D.** Individual decision trees must be optimized using backpropagation gradient descent

Page Answers: **Q61**: ABC • **Q62**: AB • **Q63**: ACD • **Q64**: BD • **Q65**: AB

Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following aspects belong to Huawei's Full-Stack AI Solution?

- **A.** ModelArts
- **B.** Compute Architecture for Neural Networks
- **C.** Ascend
- **D.** MindSpore

Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

Huawei Cloud ModelArts provides ModelBox for device-edge-cloud joint development. Which of the following are its core optimization policies?

- **A.** Mandatory cloud-only execution
- **B.** Hardware affinity
- **C.** Automatic segmentation of operators
- **D.** Operator optimization

Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Under what circumstances should Machine Learning be considered rather than traditional rule-based programming?

- **A.** When problem logic is too complex to be hand-crafted into explicit hand-crafted rules
- **B.** When large volumes of representative historical data are available to learn patterns
- **C.** When the task involves purely fixed arithmetic formulas that never change over time
- **D.** When zero data exists and the solution must execute strictly on physical relays

Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are classic application scenarios of AI in modern intelligent systems?

- **A.** Smart Homes and Internet of Things (IoT)
- **B.** Autonomous Driving and Intelligent Vehicles
- **C.** AI for Science (AI4Science) such as Pangu-Weather
- **D.** Manual paper file sorting without digital computers

Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following hardware devices belong to Huawei's Atlas AI computing platform series?

- **A.** Atlas 800 AI training/inference server
- **B.** Atlas 500 AI edge station
- **C.** Atlas 200 AI accelerator module
- **D.** Atlas 300 AI accelerator card

Page Answers: **Q66**: ABCD • **Q67**: BCD • **Q68**: AB • **Q69**: ABC • **Q70**: ABCD

Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary architectural features of Huawei's MindSpore AI framework?

- **A.** Unified front-end API enabling seamless Device-Edge-Cloud development
- **B.** Auto-Parallelism capability simplifying distributed training of large models
- **C.** Native support for both PyNative and Graph execution modes
- **D.** Requiring third-party proprietary closed-source compilers for model execution

Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are built-in immutable data types in Python?

- **A.** Tuple
- **B.** String
- **C.** List
- **D.** Dictionary

Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are built-in mutable data structures in Python?

- **A.** List
- **B.** Dictionary
- **C.** Set
- **D.** Tuple

Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following methods are commonly used in machine learning to handle imbalanced datasets?

- **A.** Adjusting class loss weights
- **B.** Undersampling majority class samples
- **C.** Duplicating majority class samples to increase class skew
- **D.** Oversampling minority class samples

Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are popular Python libraries used in machine learning and data science workflows?

- **A.** Scikit-learn
- **B.** NumPy
- **C.** Pandas
- **D.** Matplotlib or Seaborn

Page Answers: **Q71**: ABC • **Q72**: AB • **Q73**: ABC • **Q74**: ABD • **Q75**: ABCD

Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about PCA (Principal Component Analysis) are correct?

- **A.** It requires categorical ground-truth target labels for supervised training
- **B.** It is an unsupervised linear dimensionality reduction technique
- **C.** Principal components maximize the variance of the projected data
- **D.** It expands feature dimensions to infinity via the kernel trick

Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common regularization strategies in deep learning?

- **A.** Early Stopping
- **B.** Dropout
- **C.** L2 regularization
- **D.** L1 regularization

Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Huawei Ascend AI processors are correct?

- **A.** Ascend processors rely on x86 scalar instruction sets without matrix units
- **B.** Built upon the proprietary DaVinci 3D Cube computing architecture
- **C.** Ascend 910 is primarily designed for high-throughput AI model training
- **D.** Ascend chips are restricted only to consumer mobile phone microcontrollers

Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following components are part of Huawei's CANN software architecture?

- **A.** TBE
- **B.** Graph Engine
- **C.** MindSpore Hub
- **D.** AscendCL

Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary advantages of Decision Trees over complex black-box neural networks?

- **A.** Explicit rule-based tree interpretability
- **B.** Native handling of mixed feature scales
- **C.** Complete immunity to unpruned overfitting
- **D.** Zero-parameter non-linear curve fitting

Page Answers: **Q76**: BC • **Q77**: ABCD • **Q78**: BC • **Q79**: ABD • **Q80**: AB

Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements accurately describe model deployment, quantization, and framework-free LLM inference in HCIA-AI V4.0?

A. INT8 quantization drops fractional decimals to improve efficiency on Edge devices, whereas precision-critical Core servers typically retain FP32

B. PyTorch (.pt/.pth) models are commonly converted to ONNX (.onnx) middleware for heterogeneous deployment across CPUs, GPUs, and NPUs

C. llama.cpp runs LLM inference directly in C/C++ across local and cloud hardware without requiring a Python AI framework

D. Converting weights from FP32 to INT8 doubles the memory footprint on disk and slows down inference

Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common kernel functions used in Support Vector Machines (SVM)?

A. Linear Kernel

B. Sigmoid Kernel

C. Radial Basis Function

D. Polynomial Kernel

Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary stages in the Machine Learning development lifecycle?

A. Data Collection, Cleansing, and Preprocessing

B. Feature Engineering, Model Training, and Validation

C. Model Evaluation, Hyperparameter Tuning, and Deployment Serving

D. Dynamic silicon etching of host GPU instruction microcode

Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

In deep learning Foundation Models, which of the following techniques are commonly used to adapt a large pre-trained model to specific downstream tasks?

A. Parameter-Efficient Fine-Tuning (PEFT / LoRA)

B. Manual hard-coded rule rewriting

C. Supervised Fine-Tuning (SFT)

D. Random weight re-initialization of all transformer layers

Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common activation functions used in Deep Learning?

A. ReLU

B. Tanh

C. Sigmoid

D. Leaky ReLU

Page Answers: **Q81**: ABC • **Q82**: ABCD • **Q83**: ABC • **Q84**: AC • **Q85**: ABCD

Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Ensemble Learning algorithms are true?

- **A.** Boosting algorithms train models sequentially to reduce bias
- **B.** Bagging algorithms build multiple models in parallel on bootstrap samples to reduce variance
- **C.** Ensemble models can only combine decision trees and prohibit neural networks
- **D.** Random Forest is an exemplary Bagging algorithm, whereas GBDT is a Boosting algorithm

Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following evaluation metrics are specifically designed for continuous regression rather than classification?

- **A.** Classification Accuracy Rate
- **B.** Precision and Recall Score
- **C.** Mean Squared Error
- **D.** Mean Absolute Error

Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are foundational architectural principles that allow CNNs to process images efficiently compared to Fully Connected Networks?

- **A.** Local receptive fields
- **B.** Weight sharing
- **C.** Full connection
- **D.** Recurrent feedback

Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are operations supported by the Python string data type?

- **A.** string.join()
- **B.** string.strip()
- **C.** string.replace()
- **D.** string.split()

Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are core components and services of Huawei's Cloud EI (Enterprise Intelligence) ecosystem?

- **A.** ModelArts one-stop AI development and lifecycle platform
- **B.** Graph Engine Service for large-scale graph analysis
- **C.** Image Recognition, OCR, and Natural Language Processing Cloud APIs
- **D.** Relational Database Schema Migration Service

Page Answers: **Q86**: ABD • **Q87**: CD • **Q88**: AB • **Q89**: ABCD • **Q90**: ABC

Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei MindSpore framework, which of the following statements regarding neural network definition and execution are TRUE?

- **A.** Custom neural network models and custom layers must inherit from mindspore.nn.Cell
- **B.** The forward computation logic of the network must be implemented inside the construct member function
- **C.** MindSpore requires manual symbolic derivation of all backward gradient formulas
- **D.** MindSpore supports both PyNative and Graph execution modes

Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei Ascend AI computing architecture, which of the following components constitute the CANN (Compute Architecture for Neural Networks) software stack?

- **A.** Ascend Computing Language
- **B.** Tensor Boost Engine operator development and compilation tool
- **C.** Fusion Engine for automated graph and operator fusion optimization
- **D.** Graph Engine for execution graph optimization, allocation, and stream management

Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's AI processor series built on the DaVinci architecture is named ____.

Answer: _____

Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

In the Transformer Self-Attention mechanism, the dot product between query matrix Q and key matrix K requires computing the ____ of matrix K (denoted as K^T).

Answer: _____

Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's computing platform that includes AI modules, cards, edge stations, and servers is named ____.

Answer: _____

Page Answers: **Q91**: ABD • **Q92**: ABCD • **Q93**: Ascend • **Q94**: transpose • **Q95**: Atlas

Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

In Huawei Cloud, the end-to-end one-stop AI development and lifecycle management platform is called ____.

Answer: _____

Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

The tool in ModelArts that supports device-edge-cloud joint development and execution is called Model____.

Answer: _____

Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

In Python, the data structure defined with curly braces {key: value} is called a ____.

Answer: _____

Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

The DaVinci architecture features a specialized 3D ____ computing engine optimized for tensor matrix multiplications.

Answer: _____

Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's edge-cloud collaborative vision AI development platform is called Hi____.

Answer: _____

Page Answers: **Q96**: ModelArts • **Q97**: Box • **Q98**: dictionary (or dict) • **Q99**: Cube • **Q100**: Lens

HCIA-AI V4.0 Certification Practice Exam • Practice Set 2

Module 9: Practice Exam Set 2 (100 Advanced Scenario-Based Questions)

Curated by: Umed Jamal Nouri • v1.1 (Nov 2026) • github.com/UmedJaMala

Question 1

TRUE OR FALSE • 16 Points

In Decision Tree algorithms, post-pruning is executed before the decision tree begins growing, whereas pre-pruning is executed only after the tree is fully grown.

A. True
B. False

Question 2

TRUE OR FALSE • 16 Points

The K-Means clustering algorithm is highly sensitive to the initial selection of cluster centroids and can easily converge to local optima.

A. True
B. False

Question 3

TRUE OR FALSE • 16 Points

When a machine learning model performs exceptionally well on the training dataset but exhibits very poor accuracy on unseen test data, the model has high bias.

A. True
B. False

Question 4

TRUE OR FALSE • 16 Points

In Huawei MindSpore, a mindspore.Tensor object can be directly converted into a standard NumPy array using its .asnumpy() member method.

A. True
B. False

Question 5

TRUE OR FALSE • 16 Points

In Huawei Ascend AI processors, the Unified Buffer (UB) is used to temporarily store input and output tensor data for the Vector Unit and Cube Unit.

A. True
B. False

Page Answers: **Q1**: False • **Q2**: True • **Q3**: False • **Q4**: True • **Q5**: True

Question 6

TRUE OR FALSE • 16 Points

A single-layer perceptron can successfully classify linearly non-separable problems such as the XOR logic function.

A. True
B. False

Question 7

TRUE OR FALSE • 16 Points

In convolutional neural networks, Max Pooling layers contain learnable weight parameters that are updated via backpropagation during network training.

A. True
B. False

Question 8

TRUE OR FALSE • 16 Points

In Python, a dictionary key can be any mutable object, such as a Python list, dictionary, or set.

A. True
B. False

Question 9

TRUE OR FALSE • 16 Points

In a Generative Adversarial Network (GAN), the Generator aims to maximize the probability that the Discriminator makes a mistake, while the Discriminator aims to distinguish real data from fake data.

A. True
B. False

Question 10

TRUE OR FALSE • 16 Points

In Huawei ModelArts, ExeML is designed for experienced AI algorithm engineers who want to develop complex custom distributed training scripts from scratch.

A. True
B. False

Page Answers: **Q6**: False • **Q7**: False • **Q8**: False • **Q9**: True • **Q10**: False

Question 11

TRUE OR FALSE • 16 Points

Feature scaling (such as Min-Max normalization or Z-score standardization) does not alter the split decisions or prediction results of tree-based models such as Decision Trees.

A. True
B. False

Question 12

TRUE OR FALSE • 16 Points

Static word embeddings such as Word2Vec and GloVe generate dynamic, context-dependent word representations where polysemous words receive different vectors based on sentence context.

A. True
B. False

Question 13

TRUE OR FALSE • 16 Points

In deep neural networks, the standard ReLU activation function completely avoids the vanishing gradient problem across both positive and negative input domains.

A. True
B. False

Question 14

TRUE OR FALSE • 16 Points

The ATC (Ascend Tensor Compiler) tool provided by Huawei CANN can convert open-source framework models (such as ONNX, Caffe, and MindSpore) into offline models (.om) supported by Ascend processors.

A. True
B. False

Question 15

TRUE OR FALSE • 16 Points

In supervised machine learning, Logistic Regression is actually a classification algorithm used to predict continuous numeric target values.

A. True
B. False

Page Answers: **Q11**: True • **Q12**: False • **Q13**: False • **Q14**: True • **Q15**: False

Question 16

TRUE OR FALSE • 16 Points

In Huawei MindSpore, executing 'context.set_context(mode=context.GRAPH_MODE)' configures the framework to run in static graph mode for maximum operational efficiency and deployment performance.

A. True
B. False

Question 17

TRUE OR FALSE • 16 Points

The ROC curve plots the True Positive Rate (TPR) on the vertical axis against the False Positive Rate (FPR) on the horizontal axis across all possible classification decision thresholds.

A. True
B. False

Question 18

TRUE OR FALSE • 16 Points

Random Forest builds multiple decision trees using Bagging (bootstrap aggregating) and random feature subsampling, which primarily acts to reduce model variance.

A. True
B. False

Question 19

TRUE OR FALSE • 16 Points

In Huawei Ascend AI processor architecture, the Matrix Multiply Unit (Cube Core) is primarily responsible for scalar control flow execution and address calculations.

A. True
B. False

Question 20

TRUE OR FALSE • 16 Points

In Transformer architecture, Multi-Head Attention allows the network to jointly attend to information from different representation subspaces at different positions.

A. True
B. False

Page Answers: **Q16**: True • **Q17**: True • **Q18**: True • **Q19**: False • **Q20**: True

Question 21

TRUE OR FALSE • 16 Points

In deep learning multi-class classification problems, Categorical Cross-Entropy Loss is the most commonly used loss function.

A. True
B. False

Question 22

TRUE OR FALSE • 16 Points

For any real-valued input, the Sigmoid activation function maps it to an output within the interval (0, 1).

A. True
B. False

Question 23

TRUE OR FALSE • 16 Points

The classical ID3 Decision Tree algorithm uses the Gini Index as its fundamental splitting criterion.

A. True
B. False

Question 24

TRUE OR FALSE • 16 Points

The CART (Classification and Regression Trees) algorithm uses the Gini Index as its node impurity splitting criterion.

A. True
B. False

Question 25

TRUE OR FALSE • 16 Points

Dropout is a deep learning regularization technique that works by randomly setting a specified fraction of neuron activations to zero during training forward passes.

A. True
B. False

Page Answers: **Q21**: True • **Q22**: True • **Q23**: False • **Q24**: True • **Q25**: True

Question 26

SINGLE-ANSWER QUESTION • 16 Points

According to the HCIA-AI V4.0 Model File specification, which format serves as PyTorch's default file format for saving and loading complete models (structure and parameters)?

- **A.** .csv or .tsv
- **B.** .pt or .pth
- **C.** .html or .xml
- **D.** .wav or .mp3

Question 27

SINGLE-ANSWER QUESTION • 16 Points

In Convolutional Neural Networks, what is the primary role of 1x1 convolutions?

- **A.** Capturing long-range spatial context across large image dimensions
- **B.** Replacing spatial max-pooling subsampling operations completely
- **C.** Doubling height and width pixel resolution via bilinear upsampling
- **D.** Cross-channel interaction and feature dimensionality reduction

Question 28

SINGLE-ANSWER QUESTION • 16 Points

Which of the following ensemble learning algorithms is based on the Boosting methodology?

- **A.** Random Forest
- **B.** Extra Trees
- **C.** Bagging
- **D.** AdaBoost

Question 29

SINGLE-ANSWER QUESTION • 16 Points

In Huawei CANN software architecture, what is the core responsibility of TBE (Tensor Boost Engine)?

- **A.** Managing high-speed physical PCIe hardware bus communications
- **B.** Custom neural operator development, optimization, and code compilation
- **C.** Routing cloud REST web API tokens and tracking compute billing
- **D.** Ingesting raw camera video streams and decoding H.265 frames

Question 30

SINGLE-ANSWER QUESTION • 16 Points

In mathematical optimization, which optimizer maintains an exponential moving average of squared past gradients to adaptively scale the learning rate?

- **A.** Momentum
- **B.** RMSProp
- **C.** Batch GD
- **D.** Mini-batch SGD

Page Answers: **Q26**: B • **Q27**: D • **Q28**: D • **Q29**: B • **Q30**: B

Question 31

SINGLE-ANSWER QUESTION • 16 Points

Given an input image of size 32x32, a convolution kernel of size 5x5, stride of 1, and no padding (valid padding), what is the size of the output feature map?

- **A.** 30x30
- **B.** 27x27
- **C.** 28x28
- **D.** 32x32

Question 32

SINGLE-ANSWER QUESTION • 16 Points

Which of the following clustering algorithms does NOT require the practitioner to pre-specify the number of clusters (K)?

- **A.** Gaussian Mixture Models with Expectation-Maximization
- **B.** Density-Based Spatial Clustering of Applications
- **C.** Standard K-Medoids Partitioning Representative Algorithm
- **D.** Classic K-Means Centroid-Based Optimization Algorithm

Question 33

SINGLE-ANSWER QUESTION • 16 Points

In Huawei MindSpore, which dedicated package provides modules for dataset loading, data pipeline processing, and data augmentation?

- **A.** mindspore.ops
- **B.** mindspore.train
- **C.** mindspore.nn
- **D.** mindspore.dataset

Question 34

SINGLE-ANSWER QUESTION • 16 Points

What is the primary structural limitation of standard recurrent neural networks (RNNs)?

- **A.** Vanishing and exploding gradient issues over long sequential time horizons
- **B.** Inability to execute backpropagation through standard reverse autograd
- **C.** Extremely high memory fragmentation caused by static matrix dimensions
- **D.** Inability to process sequential inputs of variable token lengths

Question 35

SINGLE-ANSWER QUESTION • 16 Points

In an LSTM cell, which gating structure determines how much historical information from the previous cell state C_(t-1) should be discarded?

- **A.** Output Gate
- **B.** Input Gate
- **C.** Modulation Gate
- **D.** Forget Gate

Page Answers: **Q31**: C • **Q32**: B • **Q33**: D • **Q34**: A • **Q35**: D

Question 36

SINGLE-ANSWER QUESTION • 16 Points

Which structural mechanism in the Transformer network is designed to preserve the sequential order of words without using recurrent loops?

- **A.** Feed-Forward Network
- **B.** Multi-Head Attention
- **C.** Layer Normalization
- **D.** Positional Encoding

Question 37

SINGLE-ANSWER QUESTION • 16 Points

In Support Vector Machines (SVM), what are the critical training sample points that lie directly on the margin boundaries called?

- **A.** Support Vectors
- **B.** Cluster Centroids
- **C.** Outlier Vectors
- **D.** Boundary Vectors

Question 38

SINGLE-ANSWER QUESTION • 16 Points

In Python NumPy, what is the resulting array shape when an array of shape (4, 1) is added to an array of shape (1, 3) using broadcasting rules?

- **A.** (4, 3)
- **B.** (1, 3)
- **C.** (4, 1)
- **D.** (3, 4)

Question 39

SINGLE-ANSWER QUESTION • 16 Points

In Huawei ModelArts, what is the smart data labeling technology that uses pre-trained models to accelerate annotation?

- **A.** HiLens Edge Camera Deployment Pipeline
- **B.** ExeML Automated Training Script Parser
- **C.** Semi-Automated or Smart Annotation Engine
- **D.** Rule-based regular expression parser

Question 40

SINGLE-ANSWER QUESTION • 16 Points

In binary classification evaluation, if a model yields 90 True Positives, 10 False Negatives, 20 False Positives, and 80 True Negatives, what is the Recall?

- **A.** 0.75
- **B.** 0.90
- **C.** 0.80
- **D.** 0.82

Page Answers: **Q36**: D • **Q37**: A • **Q38**: A • **Q39**: C • **Q40**: B

Question 41

SINGLE-ANSWER QUESTION • 16 Points

Which of the following problems represents an Unsupervised Learning task?

A. Segmenting store customers into behavioral clusters based on purchasing patterns without predefined labels
B. Classifying incoming emails into spam or legitimate inbox messages using thousands of human-labeled records
C. Estimating real estate sale valuations based on historical floor area and municipal sales registries
D. Detecting fraudulent financial credit transactions based on verified historical fraudulent banking records

Question 42

SINGLE-ANSWER QUESTION • 16 Points

What is the training phenomenon called when gradient magnitudes grow exponentially during backpropagation, causing weights to oscillate or produce NaN values?

A. Vanishing Gradient
B. Dying ReLU
C. Underfitting
D. Exploding Gradient

Question 43

SINGLE-ANSWER QUESTION • 16 Points

Which landmark deep learning architecture first introduced Residual Skip Connections (x + F(x)) to enable training of networks exceeding 100 layers?

A. VGGNet
B. Inception-V3
C. ResNet
D. AlexNet

Question 44

SINGLE-ANSWER QUESTION • 16 Points

In Huawei MindSpore, what is the primary role of the 'mindspore.train.callback.ModelCheckpoint' callback?

A. Visualizing real-time loss curves on external web dashboards
B. Partitioning training tensors across multi-node GPU clusters
C. Terminating model training if intermediate loss values diverge
D. Periodically saving model weights and parameters to checkpoint files

Question 45

SINGLE-ANSWER QUESTION • 16 Points

In Huawei CANN software stack, which core engine is responsible for whole-graph compilation, memory planning, and operator stream scheduling on Ascend chips?

A. ModelArts Studio
B. Graph Engine
C. MindSpore Lite
D. TBE Compiler Engine

Page Answers: **Q41**: A • **Q42**: D • **Q43**: C • **Q44**: D • **Q45**: B

Question 46

SINGLE-ANSWER QUESTION • 16 Points

Which of the following mathematical activation functions produces outputs strictly in the range (-1, 1)?

- **A.** Softplus Activation Curve
- **B.** Hyperbolic Tangent
- **C.** Rectified Linear Unit
- **D.** Classic Sigmoid Function

Question 47

SINGLE-ANSWER QUESTION • 16 Points

In Python, what is the exact output of evaluating the expression '[1, 2, 3] * 2'?

- **A.** [1, 2, 3, 1, 2, 3]
- **B.** [2, 4, 6]
- **C.** [[1, 2, 3], [1, 2, 3]]
- **D.** TypeError: cannot multiply sequence by non-int

Question 48

SINGLE-ANSWER QUESTION • 16 Points

In Natural Language Processing, which pretrained language model uses a bidirectional Transformer encoder trained on Masked Language Modeling and Next Sentence Prediction?

- **A.** Word2Vec
- **B.** GPT-2
- **C.** BERT
- **D.** TextBlob

Question 49

SINGLE-ANSWER QUESTION • 16 Points

Which class of probabilistic machine learning models does the Naive Bayes algorithm belong to?

- **A.** Generative Model
- **B.** Reinforcement Learning Policy
- **C.** Discriminative Model
- **D.** Non-parametric Metric Model

Question 50

SINGLE-ANSWER QUESTION • 16 Points

In the Huawei Atlas AI product family, which hardware device is designed as a standard PCIe accelerator card plugged into servers for high-density inference?

- **A.** Atlas 200
- **B.** Atlas 900
- **C.** Atlas 300
- **D.** Atlas 500

Page Answers: **Q46**: B • **Q47**: A • **Q48**: C • **Q49**: A • **Q50**: C

Question 51

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of Principal Component Analysis (PCA) in machine learning?

- **A.** Reducing feature dimensionality while preserving as much data variance as possible
- **B.** Pruning deep decision branches to prevent overfitting in trees
- **C.** Normalizing raw model logits into a probability distribution summing to 1
- **D.** Measuring the harmonic mean between precision and recall

Question 52

SINGLE-ANSWER QUESTION • 16 Points

Which classification evaluation metric represents the harmonic mean of Precision and Recall?

- **A.** Area Under the ROC Curve
- **B.** Specificity
- **C.** F1-Score
- **D.** Accuracy

Question 53

SINGLE-ANSWER QUESTION • 16 Points

In Huawei MindSpore, which function is used to export a trained network model into the unified MindIR format for cross-platform deployment?

- **A.** mindspore.dump()
- **B.** mindspore.load()
- **C.** mindspore.save_checkpoint()
- **D.** mindspore.export()

Question 54

SINGLE-ANSWER QUESTION • 16 Points

In Computer Vision object detection, which of the following algorithms is a classic One-Stage (single-shot) detector?

- **A.** You Only Look Once Detector
- **B.** Region-Based Convolutional Net
- **C.** Fast R-CNN
- **D.** Faster R-CNN

Question 55

SINGLE-ANSWER QUESTION • 16 Points

What is the primary effect of adding L2 regularization (weight decay) into neural network training?

- **A.** Penalizing large weights by adding the sum of squared weights, encouraging small weights and lower variance
- **B.** Enforcing exact parameter sparsity by driving numerous individual weight coefficients precisely to zero
- **C.** Dynamically increasing the learning rate step size whenever validation accuracy plateaus during training
- **D.** Removing duplicated training data samples automatically from the training split during each epoch

Page Answers: **Q51**: A • **Q52**: C • **Q53**: D • **Q54**: A • **Q55**: A

Question 56

SINGLE-ANSWER QUESTION • 16 Points

Which of the following statements correctly highlights the core difference between Supervised and Unsupervised Learning?

A. Supervised learning optimizes continuous regression targets, whereas unsupervised predicts discrete classes

B. Unsupervised learning requires fully annotated target labels, whereas supervised operates only on unlabeled features

C. Supervised learning cannot be used for classification problems, whereas unsupervised learning handles multi-class targets

D. Supervised learning maps features to ground-truth labels, while unsupervised learning discovers intrinsic data structure

Question 57

SINGLE-ANSWER QUESTION • 16 Points

In Huawei Ascend AI processor DaVinci architecture, which specialized hardware unit executes dense 16x16x16 matrix multiplications in a single clock cycle?

A. Control Unit

B. Scalar Unit

C. Cube Unit

D. Vector Unit

Question 58

SINGLE-ANSWER QUESTION • 16 Points

In Reinforcement Learning, which function estimates the expected cumulative future reward starting from state s and following policy pi?

A. Value Function or Q-Function

B. Loss Function

C. Softmax Function

D. State Transition Matrix

Question 59

SINGLE-ANSWER QUESTION • 16 Points

In Python, which keyword is placed inside a function body to turn it into a generator function that yields values on demand?

A. pass

B. return

C. async

D. yield

Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following benefits are achieved by applying Batch Normalization in deep neural networks? (Multiple Choice)

A. It reduces internal covariate shift by normalizing layer activations across each mini-batch

B. It allows the use of greatly higher learning rates without causing training instability

C. It reduces the network's sensitive dependence on initial parameter weight values

D. It completely eliminates the need for activation functions in hidden layers

Page Answers: **Q56**: D • **Q57**: C • **Q58**: A • **Q59**: D • **Q60**: ABC

Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei ModelArts platform and MindSpore distributed architecture, which acceleration parallelism modes enhance large-scale model training efficiency?

- **A.** Data parallelism
- **B.** Pipeline parallelism
- **C.** Model pruning
- **D.** Tensor parallelism

Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

In Linear Regression modeling, which of the following conditions or assumptions are evaluated?

- **A.** Linearity between independent explanatory features and the continuous target
- **B.** Homoscedasticity
- **C.** Absence of severe multicollinearity among explanatory input features
- **D.** Residuals are assumed to be approximately normally distributed around zero

Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

In Machine Learning data preparation, which of the following approaches can be used to handle missing attribute values?

- **A.** Fabricating synthetic random continuous labels for unobserved training targets
- **B.** Predicting missing values using machine learning algorithms
- **C.** Deleting sample records or features that exhibit excessive missingness rates
- **D.** Imputing missing values with statistical measures

Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are hyperparameters configured prior to training rather than internal parameters learned via backpropagation?

- **A.** Learning Rate
- **B.** Number of Hidden Layers or Network Depth
- **C.** Weight Matrices of Convolutional Kernels
- **D.** Bias Vectors of Dense Output Neurons

Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following factors commonly contribute to Overfitting in deep neural network models?

- **A.** The training dataset is too small relative to model capacity
- **B.** Applying heavy L2 regularization and high dropout rates
- **C.** The model architecture has an excessively large number of parameters
- **D.** Training for too many epochs without early stopping mechanisms

Page Answers: **Q61**: ABD • **Q62**: ABCD • **Q63**: BCD • **Q64**: AB • **Q65**: ACD

Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei CANN software architecture, which of the following components are part of the compute software stack?

- **A.** Ascend Computing Language
- **B.** Fusion Engine
- **C.** Operator compilation tool
- **D.** Graph Engine

Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding the K-Means clustering algorithm are TRUE?

- **A.** K-Means is sensitive to extreme outlier points
- **B.** The algorithm iterates between sample assignment to centroids and updating centroid coordinates
- **C.** K-Means results are completely fixed and independent of initial centroid placement
- **D.** K-Means assumes spherical clusters of approximately equal size and variance

Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following mechanisms are combined in the Adam optimizer to accelerate convergence?

- **A.** Momentum
- **B.** RMSProp
- **C.** Lasso
- **D.** Ridge

Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei MindSpore framework, which of the following capabilities are natively supported?

- **A.** Dynamic switching between PyNative and Graph execution modes
- **B.** Automatic Differentiation based on source-to-source transformation
- **C.** Device-Edge-Cloud synergy using unified MindIR representation
- **D.** Auto-Parallelism spanning data, model, and pipeline parallel modes

Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are foundational architectural building blocks of the Transformer model?

- **A.** Multi-Head Self-Attention mechanisms
- **B.** Position-wise Feed-Forward Neural Networks
- **C.** Layer Normalization and residual shortcut connections
- **D.** Bidirectional Recurrent Gated Memory Cells

Page Answers: **Q66**: ABCD • **Q67**: ABD • **Q68**: AB • **Q69**: ABCD • **Q70**: ABC

Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

In Python, which of the following statements regarding functions are TRUE?

A. Functions are first-class citizens and can be assigned to variables
B. Functions can be passed as arguments into other functions
C. Functions can be stored in container structures like lists and dictionaries
D. Functions can return other functions as return values

Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following evaluation tools and metrics are specifically designed for continuous regression rather than classification?

A. Mean Squared Error
B. Coefficient of Determination
C. Confusion Matrix Evaluation Grid
D. Area Under the ROC Curve

Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

According to the HCIA-AI V4.0 Model File specification, which of the following statements accurately describe standard deep learning model file formats?

A. Checkpoint (.ckpt) uses Protocol Buffers to store parameter values by default and supports resuming interrupted training or fine-tuning
B. ONNX (.onnx) provides a universal model representation for migrating models across frameworks or running on inference engines like TensorRT
C. PyTorch (.pt/.pth) is PyTorch's default format for saving and loading complete models or trained weights
D. Generic Binary (.bin) is a common binary format into which PyTorch models must be converted when required

Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei Atlas AI product family, which hardware devices are specifically engineered for cloud and data center large-scale training?

A. Atlas 500
B. Atlas 200
C. Atlas 900
D. Atlas 800

Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following characteristics differentiate Deep Learning from Traditional Machine Learning?

A. Automatic end-to-end feature representation learning from raw data
B. Superior performance scaling when trained on massive big datasets
C. Supervised Gradient Backpropagation into Raw Features
D. Guaranteed execution on low-power microcontrollers with zero compute

Page Answers: **Q71**: ABCD • **Q72**: AB • **Q73**: ABCD • **Q74**: CD • **Q75**: AB

Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

In Convolutional Neural Networks, which benefits are primarily achieved by applying Pooling (Subsampling) layers?

- **A.** Reducing spatial feature map dimensionality and parameter counts
- **B.** Providing spatial translation invariance to minor input shifts
- **C.** Expanding the effective receptive field of subsequent convolutional layers
- **D.** Converting floating-point tensors into ASCII text characters

Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following operations are standard data preprocessing steps in Natural Language Processing (NLP)?

- **A.** Lowercasing and punctuation text normalization
- **B.** Morphological stemming and lemmatization
- **C.** Autoregressive beam search decoding and temperature scaling
- **D.** Lexical tokenization and stopword removal

Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei ModelArts platform, which of the following lifecycle services are provided?

- **A.** Interactive development environment with cloud-hosted JupyterLab notebooks
- **B.** Multi-modal data ingestion, dataset management, and automated labeling
- **C.** Distributed model training with automatic hyperparameter optimization
- **D.** One-click deployment to real-time cloud REST APIs or edge devices

Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Gradient Descent algorithms are TRUE?

- **A.** Momentum helps accelerate convergence along shallow dimensions and suppresses oscillations
- **B.** Stochastic Gradient Descent updates parameters based on a single training example per step
- **C.** Mini-batch Gradient Descent balances vectorization speed and gradient estimation stability
- **D.** Standard SGD is mathematically guaranteed to converge monotonically without stochastic noise

Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following tasks are typical industrial applications of Computer Vision?

- **A.** Object detection and multi-object tracking
- **B.** Optical Character Recognition
- **C.** Syntactic dependency tree parsing in text datasets
- **D.** Semantic and instance image segmentation

Page Answers: **Q76**: ABC • **Q77**: ABD • **Q78**: ABCD • **Q79**: ABC • **Q80**: ABD

Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

In Machine Learning, which of the following techniques can be used for dimensionality reduction?

- **A.** Linear Discriminant Analysis
- **B.** Principal Component Analysis
- **C.** Feature selection methods
- **D.** t-Distributed Stochastic Neighbor Embedding

Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei Ascend AI architecture, which of the following architectural properties describe the DaVinci NPU core?

- **A.** High-throughput 3D Cube computing engine optimized for matrix multiplication
- **B.** Unified Buffer architecture enabling high on-chip bandwidth data reuse
- **C.** Native hardware support for mixed-precision computation
- **D.** Dynamic clock throttling during all matrix multiplication operations

Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about the Sigmoid and Tanh activation functions are correct?

- **A.** The Sigmoid function output is zero-centered around 0
- **B.** The Sigmoid function compresses any input into the interval (0, 1)
- **C.** The Tanh function compresses input into the interval (-1, 1) and is zero-centered
- **D.** The ReLU function output is strictly confined to the range (-1, 1)

Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Decision Trees are TRUE?

- **A.** Decision trees strictly require all numerical inputs to be normalized to [0, 1]
- **B.** Small variations in training data can lead to completely different tree structures
- **C.** Deep unpruned decision trees have high variance and are prone to overfitting
- **D.** Decision trees are non-parametric and offer clear, interpretable decision logic

Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei MindSpore, which of the following statements regarding Tensors are correct?

- **A.** Tensors support slicing, tensor broadcasting, and mathematical operations
- **B.** Tensors in MindSpore cannot be converted back to NumPy ndarrays
- **C.** A Tensor is a multi-dimensional array data structure storing homogeneous data types
- **D.** Tensors can be instantiated directly from Python lists, tuples, or NumPy ndarrays

Page Answers: **Q81**: ABCD • **Q82**: ABC • **Q83**: BC • **Q84**: BCD • **Q85**: ACD

Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following architectural types represent foundational transformer attention variants in modern LLMs?

- **A.** CNN architecture
- **B.** RNN architecture
- **C.** Decoder-only architecture
- **D.** Encoder-decoder architecture

Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

In LLM text generation, which decoding hyperparameters directly modulate output randomness and candidate token diversity?

- **A.** Temperature
- **B.** Top-P or Nucleus Sampling
- **C.** Training Epoch Iteration Count
- **D.** Learning Rate Warmup Decay Steps

Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following represent recognized main methods of Ensemble Learning in machine learning?

- **A.** Bagging
- **B.** Boosting
- **C.** Stacking or Blending
- **D.** Random Single-Tree Pruning without aggregation

Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

In Python Object-Oriented Programming (OOP), which of the following statements are correct?

- **A.** The 'self' parameter explicitly references the specific class instance
- **B.** Python natively supports multiple inheritance
- **C.** The __init__ method acts as the class constructor executed during object instantiation
- **D.** Private attributes and methods are prefixed with double underscores

Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

A Generative Adversarial Network (GAN) mainly consists of which competing neural subnetworks trained in a minimax game?

- **A.** Encoder
- **B.** Decoder
- **C.** Generator
- **D.** Discriminator

Page Answers: **Q86**: CD • **Q87**: AB • **Q88**: ABC • **Q89**: ABCD • **Q90**: CD

Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei Ascend computing, which of the following capabilities are provided by the AscendCL (Ascend Computing Language) API?

- **A.** Loads, initializes, and executes offline model files
- **B.** Provides unified C/C++ and Python programming interfaces
- **C.** Provides stream management APIs for asynchronous operator dispatch and synchronization
- **D.** Manages Device memory allocation, copy, and deallocation

Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following architectural strategies help prevent or reduce the Vanishing Gradient problem in deep networks?

- **A.** Using proper weight initialization techniques
- **B.** Using Residual Skip Connections
- **C.** Using non-saturating activation functions such as ReLU and Leaky ReLU
- **D.** Initializing all network weights to identical constant zero values

Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

On a convolution layer of a convolutional neural network, if there are 128 3 x 3 convolution kernels and the size of the input feature map is 28 x 28 x 64, the depth of a convolution kernel is ____. (Enter a number.)

Answer: _____

Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

CNN stands for ____. (Enter the full name.)

Answer: _____

Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

The ____ algorithm is based on the idea that if most of a sample's k nearest neighbors in the feature space belong to the same class, the sample also belongs to that class. (Enter an abbreviation.)

Answer: _____

Page Answers: **Q91**: ABCD • **Q92**: ABC • **Q93**: 64 • **Q94**: Convolutional Neural Network • **Q95**: KNN

Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

In multi-class classification tasks, the sum of all output probability values of the Softmax function is ____. (Enter a number.)

Answer: _____

Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

In k-fold cross-validation, the existing dataset is divided into ____ groups.

Answer: _____

Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

The three essential elements for AI development are data, compute, and ____.

Answer: _____

Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

A/An ____ is a binary classification model that is based on a linear classifier defined in the feature space that maximizes the margin. (Enter an abbreviation.)

Answer: _____

Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

In deep learning optimization, the technique that normalizes layer inputs across each mini-batch during training is called ____ Normalization.

Answer: _____

Page Answers: **Q96**: 1 • **Q97**: k • **Q98**: algorithms (or algorithm) • **Q99**: SVM • **Q100**: Batch